PMATH 450 Measure Theory

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1 Measure

1.1 Introduction

Problem: Consider a random (infinite) string of 0's and 1's $(\ell_1, \ell_2, \dots, \ell_m, \dots)$ where $\ell_n \in \{0,1\}$, $\forall m \in \mathbb{N}$. What is the probability that somewhere in the string there are 100 consecutive 0's?

Comment: the statement of the problem is unclear. Lebesgue's machinary will clarify this.

Let $X = \{x = (\ell_1, \ell_2, \dots, \ell_m, \dots) \mid \ell_m \in \{0,1\}, \forall m \in \mathbb{N}\}$. This is, in a natural way, a compact metric space (use "Hamming-style" distance).

For $x = (\ell_1, \ell_2, \dots, \ell_m, \dots)$ and $x' = (\ell_1', \ell_2', \dots, \ell_m', \dots)$ in X. Define

$$d(x, x') = \frac{|\ell_1 - \ell_1'|}{2} + \frac{|\ell_2 - \ell_2'|}{4} + \cdots + \frac{|\ell_n - \ell_n'|}{2^n} + \cdots$$

In a metric space, we can consider

$$\mathcal{T} = \{G \subseteq X \mid G \text{ is open}\} \subseteq 2^X, \quad \text{where } 2^X = \{A \mid A \subseteq X\}.$$

Then, we can consider

$$\mathcal{B} = \sigma\text{- Alg}(\mathcal{T}),$$

the **Borel σ-algebra**.

Example. Probability is $\mathbb{P} : \mathcal{B} \to [0,1]$ such that $\mathbb{P}(B) \in [0,1]\ \forall B \in \mathcal{B}$.

"Problem" says: let $S = \{x = (\ell_1, \ell_2, \dots, \ell_m, \dots) \in X \mid \exists m \in \mathbb{N} \text{ s.t. } \ell_m = \ell_{m+1} = \cdots = \ell_{m+99} = 0\}$, then check that $S \in \mathcal{B}$, and compute $\mathbb{P}(S) = ?$

1.2 Additive Set-Function on An Algebra of Sets

Definition 1.1 (**Algebra of Sets**).

Let $\mathfrak{A} \subseteq 2^X$. We say that $\mathfrak{A}$ is an **algebra of sets** if it satisfies:

(AS1) $\varnothing \in \mathfrak{A}$.

(AS2) If $A \in \mathfrak{A}$, then $X \setminus A \in \mathfrak{A}$.

(AS3) If $A, B \in \mathfrak{A}$, then $A \cap B \in \mathfrak{A}$.

Proposition 1.1. Let $\mathfrak{A}$ be an algebra of sets of X. Then,

(1) $X \in \mathfrak{A}$.

(2) If $A, B \in \mathfrak{A}$, then $A \cup B \in \mathfrak{A}$.

(3) If $A, B \in \mathfrak{A}$, then $A \setminus B \in \mathfrak{A}$.

(4) For every $n \geq 2$ and every $A_1, \dots, A_n \in \mathfrak{A}$, we have $\bigcap_{i=1}^n A_i \in \mathfrak{A}$.

(5) For every $n \geq 2$ and every $A_1, \dots, A_n \in \mathfrak{A}$, we have $\bigcup_{i=1}^n A_i \in \mathfrak{A}$.

Proof.

(1) By AS1, $\varnothing \in \mathfrak{A}$. By AS2, $X \setminus \varnothing \in \mathfrak{A}$. Thus, $X \in \mathfrak{A}$.

(2) Suppose $A, B \in \mathfrak{A}$. By AS2, $X \setminus A, X \setminus B \in \mathfrak{A}$. By AS3, $(X \setminus A) \cap (X \setminus B) \in \mathfrak{A}$. By De Morgan's law, $X \setminus (A \cup B) \in \mathfrak{A}$. By AS2, $X \setminus (X \setminus (A \cup B)) = A \cup B \in \mathfrak{A}$.

(3) Exercise.

(4) Exercise.

(5) Exercise.

□

Example. $\mathfrak{A} = 2^X$ is an algebra of sets.

Example. $\mathfrak{A} = \{\varnothing, X\}$ is an algebra of sets.

Using the following lemma and proposition, given any $\mathfrak{C} \subseteq 2^X$, we have an algebra of sets generated by $\mathfrak{C}$.

Lemma 1.2. Let I be an index set (finite or infinite). Let $(\mathfrak{A}_i)_{i \in I}$ be a family of algebras of subsets of X. Then,

$$\mathfrak{A} = \bigcap_{i \in I} \mathfrak{A}_i \subseteq 2^X$$

is an algebra of subsets of X.

Proof. Since $\mathfrak{A}_i$ is an algebra of sets, $\varnothing \in \mathfrak{A}_i$, $\forall i \in I$. Hence, $\varnothing \in \mathfrak{A}$. Let $A \in \mathfrak{A}$. Then, $[A \in \mathfrak{A}_i, \forall i \in I] \implies [X \setminus A \in \mathfrak{A}_i, \forall i \in I] \implies [X \setminus A \in \mathfrak{A}]$. Let $A, B \in \mathfrak{A}$. Then, $[A, B \in \mathfrak{A}_i, \forall i \in I] \implies [A \cap B \in \mathfrak{A}_i, \forall i \in I] \implies [A \cap B \in \mathfrak{A}]$. □

Proposition 1.3. Let $\mathfrak{C} \subseteq 2^X$. There exists $\mathfrak{A}_\circ \subseteq 2^X$, uniquely determined, such that

(1) $\mathfrak{A}_\circ$ is an algebra of sets, and $\mathfrak{A}_\circ \supseteq \mathfrak{C}$.

(2) Whenever $\mathfrak{A} \subseteq 2^X$ is an algebra of sets such that $\mathfrak{A} \supseteq \mathfrak{C}$, we have $\mathfrak{A} \supseteq \mathfrak{A}_\circ$.

Note. $\mathfrak{A}_\circ$ is the smallest algebra of sets which contains $\mathfrak{C}$.

Proof. For existence, let $(\mathfrak{A}_i)_{i \in I}$ be the family of all algebras of sets of X which contains $\mathfrak{C}$ (such algebras exist, e.g. 2^X is one of them). Put $\mathfrak{A}_\circ := \bigcap_{i \in I} \mathfrak{A}_i \subseteq 2^X$. Then, $\mathfrak{A}_\circ$ is an algebra of sets by the previous lemma. Also, $\mathfrak{A}_\circ \supseteq \mathfrak{C}$ because $\mathfrak{A}_i \supseteq \mathfrak{C}$, $\forall i \in I$. Hence, $\mathfrak{A}_\circ$ satisfies condition (1).

Verification of condition (2): Let $\mathfrak{A} \subseteq 2^X$ be an algebra of sets such that $\mathfrak{A} \supseteq \mathfrak{C}$. Then, $\exists j \in I$ such that $\mathfrak{A} = \mathfrak{A}_j$. But then,

$$\mathfrak{A} = \mathfrak{A}_j \supseteq \bigcap_{i \in I} \mathfrak{A}_i = \mathfrak{A}_\circ.$$

So $\mathfrak{A} \supseteq \mathfrak{A}_\circ$.

For uniqueness, let $\mathfrak{A}'_\circ$ be another algebra of sets which satisfies (1) and (2). Then, $\mathfrak{A}'_\circ \supseteq \mathfrak{A}_\circ$ by (2) applied to $\mathfrak{A}'_\circ$. Also, $\mathfrak{A}_\circ \supseteq \mathfrak{A}'_\circ$ by (2) applied to $\mathfrak{A}_\circ$. Hence, $\mathfrak{A}_\circ = \mathfrak{A}'_\circ$. □

Definition 1.2 (Algebra of Sets Generated by $\mathfrak{C}$).
Let $\mathfrak{C} \subseteq 2^X$. The algebra of sets $\mathfrak{A}_\circ$ from Proposition 1.3 is called the **algebra of sets generated by** $\mathfrak{C}$, denoted by $\mathrm{Alg}(\mathfrak{C})$.

Lecture 2, 2026/05/13

Definition 1.3 (**Additive Set-Function**).

Let $\mathfrak{A} \subseteq 2^X$ be an algebra of sets. A function $\mu : \mathfrak{A} \to [0, \infty]$ is called an **additive set-function** if $\mu(\varnothing) = 0$ and $\mu(A \cup B) = \mu(A) + \mu(B)$ whenever $A, B \in \mathfrak{A}$ s.t. $A \cap B = \varnothing$.

Remark. We use the arithmetic of $[0, \infty]$. By convention in measure theory,

$$0 \cdot \infty = 0 = \infty \cdot 0.$$

Also, we avoid the subtraction $\infty - \infty$, which is undefined.

Remark (**Why do we require** $\mu(\varnothing) = 0$**?**).

It is to avoid the case when $\mu(A) = \infty, \forall A \in \mathfrak{A}$.

Proposition 1.4 (**Properties of an Additive Set-Function**).

Let $\mathfrak{A} \subseteq 2^X$ be an algebra of sets and $\mu : \mathfrak{A} \to [0, \infty]$ be an additive set-function. Then,

(1) μ is an increasing function, i.e. $[A, C \in \mathfrak{A}$ and $A \subseteq C] \implies [\mu(A) \leq \mu(C)]$.

(2) Sub-additivity: $\mu(A \cup B) \leq \mu(A) + \mu(B), \forall A, B \in \mathfrak{A}$.

(3) General finite additivity: For every $n \geq 2$ and $A_1, \dots, A_n \in \mathfrak{A}$ s.t. $A_i \cap A_j = \varnothing, \forall i \neq j$, we have $\mu\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i)$.

(4) General finite sub-additivity: For every $n \geq 2$ and $A_1, \dots, A_n \in \mathfrak{A}$, we have $\mu\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mu(A_i)$.

Proof.

(1) Note that $C = A \cup (C \setminus A)$, where $A, C \setminus A \in \mathfrak{A}$ and $A \cap (C \setminus A) = \varnothing$. Hence, $\mu(C) = \mu(A) + \mu(C \setminus A) \geq \mu(A)$.

(2) Note that $A \cup B = A \cup (B \setminus A)$, where $A, B \setminus A \in \mathfrak{A}$ and $A \cap (B \setminus A) = \varnothing$. Hence, $\mu(A \cup B) = \mu(A) + \mu(B \setminus A) \leq \mu(A) + \mu(B)$.

(3) Exercise.

(4) Exercise.

$\square$

Definition 1.4 (**σ-Algebra**).

We say that $\mathfrak{M} \subseteq 2^X$ is a **σ-algebra** if

(Sigma-AS1) $\varnothing \in \mathfrak{M}$.

(Sigma-AS2) If $A \in \mathfrak{M}$, then $X \setminus A \in \mathfrak{M}$.

(Sigma-AS3) If $A_1, A_2, \ldots \in \mathfrak{M}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathfrak{M}$.

Note. Sigma-AS3 is stronger than AS3.

Remark. Let $\mathfrak{M} \subseteq 2^X$ be a σ-algebra.

(1) $\mathfrak{M}$ is an algebra of sets.

(2) If $A_1, A_2, \ldots \in \mathfrak{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathfrak{M}$.

Lemma 1.5. Let I be an index set (finite or infinite). Let $(\mathfrak{M}_i)_{i \in I}$ be a family of σ-algebras of subsets of X. Then,

$$\mathfrak{M} = \bigcap_{i \in I} \mathfrak{M}_i \subseteq 2^X$$

is a σ-algebra of subsets of X.

Proposition 1.6. Let $\mathfrak{C} \subseteq 2^X$. There exists $\mathfrak{M}_\circ \subseteq 2^X$, uniquely determined, such that

(1) $\mathfrak{M}_\circ$ is a σ-algebra, and $\mathfrak{M}_\circ \supseteq \mathfrak{C}$.

(2) Whenever $\mathfrak{M} \subseteq 2^X$ is a σ-algebra such that $\mathfrak{M} \supseteq \mathfrak{C}$, we have $\mathfrak{M} \supseteq \mathfrak{M}_\circ$.

Definition 1.5 (**σ-Algebra Generated by $\mathfrak{C}$**).

Let $\mathfrak{C} \subseteq 2^X$. The σ-algebra $\mathfrak{M}_\circ$ from Proposition 1.6 is called the **σ-algebra generated by** $\mathfrak{C}$, denoted by $\sigma\text{-}\operatorname{Alg}(\mathfrak{C})$.

Note. This is the smallest σ-algebra which contains $\mathfrak{C}$.

Definition 1.6 (**Borel σ-algebra**).

Let (X, d) be a metric space and let $\mathcal{T}$ denote the collection of all open subsets of X. The σ-algebra $\mathcal{B}_X = \sigma\text{-}\operatorname{Alg}(\mathcal{T})$ is called the **Borel σ-algebra** of (X, d).

Note. $\mathcal{T} = \{G \subseteq X \mid G \text{ is open}\}$.

Problem P2.1: Let X be an infinite uncountable set. Let $\mathcal{F} \coloneqq \{F \subseteq X \mid F \text{ is finite}\}$ and let $\sigma\text{-}\operatorname{Alg}(\mathcal{F})$ be the σ-algebra generated by $\mathcal{F}$. Also, let

$$\mathcal{U} = \underbrace{\{B \subseteq X \mid B \text{ is countable}\}}_{\mathcal{U}_1} \cup \underbrace{\{B \subseteq X \mid X \setminus B \text{ is countable}\}}_{\mathcal{U}_2}.$$

(a) Prove that $\mathcal{U}$ is a σ-algebra of subsets of X.

(b) Prove that $\mathcal{U} = \sigma\text{-}\operatorname{Alg}(\mathcal{F})$.

Proof.

(a) Note that $\emptyset \in \mathcal{U}$. Let $A \in \mathcal{U}_1$. Then, $X \setminus A$ is in $\mathcal{U}_2$, hence $X \setminus B \in \mathcal{U}$. Let $B \in \mathcal{U}_2$. Then, $X \setminus B$ is in $\mathcal{U}_1$, hence $X \setminus B \in \mathcal{U}$. Now, let $S_1, S_2, \dots \in \mathcal{U}$. We have two cases:

Case α: If $\exists m \in \mathbb{N}$ such that $S_m \in \mathcal{U}_1$ (i.e. S_m is countable), then $\bigcap_{n=1}^{\infty} S_n \subseteq S_m$ is countable, hence $\bigcap_{n=1}^{\infty} S_n \in \mathcal{U}_1 \subseteq \mathcal{U}$.

Case β: If $S_n \in \mathcal{U}_2$ for all $n \in \mathbb{N}$ (i.e. $X \setminus S_n$ is countable for all $n \in \mathbb{N}$), then

$$X \setminus \bigcap_{n=1}^{\infty} S_n = \bigcup_{n=1}^{\infty} (X \setminus S_n)$$

is countable, hence $\bigcap_{n=1}^{\infty} S_n \in \mathcal{U}_2 \subseteq \mathcal{U}$.

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(b) We prove the following two claims.

Claim (1). $\mathcal{U} \supseteq \mathcal{F}$.

Proof of Claim 1. Let $F \in \mathcal{F}$. Then, F is finite, hence countable. So $F \in \mathcal{U}_1 \subseteq \mathcal{U}$. □

Claim (2). Let $\mathfrak{M} \subseteq 2^X$ be a σ-algebra such that $\mathfrak{M} \supseteq \mathcal{F}$. Then, $\mathfrak{M} \supseteq \mathcal{U}$.

Proof of Claim 2. We first check that every $A \in \mathcal{U}_1$ is in $\mathfrak{M}$. If A is finite, then $A \in \mathcal{F} \subseteq \mathfrak{M}$. If A is infinite countable, then write $A = \{x_1, x_2, \dots, x_n, \dots\} = \bigcup_{n=1}^{\infty} \{x_n\}$ and observe that $\{x_n\} \in \mathfrak{M}$, $\forall n \in \mathbb{N}$. Hence, $A = \bigcup_{n=1}^{\infty} \{x_n\} \in \mathfrak{M}$ since $\mathfrak{M}$ is stable under countable unions.

We now check that every $B \in \mathcal{U}_2$ is in $\mathfrak{M}$. Let $B \in \mathcal{U}_2$. Then, $X \setminus B \in \mathcal{U}_1$, hence $X \setminus B \in \mathfrak{M}$. By (Sigma-AS2), $B = X \setminus (X \setminus B) \in \mathfrak{M}$.

Hence, $\mathcal{U} = \mathcal{U}_1 \cup \mathcal{U}_2 \subseteq \mathfrak{M}$. □

Thus, $\mathcal{U}$ satisfies conditions (1) and (2) in Proposition 1.6. By the uniqueness part of Proposition 1.6, we have $\mathcal{U} = \sigma\text{- Alg}(\mathcal{F})$. □

1.3 Positive Measure

Remark.

(1) An increasing non-negative sequence always converges in $[0, \infty]$.

(2) Let $(t_n)_{n=1}^{\infty}$ be a sequence in $[0, \infty]$. Then, $\sum_{n=1}^{\infty} t_n = \lim_{k\to\infty} \sum_{n=1}^{k} t_n$ is well-defined in $[0, \infty]$.

Definition 1.7 (Positive Measure).
Let $\mathfrak{M}$ be a σ-algebra of subsets of X. A **positive measure** on $\mathfrak{M}$ is a function $\mu : \mathfrak{M} \to [0, \infty]$ such that $\mu(\varnothing) = 0$ and

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

whenever $A_1, A_2, \ldots \in \mathfrak{M}$ are such that $A_i \cap A_j = \varnothing, \forall i \neq j$.

Note. The two conditions are $\mu(\varnothing) = 0$ and sigma-additivity (Sigma-Add).

Remark. A positive measure is, in particular, a finitely additive set-function on $\mathfrak{M}$. Thus, it satisfies all the properties in Proposition 1.4.

Definition 1.8 (Measurable Space, Measure Space).
Let X be a set and $\mathfrak{M} \subseteq 2^X$ be a σ-algebra. Then,

(1) $(X, \mathfrak{M})$ is called a **measurable space**.

(2) If $\mu : \mathfrak{M} \to [0, \infty]$ is a positive measure, then $(X, \mathfrak{M}, \mu)$ is called a **measure space**.

Definition 1.9 (Finite Measure).
Let $(X, \mathfrak{M}, \mu)$ be a measure space. If $\mu(X) < \infty$ (which implies $\mu(A) \leq \mu(X) < \infty$ for all $A \in \mathfrak{M}$), then μ is called a **finite measure**.

Note. If $\mu(X) = 1$, then μ is a **probability measure** and $(X, \mathfrak{M}, \mu)$ is a **probability space**.

Definition 1.10 (Sigma-Finite Measure).

Let $(X, \mathfrak{M}, \mu)$ be a measure space. We say that μ is **sigma-finite** if $\exists$ a sequence of sets $(X_n)_{n=1}^{\infty}$ in $\mathfrak{M}$ such that

- $X_1 \subseteq X_2 \subseteq \cdots \subseteq X_n \subseteq \cdots$;
- $X = \bigcup_{n=1}^{\infty} X_n$;
- $\mu(X_n) < \infty, \forall n \in \mathbb{N}$.

Note. A finite measure is sigma-finite by taking $X_n = X$ for all $n \in \mathbb{N}$.

Proposition 1.7. Let $\mathfrak{M} \subseteq 2^X$ be a σ-algebra and let $\mu : \mathfrak{M} \to [0, \infty]$ be a positive measure. Then,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n), \quad \forall A_1, A_2, \ldots \in \mathfrak{M}.$$

Note. This property is called **countable sub-additivity** (Sigma-SubAdd). Also, $\sum_{n=1}^{\infty} \mu(A_n)$ is defined as the increasing limit

$$\sum_{n=1}^{\infty} \mu(A_n) = \lim_{k \to \infty} \sum_{n=1}^{k} \mu(A_n) = \sup\left\{\sum_{n=1}^{k} \mu(A_n) \mid k \in \mathbb{N}\right\}.$$

Proposition 1.8 (Continuity of a Positive Measure).

Let $\mathfrak{M} \subseteq 2^X$ be a σ-algebra and let $\mu : \mathfrak{M} \to [0, \infty]$ be a positive measure.

(1) Let $(B_n)_{n=1}^{\infty}$ be a sequence of sets in $\mathfrak{M}$ such that $B_1 \subseteq B_2 \subseteq \cdots \subseteq B_n \subseteq \cdots$. Then,

$$\mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \lim_{n \to \infty} \mu(B_n)$$

(2) Let $(C_n)_{n=1}^{\infty}$ be a sequence of sets in $\mathfrak{M}$ such that $C_1 \supseteq C_2 \supseteq \cdots \supseteq C_n \supseteq \cdots$ and $\mu(C_1) < \infty$ (which implies $\mu(C_n) \leq \mu(C_1) < \infty, \forall n \in \mathbb{N}$). Then,

$$\mu\left(\bigcap_{n=1}^{\infty} C_n\right) = \lim_{n \to \infty} \mu(C_n).$$

Proof.

(1) Let $B = \bigcup_{n=1}^{\infty} B_n \in \mathfrak{M}$. We want to show that $\mu(B) = \lim_{n\to\infty} \mu(B_n)$. Define $A_1 = B_1, A_2 = B_2 \setminus B_1, A_3 = B_3 \setminus B_2, \dots, A_n = B_n \setminus B_{n-1}, \dots$. Then, $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$ such that $A_i \cap A_j = \emptyset, \forall i \neq j$, and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n = B$. Hence,

$$\mu(B) = \mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n) = \lim_{n\to\infty} \sum_{i=1}^{n} \mu(A_i) = \lim_{n\to\infty} \mu\left(\bigcup_{i=1}^{n} A_i\right) = \lim_{n\to\infty} \mu(B_n).$$

(2) Let $C = \bigcap_{n=1}^{\infty} C_n \in \mathfrak{M}$. We want to show that $\mu(C) = \lim_{n\to\infty} \mu(C_n)$. Define $B_n = C_1 \setminus C_n$ for all $n \in \mathbb{N}$. Then, $(B_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$ such that $B_1 \subseteq B_2 \subseteq \cdots \subseteq B_n \subseteq \cdots$ and

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (C_1 \setminus C_n) = C_1 \setminus \bigcap_{n=1}^{\infty} C_n = C_1 \setminus C.$$

By part (1),

$$\lim_{n\to\infty} \mu(B_n) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \mu\left(C_1 \setminus C\right).$$

Also, $C_1 = B_n \sqcup C_n$ for all $n \in \mathbb{N}$, hence $\mu(C_1) = \mu(B_n) + \mu(C_n), \forall n \in \mathbb{N}$. Since $\mu(C_1) < \infty$, we have $\mu(B_n) = \mu(C_1) - \mu(C_n), \forall n \in \mathbb{N}$. Similarly,

$$\mu\left(C_1 \setminus C\right) = \mu(C_1) - \mu(C).$$

Hence,

$$\lim_{n\to\infty} \mu(B_n) = \lim_{n\to\infty} [\mu(C_1) - \mu(C_n)] = \mu(C_1) - \mu(C).$$

Therefore, $\mu(C) = \mu(C_1) - \lim_{n\to\infty}[\mu(C_1) - \mu(C_n)] = \lim_{n\to\infty} \mu(C_n)$.

□

Note. **Problem P2.3** shows if we allow $\mu(C_n) = \infty \ \forall n \in \mathbb{N}$, then the conclusion in (2) may fail.

Recall the following property.

Proposition 1.7. Let $\mathfrak{M} \subseteq 2^X$ be a σ-algebra and let $\mu : \mathfrak{M} \to [0, \infty]$ be a positive measure. Then,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n), \quad \forall A_1, A_2, \ldots \in \mathfrak{M}.$$

Proof. Let $A = \bigcup_{n=1}^{\infty} A_n \in \mathfrak{M}$. We want to show that $\mu(A) \leq \sum_{n=1}^{\infty} \mu(A_n)$. For every $m \in \mathbb{N}$, define $B_m = \bigcup_{n=1}^{m} A_n \in \mathfrak{M}$. Then, $(B_n)_{n=1}^{\infty}$ is an increasing chain with $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n = A$. Hence, by the continuity of μ,

$$\mu(A) = \lim_{n\to\infty} \mu(B_n) = \sup\{\mu(B_n) \mid n \in \mathbb{N}\} \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Exercise: check that $\mu(B_m) \leq \sum_{n=1}^{m} \mu(A_n)$ for all $m \in \mathbb{N}$. $\square$

Problem P2.4 (Borel-Cantelli)

Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $(A_n)_{n=1}^{\infty}$ be a family of sets in $\mathfrak{M}$. Consider the set

$$T := \{x \in X \mid x \text{ belongs to infinitely many of the } A_n\text{'s}\}.$$

(a) Prove that $T = \bigcap_{n=1}^{\infty}\left(\bigcup_{k=n}^{\infty} A_k\right)$ and that, as a consequence, $T \in \mathfrak{M}$.

(b) Can you identify a decreasing chain of sets which appeared in the description of T from part (1)?

(c) Suppose that $\sum_{n=1}^{\infty} \mu(A_n) < \infty$. Prove that $\mu(T) = 0$.

Proof.

(a) We can rewrite T:

$$\begin{aligned} T &= \{x \in X \mid \forall n \in \mathbb{N}, \exists k \geq n \text{ s.t. } x \in A_k\} \\ &= \{x \in X \mid \forall n \in \mathbb{N}, x \in \bigcup_{k=n}^{\infty} A_k\} \\ &= \bigcap_{n=1}^{\infty}\left(\bigcup_{k=n}^{\infty} A_k\right). \end{aligned}$$

Observe that $\bigcup_{k=n}^{\infty} A_k \in \mathfrak{M}$ for all $n \in \mathbb{N}$. Hence, $T \in \mathfrak{M}$.

(b) Note that $\left(\bigcup_{k=n}^{\infty} A_k\right)_{n=1}^{\infty}$ is a decreasing chain of sets in $\mathfrak{M}$.

(c) Exercise.

$\square$

Lecture 5, 2026/05/22

Example. Let $\mathfrak{M} = 2^X$ be the σ-algebra. Suppose $w : X \to [0, \infty]$ is a function that assigns some "weight" to each $x \in X$. Then, define $\mu : \mathfrak{M} \to [0, \infty]$ by

$$\mu(A) := \sup\left\{\sum_{x \in F} w(x) \mid F \subseteq A, F \text{ is finite}\right\} \in [0, \infty], \quad \forall A \in \mathfrak{M}.$$

If $F = \varnothing$, then $\sum_{x \in F} w(x) := 0$ by convention.

Exercise: Check that μ is a positive measure on $\mathfrak{M}$.

Example (**Two Special Cases of the Previous Example**).

(1) Suppose $w : X \to [0, \infty]$ is defined by $w(x) = 1$ for all $x \in X$. Then,

$$\mu(A) = \begin{cases} k, & \text{if } A \text{ is a finite set with } k \text{ elements} \\ \infty, & \text{if } A \text{ is an infinite set.} \end{cases}$$

In this case, μ is called the **counting measure** on 2^X.

(2) Suppose we have singled out one element $x_\circ \in X$ and define $w : X \to [0, \infty]$ by $w(x_\circ) = 1$ and $w(x) = 0$ for all $x \in X \setminus \{x_\circ\}$. Then,

$$\mu(A) = \begin{cases} 1, & \text{if } x_\circ \in A \\ 0, & \text{if } x_\circ \notin A. \end{cases}$$

In this case, μ is called the **Dirac measure concentrated at** $x_\circ$.

1.4 Lebesgue Measure on $\mathbb{R}$

Let $X = \mathbb{R}$, $\mathfrak{M} = \mathcal{B}_{\mathbb{R}} = \sigma\text{-}\operatorname{Alg}(\mathcal{T}_{\mathbb{R}})$, where $\mathcal{T}_{\mathbb{R}} = \{G \subseteq \mathbb{R} \mid G \text{ is open}\}$.

Fact 1: We can view $\mathcal{B}_{\mathbb{R}}$ as $\sigma\text{-}\operatorname{Alg}(\mathcal{J})$ where $\mathcal{J}$ is the collection of half-open intervals considered in Problem P1.3.

Fact 2: Let $\mu, \nu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ be positive measures such that

$$\mu((a, b]) = \nu((a, b]) < \infty, \quad \forall a < b \in \mathbb{R}.$$

Then, it follows that $\mu = \nu$ (i.e. $\mu(B) = \nu(B)$ for every $B \in \mathcal{B}_{\mathbb{R}}$).

Exercise: Do Problem P2.6 and P2.7.

Definition 1.11 (**Lebesgue Measure**).
The **Lebesgue measure** is the positive measure $\lambda : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ determined by the condition that

$$\lambda((a, b]) = b - a, \quad \forall a < b \in \mathbb{R}.$$

Remark (**Another Approach**).
λ is uniquely determined by the condition that

$$\lambda(B + t) = \lambda(B), \quad \forall B \in \mathcal{B}_{\mathbb{R}}, t \in \mathbb{R},$$

with the normalization condition $\lambda((0, 1]) = 1$. These two conditions imply that $\lambda((a, b]) = b - a$ for all $a < b \in \mathbb{R}$.

Example. Note that

$$1 = \lambda((0, 1]) = \lambda\Big((0, \tfrac{1}{2}] \cup (\tfrac{1}{2}, 1]\Big) = 2\lambda\Big((0, \tfrac{1}{2}]\Big) \implies \lambda\Big((0, \tfrac{1}{2}]\Big) = \frac{1}{2}.$$

For the existence of λ, we will invoke an extension theorem of Caratheodory (do be done later).

Fact 3: There exists no positive measure $\mu : 2^{\mathbb{R}} \to [0, \infty]$ such that

(1) μ is invariant under translations (i.e $\mu(B + t) = \mu(B)$ for all $B \subseteq \mathbb{R}$ and $t \in \mathbb{R}$).

(2) μ satisfies the normalization condition $\mu((0, 1]) = 1$.

Remark. This shows that $\mathcal{B}_{\mathbb{R}} \neq 2^{\mathbb{R}}$.

2 Measurable Functions and Integration

2.1 Measurable Functions

A key point of the Lebesgue machinery is that from measuring sets one can go in a rather natural way to integrating functions.

Definition 2.1 (**Measurable Function**).
Let $(X, \mathfrak{M})$ and $(Y, \mathfrak{N})$ be measurable spaces. A function $f : X \to Y$ is said to be $\mathfrak{M}/\mathfrak{N}$-**measurable** if $f^{-1}(N) \in \mathfrak{M}$ for every $N \in \mathfrak{N}$.

Remark. The pre-image under f:

$$f^{-1}(S) \coloneqq \{x \in X \mid f(x) \in S\} \subseteq X, \quad \forall S \subseteq Y.$$

Proposition 2.1. Let $(X, \mathfrak{M})$, $(Y, \mathfrak{N})$ and $(Z, \mathfrak{P})$ be measurable spaces. Let $f : X \to Y$ be $\mathfrak{M}/\mathfrak{N}$-measurable and $g : Y \to Z$ be $\mathfrak{N}/\mathfrak{P}$-measurable. Then, the composition $h \coloneqq g \circ f : X \to Z$ is $\mathfrak{M}/\mathfrak{P}$-measurable.

Proof. For every $C \in \mathfrak{P}$, we have

$$\begin{aligned} g^{-1}(C) \in \mathfrak{N} &\implies f^{-1}(g^{-1}(C)) \in \mathfrak{M} \\ &\implies h^{-1}(C) \in \mathfrak{M}. \end{aligned}$$

For the last implication, note that

$$h^{-1}(C) = (g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C)).$$

Hence, h is $\mathfrak{M}/\mathfrak{P}$-measurable. $\square$

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Proposition 2.2. Let $(X, \mathfrak{M})$ and $(Y, \mathfrak{N})$ be measurable spaces and let $\mathfrak{C} \subseteq \mathfrak{N}$ be a collection of subsets of Y such that $\sigma\text{-}\operatorname{Alg}(\mathfrak{C}) = \mathfrak{N}$. Let $f : X \to Y$ be a function such that $f^{-1}(C) \in \mathfrak{M}$ for every $C \in \mathfrak{C}$. Then, f is $\mathfrak{M}/\mathfrak{N}$-measurable.

Proof. We want to show that $f^{-1}(N) \in \mathfrak{M}$ for every $N \in \mathfrak{N}$. We use the method of "civilized sets". Let $\mathfrak{Q} = \{Q \subseteq Y \mid f^{-1}(Q) \in \mathfrak{M}\} \subseteq 2^Y$.

Claim (1)**.** $\mathfrak{Q}$ is a σ-algebra of subsets of Y.

Proof of Claim (1). Note that $\varnothing \in \mathfrak{Q}$ since $f^{-1}(\varnothing) = \varnothing \in \mathfrak{M}$, so (Sigma-AS1) holds. Let $Q \in \mathfrak{Q}$. Then, $f^{-1}(Q) \in \mathfrak{M}$, hence $X \setminus f^{-1}(Q) = f^{-1}(Y \setminus Q) \in \mathfrak{M}$, so $Y \setminus Q \in \mathfrak{Q}$, and (Sigma-AS2) holds. Let $Q_1, Q_2, \ldots \in \mathfrak{Q}$. Then, $f^{-1}(Q_n) \in \mathfrak{M}$ for all $n \in \mathbb{N}$, hence

$$f^{-1}\left(\bigcap_{n=1}^{\infty} Q_n\right) = \bigcap_{n=1}^{\infty} f^{-1}(Q_n) \in \mathfrak{M},$$

so $\bigcap_{n=1}^{\infty} Q_n \in \mathfrak{Q}$, and (Sigma-AS3) holds. Thus, $\mathfrak{Q}$ is a σ-algebra of subsets of Y. □

Claim (2)**.** $\mathfrak{Q} \supseteq \mathfrak{N}$.

Proof of Claim (2). Since $\sigma\text{-}\operatorname{Alg}(\mathfrak{C}) = \mathfrak{N}$, it suffices to show that $\mathfrak{Q} \supseteq \mathfrak{C}$. Let $C \in \mathfrak{C}$. Then, $f^{-1}(C) \in \mathfrak{M}$ by assumption, hence $C \in \mathfrak{Q}$. Thus, $\mathfrak{Q} \supseteq \mathfrak{C}$. □

By Claim (2), every $N \in \mathfrak{N}$ is civilized. Hence, $f^{-1}(N) \in \mathfrak{M}, \forall N \in \mathfrak{N}$, and f is $\mathfrak{M}/\mathfrak{N}$-measurable. □

Corollary 2.3. Let (X, d) and (Y, d') be metric spaces and let $\mathcal{B}_X$ and $\mathcal{B}_Y$ be the Borel σ-algebras of X and Y, respectively. Let $f : X \to Y$ be continuous. Then, f is $\mathcal{B}_X/\mathcal{B}_Y$-measurable.

Proof. Consider $\mathcal{T}_Y = \{G \subseteq Y \mid G \text{ is open}\}$. Observe that

$$G \in \mathcal{T}_Y \implies f^{-1}(G) \text{ is an open subset of } X \implies f^{-1}(G) \in \mathcal{B}_X.$$

Note that $\sigma\text{-}\operatorname{Alg}(\mathcal{T}_Y) = \mathcal{B}_Y$. Then, $f^{-1}(G) \in \mathcal{B}_X$ for every $G \in \mathcal{T}_Y$. Hence, by Proposition 2.2, f is $\mathcal{B}_X/\mathcal{B}_Y$-measurable. □

2.2 The Space of Functions $\mathrm{Bor}(X, \mathbb{R})$

Definition 2.2 (Notation).
Let $(X, \mathfrak{M})$ be a measurable space. We will denote

$$\mathrm{Bor}(X, \mathbb{R}) = \{f : X \to \mathbb{R} \mid f \text{ is } \mathfrak{M}/\mathcal{B}_{\mathbb{R}}\text{-measurable}\}.$$

The functions in $\mathrm{Bor}(X, \mathbb{R})$ are called **Borel functions** on X.

Proposition 2.4 (Stability Properties of $\mathrm{Bor}(X, \mathbb{R})$).
Let $(X, \mathfrak{M})$ be a measurable space and let $f, g \in \mathrm{Bor}(X, \mathbb{R})$. Then, the following functions (constructed from f and g) are also in $\mathrm{Bor}(X, \mathbb{R})$:

(1) $f + g \in \mathrm{Bor}(X, \mathbb{R})$, where $(f + g)(x) \coloneqq f(x) + g(x)$, $\forall x \in X$.

(2) More generally, for any $\alpha, \beta \in \mathbb{R}$, $\alpha f + \beta g \in \mathrm{Bor}(X, \mathbb{R})$, where $(\alpha f + \beta g)(x) \coloneqq \alpha f(x) + \beta g(x)$, $\forall x \in X$.

(3) $f \cdot g \in \mathrm{Bor}(X, \mathbb{R})$, where $(f \cdot g)(x) \coloneqq f(x) \cdot g(x)$, $\forall x \in X$.

(4) $f \vee g$ and $f \wedge g$ are in $\mathrm{Bor}(X, \mathbb{R})$, where $(f \vee g)(x) \coloneqq \max\{f(x), g(x)\}$ and $(f \wedge g)(x) \coloneqq \min\{f(x), g(x)\}$, $\forall x \in X$.

Proof. All parts are proved the same way, so we will only prove (3). Let $f, g \in \mathrm{Bor}(X, \mathbb{R})$. We want to show that $f \cdot g$ is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable. Let $h : X \to \mathbb{R}^2$ be defined by

$$h(x) = (f(x), g(x)), \quad \forall x \in X.$$

Let $p : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$p(s, t) = s \cdot t, \quad \forall s, t \in \mathbb{R}.$$

Claim (1). h is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}^2}$-measurable.

Claim (2). p is $\mathcal{B}_{\mathbb{R}^2}/\mathcal{B}_{\mathbb{R}}$-measurable.

Claim (3). $f \cdot g = p \circ h$.

$$\begin{tikzcd} X \arrow[r, "h"] \arrow[rr, bend right, "p\circ h"'] & \mathbb{R}^2 \arrow[r, "p"] & \mathbb{R} \end{tikzcd}$$

Observe that the three claims give us that $f \cdot g \in \mathrm{Bor}(X, \mathbb{R})$. Indeed, Claim (1) + Claim (2) + Proposition 2.1 $\implies$ $p \circ h$ is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable, and Claim (3) identifies $f \cdot g = p \circ h$.

Proof of Claim (3). By direct inspection, we have

$$\begin{aligned}(p \circ h)(x) &= p(h(x)) \\ &= p((f(x), g(x))) \\ &= f(x) \cdot g(x) = (f \cdot g)(x), \quad \forall x \in X. \end{aligned}$$

□

Proof of Claim (2). Observe that $p : \mathbb{R}^2 \to \mathbb{R}$ is continuous (between usual metric spaces). Then, by Corollary 2.3, p is $\mathcal{B}_{\mathbb{R}^2}/\mathcal{B}_{\mathbb{R}}$-measurable. □

Proof of Claim (1). Use Proposition 2.2 with $\mathcal{B}_{\mathbb{R}^2} = \sigma\text{- Alg}(\mathcal{J})$, where

$$\mathcal{J} = \text{set of all open squares in } \mathbb{R}^2,$$

it suffices to check that

$$h^{-1}(S) \in \mathfrak{M}, \quad \forall S = (\alpha_1, \alpha_2) \times (\beta_1, \beta_2) \in \mathcal{J}.$$

And indeed, for such S, we have

$$\begin{aligned} h^{-1}(S) &= \{x \in X \mid h(x) \in S\} \\ &= \{x \in X \mid (f(x), g(x)) \in (\alpha_1, \alpha_2) \times (\beta_1, \beta_2)\} \\ &= \{x \in X \mid f(x) \in (\alpha_1, \alpha_2) \text{ and } g(x) \in (\beta_1, \beta_2)\} \\ &= f^{-1}((\alpha_1, \alpha_2)) \cap g^{-1}((\beta_1, \beta_2)) \in \mathfrak{M}. \end{aligned}$$

So, h is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}^2}$-measurable. □

□

Exercise: Do Exercise 3.8.

Remark.

- By (1) and (2), $\mathrm{Bor}(X,\mathbb{R})$ is a **linear space** of functions.
- A linear space that is also closed under pointwise multiplication is called an **algebra** of functions (i.e. $\mathrm{Bor}(X,\mathbb{R})$ is an algebra of functions by (3)).
- A linear space that is also closed under $\wedge$ and $\vee$ is called a **lattice** of functions (i.e. $\mathrm{Bor}(X,\mathbb{R})$ is a lattice of functions by (4)).

Remark. Let $(X,\mathfrak{M})$ be a measurable space and let $f : X \to \mathbb{R}$ be a Borel function. One can create new Borel functions by just applying one-variable functions to f.

Example. Consider $|f| : X \to \mathbb{R}$ defined by $|f|(x) = |f(x)|$, $\forall x \in X$.

$$\Big[f \in \mathrm{Bor}(X,\mathbb{R})\Big] \implies \Big[|f| \in \mathrm{Bor}(X,\mathbb{R})\Big].$$

Note that this is a composition

$$X \xrightarrow{f} \mathbb{R} \xrightarrow{|\cdot|} \mathbb{R}.$$

So $|f| = |\cdot| \circ f$. Since f is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable and $|\cdot|$ is $\mathcal{B}_{\mathbb{R}}/\mathcal{B}_{\mathbb{R}}$-measurable (by Corollary 2.3 and $|\cdot|$ is continuous), by Proposition 2.1, $|f|$ is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable, i.e. $|f| \in \mathrm{Bor}(X,\mathbb{R})$.

Problem P3.1: Let $(X,\mathfrak{M})$ be a measurable space and let $f : X \to \mathbb{R}$ be a function about which the following is known:

$$\text{For every } t \in \mathbb{R},\ \{x \in X \mid f(x) \le t\} \in \mathfrak{M}.$$

Prove that $f \in \mathrm{Bor}(X,\mathbb{R})$.

Fact: Let $\mathfrak{C} = \{(-\infty,t] \mid t \in \mathbb{R}\} \subseteq \mathcal{B}_{\mathbb{R}}$. Then, $\sigma\text{-}\mathrm{Alg}(\mathfrak{C}) = \mathcal{B}_{\mathbb{R}}$ (i.e. $\mathfrak{C}$ is a generator for $\mathcal{B}_{\mathbb{R}}$).

Proof. We assume the fact above. The hypothesis says that

$$f^{-1}((-\infty,t]) \in \mathfrak{M}, \quad \forall t \in \mathbb{R}.$$

That is, $f^{-1}(C) \in \mathfrak{M}$, $\forall C \in \mathfrak{C}$. By Proposition 2.2, $f^{-1}(B) \in \mathfrak{M}$, $\forall B \in \mathcal{B}_{\mathbb{R}}$. Hence, f is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable, i.e. $f \in \mathrm{Bor}(X,\mathbb{R})$. $\square$

Question: How to verify that fact above?

Idea: Let's denote $\sigma\text{- Alg}(\mathfrak{C}) = \mathfrak{P}$. Clearly, $\mathfrak{P} \subseteq \mathcal{B}_{\mathbb{R}}$. To show $\mathfrak{P} = \mathcal{B}_{\mathbb{R}}$, it suffices to show that $\mathfrak{P}$ contains all open intervals in $\mathbb{R}$, i.e., $G \in \mathfrak{P}$, $\forall G \subseteq \mathbb{R}$ open, because we will get $\mathfrak{P}$ is a σ-algebra and $\mathfrak{P} \supseteq \mathcal{T}_{\mathbb{R}} = \{G \subseteq \mathbb{R} \mid G \text{ is open}\}$. Then, $\mathfrak{P} \supseteq \sigma\text{- Alg}(\mathcal{T}_{\mathbb{R}}) = \mathcal{B}_{\mathbb{R}}$.

Now, every $\emptyset \neq G \subseteq \mathbb{R}$ open can be written as a countable union of open intervals. Hence, it suffices to check that $\mathfrak{P}$ contains all the open intervals of $\mathbb{R}$.

For illustration, let's check that $(0,1) \in \mathfrak{P}$. Check that $(-\infty, 0], (-\infty, 1] \in \mathfrak{P}$. Hence,

$$(0,1] = (-\infty, 1] \setminus (-\infty, 0] \in \mathfrak{P}.$$

For every $n \geq 2$, we have $(-\infty, 0], (-\infty, 1 - \frac{1}{n}] \in \mathfrak{P}$, hence

$$\left(0, \frac{n-1}{n}\right] = \left(-\infty, 1 - \frac{1}{n}\right] \setminus (-\infty, 0] \in \mathfrak{P}.$$

Then, $(0,1) = \bigcup_{n=2}^{\infty} \left(0, \frac{n-1}{n}\right] \in \mathfrak{P}$.

2.3 Convergence Properties of $\mathrm{Bor}(X, \mathbb{R})$

In the previous subsection, we have seen that $\mathrm{Bor}(X, \mathbb{R})$ has good stability properties under various algebraic operations. Now, we will see that $\mathrm{Bor}(X, \mathbb{R})$ is also closed under pointwise convergence of sequences of functions. For this subsection, we fix a measurable space $(X, \mathfrak{M})$.

Proposition 2.5.

(1) Let $f : X \to \mathbb{R}$ be a function. Suppose $\exists$ a sequence $(f_n)_{n=1}^{\infty}$ of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that

$$\sup\{f_n(x) \mid n \in \mathbb{N}\} = f(x), \quad \forall x \in X.$$

Then, $f \in \mathrm{Bor}(X, \mathbb{R})$.

(2) Let $g : X \to \mathbb{R}$ be a function. Suppose $\exists$ a sequence $(g_n)_{n=1}^{\infty}$ of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that

$$\inf\{g_n(x) \mid n \in \mathbb{N}\} = g(x), \quad \forall x \in X.$$

Then, $g \in \mathrm{Bor}(X, \mathbb{R})$.

Proof.

(1) We use the criterion for measurability given in Problem P3.1. Then, it suffices to check that

$$\forall t \in \mathbb{R}, \{x \in X \mid f(x) \le t\} \in \mathfrak{M}.$$

Fix $t \in \mathbb{R}$. Then,

$$\begin{aligned}
\Big[f(x) \le t\Big] &\iff \Big[\sup\{f_n(x) \mid n \in \mathbb{N}\} \le t\Big] \\
&\iff \Big[f_n(x) \le t, \forall n \in \mathbb{N}\Big] \\
&\iff \Big[x \in f_n^{-1}((-\infty, t]), \forall n \in \mathbb{N}\Big] \\
&\iff \Big[x \in \bigcap_{n=1}^{\infty} f_n^{-1}((-\infty, t])\Big].
\end{aligned}$$

Hence,

$$\{x \in X \mid f(x) \le t\} = \bigcap_{n=1}^{\infty} f_n^{-1}((-\infty, t]) \in \mathfrak{M}.$$

(2) Exercise.

□

Remark. Let's look at $\lim_{n\to\infty} \sup t_n$ where $(t_n)_{n=1}^{\infty}$ is a sequence in $\mathbb{R}$, bounded from above. We can show that $\exists \ell_+ \in \mathbb{R}$ such that

(1) $\exists$ a subsequence of $(t_n)_{n=1}^{\infty}$ that converges to ℓ_+.

(2) $\nexists$ a subsequence of $(t_n)_{n=1}^{\infty}$ that converges to some $\ell > \ell_+$.

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Let $(t_n)_{n=1}^{\infty}$ be a bounded sequence in $\mathbb{R}$ (i.e $\exists a, b \in \mathbb{R}$ s.t. $a \le t_n \le b$). In words, $\ell_+ \in [a, b]$ is the biggest value which can be obtained as a limit of a subsequence of $(t_n)_{n=1}^{\infty}$. Alternatively,

$$\ell_+ = \lim_{k\to\infty}\left(\sup_{n\ge k} t_n\right) = \inf_{k\ge 1}\left(\sup_{n\ge k} t_n\right).$$

More precisely, if we let

$$\beta_1 = \sup\{t_n \mid n \geq 1\}, \quad \beta_2 = \sup\{t_n \mid n \geq 2\}, \quad \beta_3 = \sup\{t_n \mid n \geq 3\}, \dots,$$

then we get a decreasing sequence of β_k's, and we have

$$\ell_+ = \lim_{k\to\infty} \beta_k = \inf\{\beta_k \mid k \in \mathbb{N}\}.$$

Informally, ℓ_+ is the highest value the sequence keeps getting close to infinitely many times, ignoring large values that only appear finitely many times.

Similarly, we can define $\liminf_{n\to\infty} t_n$ (a number $\ell_- \in \mathbb{R}$) as the smallest possible value reached as a limit of a subsequence of $(t_n)_{n=1}^{\infty}$. We can also write

$$\ell_- = \lim_{k\to\infty} \left(\inf_{n\geq k} t_n\right) = \sup_{k\geq 1} \left(\inf_{n\geq k} t_n\right).$$

Proposition 2.6. Let $(X, \mathfrak{M})$ be a measurable space.

(1) Let $f : X \to \mathbb{R}$ be a function. Suppose $\exists$ a sequence $(f_n)_{n=1}^{\infty}$ of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that, $\forall x \in X$, the sequence $(f_n(x))_{n=1}^{\infty}$ is bounded and has

$$\limsup_{n\to\infty} f_n(x) = f(x).$$

Then, $f \in \mathrm{Bor}(X, \mathbb{R})$.

(2) Let $g : X \to \mathbb{R}$ be a function. Suppose $\exists$ a sequence $(g_n)_{n=1}^{\infty}$ of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that, $\forall x \in X$, the sequence $(g_n(x))_{n=1}^{\infty}$ is bounded and has

$$\liminf_{n\to\infty} g_n(x) = g(x).$$

Then, $g \in \mathrm{Bor}(X, \mathbb{R})$.

Proof.

(1) For every $k \in \mathbb{N}$, define $h_k : X \to \mathbb{R}$ by

$$h_k(x) = \sup\{f_n(x) \mid n \geq k\}, \quad \forall x \in X.$$

Note that $\{f_n(x) \mid n \geq k\}$ is a bounded set, so it has sup in $\mathbb{R}$ (i.e. h_k is well-defined). Observe that $f_k, f_{k+1}, \ldots, f_n, \ldots \in \mathrm{Bor}(X, \mathbb{R})$. By Proposition 2.5 (1), $h_k \in \mathrm{Bor}(X, \mathbb{R})$ for every $k \in \mathbb{N}$. Also observe that $h_1(x) \geq h_2(x) \geq \cdots \geq h_k(x) \geq \cdots$ with

$$f(x) = \lim_{k\to\infty} h_k(x) = \inf\{h_k(x) \mid k \in \mathbb{N}\}.$$

This is because $f(x) = \limsup_{n\to\infty} f_n(x)$, where

$$\limsup_{n\to\infty} f_n(x) = \inf_{k\geq 1}\left(\sup_{n\geq k} f_n(x)\right) = \inf_{k\geq 1} h_k(x).$$

By Proposition 2.5 (2), $f \in \mathrm{Bor}(X, \mathbb{R})$.

(2) We can reduce this to (1) by introducing $f : X \to \mathbb{R}$, defined by

$$f(x) = -g(x), \quad \forall x \in X.$$

Then, introduce $f_n = -g_n \in \mathrm{Bor}(X, \mathbb{R})$, $\forall n \in \mathbb{N}$. Applying (1) to $f(x) = \limsup_{n\to\infty} f_n(x)$, we get $f \in \mathrm{Bor}(X, \mathbb{R})$. Hence, $g = -f \in \mathrm{Bor}(X, \mathbb{R})$.

□

Corollary 2.7. Let $f : X \to \mathbb{R}$ be a function. Suppose $\exists$ a sequence $(f_n)_{n=1}^{\infty}$ of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that

$$f(x) = \lim_{n\to\infty} f_n(x), \quad \forall x \in X.$$

Then, $f \in \mathrm{Bor}(X, \mathbb{R})$.

Proof. Exercise. □

Exercise: Suppose that $(X, \mathfrak{M})$ is $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable and consider the function $f' : \mathbb{R} \to \mathbb{R}$ (may not be continuous). Prove that $f' \in \mathrm{Bor}(\mathbb{R}, \mathbb{R})$.

Idea: produce $(f_n)_{n=1}^{\infty}$ in $\mathrm{Bor}(\mathbb{R}, \mathbb{R})$ (in fact continuous) such that $\lim_{n\to\infty} f_n(x) = f'(x)$, $\forall x \in \mathbb{R}$. If we find the f_n's, then we get $f' \in \mathrm{Bor}(\mathbb{R}, \mathbb{R})$ by Corollary 2.7.

Proof. For $x \in \mathbb{R}$, write

$$f'(x) = \lim_{n\to\infty} \frac{f\left(x+\frac{1}{n}\right) - f(x)}{\frac{1}{n}} = \lim_{n\to\infty} n \cdot (u_n(x) - f(x)), \quad \text{where } u_n(x) = f\left(x + \frac{1}{n}\right).$$

Then, for every $n \in \mathbb{N}$, let $u_n : \mathbb{R} \to \mathbb{R}$ be defined by

$$u_n(x) = f\left(x + \frac{1}{n}\right), \quad \forall x \in \mathbb{R}$$

and let $f_n = n \cdot (u_n - f)$. Note that f_n is continuous (hence in $\mathrm{Bor}(\mathbb{R}, \mathbb{R})$) and $\lim_{n\to\infty} f_n(x) = f'(x)$, $\forall x \in \mathbb{R}$. By Corollary 2.7, $f' \in \mathrm{Bor}(\mathbb{R}, \mathbb{R})$. $\square$

2.4 Approximation by Simple Functions

Definition 2.3 (**Simple Function**).
A function $f \in \mathrm{Bor}(X, \mathbb{R})$ is said to be **simple** if it takes finitely many values, i.e. the image $f(X)$ is a finite subset of $\mathbb{R}$. We denote

$$\mathrm{Bor}_s(X, \mathbb{R}) = \{f \in \mathrm{Bor}(X, \mathbb{R}) \mid f \text{ is simple}\}.$$

Note. We also denote the following spaces:

$$\begin{aligned} \mathrm{Bor}^+(X, \mathbb{R}) &:= \{f \in \mathrm{Bor}(X, \mathbb{R}) \mid f(x) \geq 0, \forall x \in X\}, \\ \mathrm{Bor}_s^+(X, \mathbb{R}) &:= \mathrm{Bor}_s(X, \mathbb{R}) \cap \mathrm{Bor}^+(X, \mathbb{R}) \\ &= \{f \in \mathrm{Bor}_s(X, \mathbb{R}) \mid f(x) \geq 0, \forall x \in X\}. \end{aligned}$$

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Definition 2.4 (**Indicator Function**).
Let $A \in \mathfrak{M}$. The **indicator function** of A is $\chi_A : X \to \mathbb{R}$ defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \in X \setminus A. \end{cases}$$

Note. We have $\chi_A \in \mathrm{Bor}_s^+(X,\mathbb{R})$. We can check that χ_A is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable. For every $S \in \mathcal{B}_{\mathbb{R}}$, we have $\chi_A^{-1}(S)$ is one of $\varnothing, A, X \setminus A$, or X, and all these sets are in $\mathfrak{M}$.

One can show that $\mathrm{Bor}_s^+(X,\mathbb{R}) = \mathrm{Span}\{\chi_A \mid A \in \mathfrak{M}\}$, the set of all linear combinations of indicators χ_A (see Assignment 2).

Remark. Let $f \in \mathrm{Bor}^+(X,\mathbb{R})$. For every $n \in \mathbb{N}$, define the **n-th binning of the values of** f by $f_n : X \to \mathbb{R}$,

- If $x \in X$ with $f(x) \geq n$, define $f_n(x) \coloneqq n$.
- If $x \in X$ with $f(x) < n$, then pick the unique $k \in \{1, 2, \dots, n2^n\}$ such that $\frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}$, and define $f_n(x) \coloneqq \frac{k-1}{2^n}$.

There are $1 + n2^n$ bins in total. For $f(x) < n$, divide the interval $[0, n)$ into $n2^n$ small bins of length $\frac{1}{2^n}$, and assign $f_n(x)$ to be the min value of the bin that $f(x)$ falls into.

Example. Consider f_1 (3 bins). Define

$$f_1(x) = \begin{cases} 0 & \text{if } f(x) \in [0, \frac{1}{2}), \\ \frac{1}{2} & \text{if } f(x) \in [\frac{1}{2}, 1), \\ 1 & \text{if } f(x) \in [1, \infty). \end{cases}$$

Observe that $f_1 \leq f$. This holds because the definition of f_1 uses the minimum point of every bin. Also, $f_1 \in \mathrm{Bor}_s^+(X,\mathbb{R})$. Why? Let

$$A_0 = f^{-1}([0, \tfrac{1}{2})), \quad A_1 = f^{-1}([\tfrac{1}{2}, 1)), \quad A_2 = f^{-1}([1, \infty)),$$

and note that $A_0, A_1, A_2 \in \mathfrak{M}$ because f is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable. Then,

$$f_1(x) = 0 \cdot \chi_{A_0}(x) + \frac{1}{2} \cdot \chi_{A_1}(x) + 1 \cdot \chi_{A_2}(x), \quad \forall x \in X.$$

Hence,

$$f_1 = \frac{1}{2}\chi_{A_1} + \chi_{A_2} \in \mathrm{Span}\{\chi_A \mid A \in \mathfrak{M}\} = \mathrm{Bor}_s^+(X,\mathbb{R}).$$

Example. Consider f_2 (9 bins). For example, if $x \in X$ with $f(x) \in [\frac{5}{4}, \frac{3}{2})$, then $f_2(x) = \frac{5}{4}$.

One can write f_n as a linear combination of χ_A's to get that $f_n \in \mathrm{Bor}_s^+(X, \mathbb{R})$. For every $n \in \mathbb{N}$ and every $x \in X$, we have $f_n(x) \leq f_{n+1}(x)$. Fix $x \in X$, $\exists n_0 \in \mathbb{N}$ such that $f_n(x) \leq f(x) < f_n(x) + \frac{1}{2^n}$, $\forall n \geq n_0$. Hence, $\lim_{n\to\infty} f_n(x) = f(x)$ (since $|f_n(x) - f(x)| < \frac{1}{2^n}$, for n large enough). Altogether, we get the following proposition.

Proposition 2.8. Let f be a function in $\mathrm{Bor}^+(X, \mathbb{R})$. For every $n \in \mathbb{N}$, let f_n be the n-th binning of the values of f. Then,

(1) For every $n \in \mathbb{N}$, $f_n \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

(2) For every $n \in \mathbb{N}$, $f_n \leq f_{n+1}$ (i.e. $f_n(x) \leq f_{n+1}(x)$, $\forall x \in X$).

(3) For every $x \in X$, the increasing sequence $(f_n(x))_{n=1}^\infty$ has $\lim_{n\to\infty} f_n(x) = f(x)$.

Proof. Exercise. □

Proposition 2.9. Let $f \in \mathrm{Bor}^+(X, \mathbb{R})$. There exists a sequence $(f_n)_{n=1}^\infty$ of functions in $\mathrm{Bor}_s^+(X, \mathbb{R})$ such that $\forall x \in X$, $0 \leq f_1(x) \leq f_2(x) \leq \cdots \leq f_n(x) \leq \cdots$ and $\lim_{n\to\infty} f_n(x) = f(x)$.

Proof. For every $n \in \mathbb{N}$, let f_n be the n-th binning of the values of f. By the previous proposition, $f_n \in \mathrm{Bor}_s^+(X, \mathbb{R})$, $f_n \leq f_{n+1}$, and $\lim_{n\to\infty} f_n(x) = f(x)$, $\forall x \in X$. □

2.5 Integration of Functions in $\mathrm{Bor}(X, \mathbb{R})$

For this part, we fix a measure space $(X, \mathfrak{M}, \mu)$. For didactical reasons, it is convenient to discuss separately the integration of non-negative Borel functions and the integration of general Borel functions.

- Integral is always defined but may be ∞.

$$\int_X f(x)\, d\mu(x) \quad \text{for } f \in \mathrm{Bor}^+(X, \mathbb{R}).$$

- Get a number in $\mathbb{R}$, but not always defined, need $f \in \mathcal{L}^1(\mu)$.

$$\int_X f(x)\, d\mu(x) \quad \text{for general } f \in \mathrm{Bor}(X, \mathbb{R}).$$

Moreover, it will be convenient to introduce the integral on $\mathrm{Bor}^+(X, \mathbb{R})$ as a map

$$L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \to [0, \infty]$$

Note. $L^+(f)$ will be defined for every $f \in \mathrm{Bor}^+(X, \mathbb{R})$, but it may be ∞.

Get L^+ in two parts:

(1) Define $L_s^+ : \mathrm{Bor}_s^+(X, \mathbb{R}) \to [0, \infty]$.

(2) Extend L_s^+ to a map $L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \to [0, \infty]$.

Lecture 10, 2026/06/03

Remark (**Properties of** $\mathrm{Bor}_s^+(X, \mathbb{R})$).

(1) $\left[f, g \in \mathrm{Bor}_s^+(X, \mathbb{R})\right] \implies \left[f + g \in \mathrm{Bor}_s^+(X, \mathbb{R})\right]$.

(2) $\left[f \in \mathrm{Bor}_s^+(X, \mathbb{R}) \text{ and } \alpha \in [0, \infty)\right] \implies \left[\alpha f \in \mathrm{Bor}_s^+(X, \mathbb{R})\right]$.

(3) For every $A \in \mathfrak{M}$, the indicator function $\chi_A \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

Note. Special cases when $A = \varnothing$ or $A = X$: the indicators $\chi_\varnothing(x) = \underline{0}$ and $\chi_X(x) = \underline{1}$, $\forall x \in X$, are in $\mathrm{Bor}_s^+(X, \mathbb{R})$.

Theorem 2.10. $\exists$ a map $L_s^+ : \mathrm{Bor}_s^+(X, \mathbb{R}) \to [0, \infty]$, uniquely determined, such that

(1) Additivity: $L_s^+(f + g) = L_s^+(f) + L_s^+(g)$, $\forall f, g \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

(2) Positive homogeneity: $L_s^+(\alpha f) = \alpha L_s^+(f)$, $\forall f \in \mathrm{Bor}_s^+(X, \mathbb{R})$ and $\alpha \in [0, \infty)$.

(3) Relation to the measure: $L_s^+(\chi_A) = \mu(A)$, $\forall A \in \mathfrak{M}$.

Definition 2.5 (**Measurable Division**).
Let $(X, \mathfrak{M}, \mu)$ be a measure space. A **measurable division** of X is a set $\Delta = \{A_1, A_2, \dots, A_n\}$ with $A_1, \dots, A_n \in \mathfrak{M}$ non-empty such that $A_i \cap A_j = \varnothing$ for $i \neq j$ and $\bigcup_{i=1}^n A_i = X$.

Remark. If $\Delta = \{A_1, \dots, A_n\}$ and $\Gamma = \{B_1, \dots, B_m\}$ are two measurable divisions of X, then we can form a new measurable division by taking all the non-empty intersections:

$$\Delta \wedge \Gamma = \{A_i \cap B_j \mid 1 \leq i \leq n,\ 1 \leq j \leq m,\ A_i \cap B_j \neq \varnothing\}.$$

Lemma 2.11. Let $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

(1) $\exists$ a measurable division $\Delta = \{A_1, \dots, A_n\}$ of X and some numbers $\alpha_1, \dots, \alpha_n \in [0, \infty)$ such that

$$f = \alpha_1 \chi_{A_1} + \alpha_2 \chi_{A_2} + \cdots + \alpha_n \chi_{A_n}.$$

(2) Suppose that $A_1, \dots, A_n$ and $\alpha_1, \dots, \alpha_n$ are as in (1). Given $\Gamma = \{B_1, \dots, B_m\}$ another measurable division of X. Then, we can further write

$$f = \sum_{\substack{1 \le i \le n,\ 1 \le j \le m \\ \text{s.t. } A_i \cap B_j \neq \varnothing}} \alpha_i \chi_{A_i \cap B_j}.$$

Proof.

(1) Exercise, this proof is very similar to A2Q2.

(2) For every $1 \le i \le n$, write

$$\begin{aligned} \chi_{A_i} &= \chi_{A_i} \cdot \underline{1} \\ &= \chi_{A_i} \left(\chi_{B_1} + \chi_{B_2} + \cdots + \chi_{B_m} \right) \\ &= \sum_{j=1}^{m} \chi_{A_i \cap B_j} = \sum_{\substack{1 \le j \le m \\ \text{s.t. } A_i \cap B_j \neq \varnothing}} \chi_{A_i \cap B_j}. \end{aligned}$$

Replace χ_{A_i} in the expression of f in (1) by the above,

$$f = \sum_{i=1}^{n} \alpha_i \chi_{A_i} = \sum_{i=1}^{n} \alpha_i \left(\sum_{\substack{1 \le j \le m \\ \text{s.t. } A_i \cap B_j \neq \varnothing}} \chi_{A_i \cap B_j} \right) = \sum_{\substack{1 \le i \le n,\ 1 \le j \le m \\ \text{s.t. } A_i \cap B_j \neq \varnothing}} \alpha_i \chi_{A_i \cap B_j}.$$

$\square$

Lemma 2.12. Let $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$. Suppose there are two ways of writing f as a linear combination:

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n} \quad \text{and} \quad f = \beta_1 \chi_{B_1} + \cdots + \beta_m \chi_{B_m},$$

where $\Delta = \{A_1, \dots, A_n\}$ and $\Gamma = \{B_1, \dots, B_m\}$ are two measurable divisions of X and $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m \in [0, \infty)$. Then,

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{j=1}^{m} \beta_j \mu(B_j) \in [0, \infty].$$

Proof. For every $1 \leq i \leq n$, write

$$A_i = A_i \cap X = A_i \cap (B_1 \cup B_2 \cup \cdots \cup B_m) = \bigcup_{j=1}^{m} (A_i \cap B_j) = \bigcup_{\substack{1 \leq j \leq m \\ \text{s.t. } A_i \cap B_j \neq \emptyset}} (A_i \cap B_j).$$

Then,

$$\mu(A_i) = \sum_{\substack{1 \leq j \leq m \\ \text{s.t. } A_i \cap B_j \neq \emptyset}} \mu(A_i \cap B_j).$$

Hence, the LHS becomes

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{i=1}^{n} \alpha_i \left(\sum_{\substack{1 \leq j \leq m \\ \text{s.t. } A_i \cap B_j \neq \emptyset}} \mu(A_i \cap B_j) \right) = \sum_{\substack{1 \leq i \leq n,\, 1 \leq j \leq m \\ \text{s.t. } A_i \cap B_j \neq \emptyset}} \alpha_i \mu(A_i \cap B_j).$$

Similarly, the RHS is

$$\sum_{j=1}^{m} \beta_j \mu(B_j) = \sum_{j=1}^{m} \beta_j \left(\sum_{\substack{1 \leq i \leq n \\ \text{s.t. } A_i \cap B_j \neq \emptyset}} \mu(A_i \cap B_j) \right) = \sum_{\substack{1 \leq j \leq m,\, 1 \leq i \leq n \\ \text{s.t. } B_j \cap A_i \neq \emptyset}} \beta_j \mu(B_j \cap A_i).$$

Claim. If $A_i \cap B_j \neq \emptyset$, then $\alpha_i = \beta_j$.

Proof of Claim. Suppose that $A_i \cap B_j \neq \emptyset$. Pick $x \in A_i \cap B_j$ and calculate $f(x)$ using two ways.

$$f(x) = \alpha_1 \chi_{A_1}(x) + \cdots + \alpha_n \chi_{A_n}(x) = \alpha_i,$$
$$f(x) = \beta_1 \chi_{B_1}(x) + \cdots + \beta_m \chi_{B_m}(x) = \beta_j.$$

Hence, $\alpha_i = f(x) = \beta_j$. □

By the claim, the LHS and the RHS are equal. □

Proof of Theorem 2.10. Proof of the existence part.

For $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$, we define $L_s^+(f)$ by the following procedure. Consider a representation

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n},$$

with $\Delta = \{A_1, \ldots, A_n\}$ a measurable division of X and $\alpha_1, \ldots, \alpha_n \in [0, \infty)$. Then, define

$$L_s^+(f) := \sum_{i=1}^{n} \alpha_i \mu(A_i) \in [0, \infty].$$

Lemma 2.12 guarantees that $L_s^+(f)$ is well-defined, in the sense that the proposed quantity $L_s^+(f)$ only depends on f (and not the choice of how we write f as a linear combination of the indicators of sets from Δ). We will verify (1), and the verification of (2) and (3) is for exercise.

Fix $f, g \in \mathrm{Bor}_s^+(X, \mathbb{R})$ for which we check

$$L_s^+(f + g) = L_s^+(f) + L_s^+(g).$$

Suppose $\Gamma = \{B_1, \ldots, B_m\}$ is another measurable division of X such that $g = \beta_1 \chi_{B_1} + \cdots + \beta_m \chi_{B_m}$, where $\beta_1, \ldots, \beta_m \in [0, \infty)$. Use the definition of L_s^+ for f and g, by Lemma 2.11 (1),

$$\left[f = \sum_{i=1}^{n} \alpha_i \chi_{A_i} \right] \implies \left[L_s^+(f) = \sum_{i=1}^{n} \alpha_i \mu(A_i) \right].$$

and

$$\left[g = \sum_{j=1}^{m} \beta_j \chi_{B_j} \right] \implies \left[L_s^+(g) = \sum_{j=1}^{m} \beta_j \mu(B_j) \right].$$

By Lemma 2.11 (2), we have

$$\left[f = \sum_{\substack{1 \le i \le n,\ 1 \le j \le m \\ \text{s.t. } A_i \cap B_j \neq \emptyset}} \alpha_i \chi_{A_i \cap B_j} \right] \overset{\text{Lemma 2.12}}{\implies} \left[L_s^+(f) = \sum_{\substack{1 \le i \le n,\ 1 \le j \le m \\ \text{s.t. } A_i \cap B_j \neq \emptyset}} \alpha_i \mu(A_i \cap B_j) \right].$$

Similarly,

$$\left[g = \sum_{\substack{1\le j\le m,\, 1\le i\le n \\ \text{s.t. } B_j\cap A_i\neq\varnothing}} \beta_j \chi_{B_j\cap A_i}\right] \overset{\text{Lemma 2.12}}{\Longrightarrow} \left[L_s^+(g) = \sum_{\substack{1\le j\le m,\, 1\le i\le n \\ \text{s.t. } B_j\cap A_i\neq\varnothing}} \beta_j \mu(B_j\cap A_i)\right].$$

Adding the above two equations, we get

$$\left[f+g = \sum_{\substack{1\le i\le n,\, 1\le j\le m \\ \text{s.t. } A_i\cap B_j\neq\varnothing}} (\alpha_i+\beta_j) \chi_{A_i\cap B_j}\right] \overset{\text{Lemma 2.12}}{\Longrightarrow} \left[L_s^+(f+g) = \sum_{\substack{1\le i\le n,\, 1\le j\le m \\ \text{s.t. } A_i\cap B_j\neq\varnothing}} (\alpha_i+\beta_j) \mu(A_i\cap B_j)\right].$$

Compare the above with the sum of the expressions for $L_s^+(f)$ and $L_s^+(g)$, we have

$$L_s^+(f+g) = L_s^+(f) + L_s^+(g).$$

Proof of the uniqueness part is for exercise. □

Now, we extend L_s^+ to a map $L^+ : \mathrm{Bor}^+(X,\mathbb{R}) \to [0,\infty]$.

Definition 2.6 (L^+)**.**

For every $u \in \mathrm{Bor}^+(X,\mathbb{R})$, define

$$L^+(u) := \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f\le u\} \in [0,\infty].$$

Remark.

(1) It is didactically convenient to use letters such as $u, v, \ldots$ to denote general functions from $\mathrm{Bor}^+(X,\mathbb{R})$ as we used such as f, g to denote functions from $\mathrm{Bor}_s^+(X,\mathbb{R})$.

(2) $f\le u \iff (f(x)\le u(x),\ \forall x\in X)$.

(3) The RHS is always non-empty because $\underline{0} \in \mathrm{Bor}_s^+(X,\mathbb{R})$ and $\underline{0}\le u$.

(4) If $\nexists c \in [0,\infty)$ such that $c\ge s$ for all $s\in S$, then $\sup S = \infty$. For example, $\sup(\mathbb{N}) = \infty$ and $\sup\{1,2,\infty\} = \infty$.

Proposition 2.13. L^+ is an extension of L_s^+, i.e. if $u \in \mathrm{Bor}_s^+(X,\mathbb{R})$, then $L^+(u) = L_s^+(u)$.

Proof. We must show that

$$\sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f \le u\} = L_s^+(u).$$

Note that $\sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f \le u\} \ge L_s^+(u)$ because $L_s^+(u)$ is an element of $\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f \le u\}$ by taking $f = u$. On the other hand, we must show that $L_s^+(u)$ is an upper bound for $\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f \le u\}$. If $f \in \mathrm{Bor}_s^+(X,\mathbb{R})$ and $f \le u$, then

$$L_s^+(u) = L_s^+(f + (u - f)) = L_s^+(f) + L_s^+(u - f) \ge L_s^+(f)$$

by Theorem 2.10 (1) and note that $u - f \in \mathrm{Bor}_s^+(X,\mathbb{R})$. ☐

Lecture 12, 2026/06/08

Proposition 2.14 (**Basic Properties of** L^+).

(1) L^+ is an increasing map, i.e. $\left[u, v \in \mathrm{Bor}^+(X,\mathbb{R}),\ u \le v\right] \implies \left[L^+(u) \le L^+(v)\right]$.

(2) L^+ is homogeneous, i.e. $L^+(\alpha u) = \alpha L^+(u)$, $\forall u \in \mathrm{Bor}^+(X,\mathbb{R})$ and $\alpha \in [0,\infty)$.

Proof.

(1) Suppose that $u, v \in \mathrm{Bor}^+(X,\mathbb{R})$ and $u \le v$. Note that

$$\{f \in \mathrm{Bor}_s^+(X,\mathbb{R}) \mid f \le u\} \subseteq \{f \in \mathrm{Bor}_s^+(X,\mathbb{R}) \mid f \le v\}.$$

Hence,

$$L^+(u) = \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f \le u\} \le \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f \le v\} = L^+(v).$$

(2) If $\alpha = 0$, then $L^+(\alpha u) = 0 = \alpha L^+(u)$. The second equality follows from $0 \cdot \infty = 0$. If $\alpha > 0$, then it is easy to check that

$$\{g \in \mathrm{Bor}_s^+(X,\mathbb{R}) \mid g \le \alpha u\} = \{\alpha f \mid f \in \mathrm{Bor}_s^+(X,\mathbb{R}),\ f \le u\}.$$

By the definition of $L^+(\alpha u)$,

$$\begin{aligned} L^+(\alpha u) &= \sup\{L_s^+(\alpha f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}),\ f \le u\} \\ &= \sup\{\alpha L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}),\ f \le u\} \quad \text{by Theorem 2.10 (2)} \\ &= \alpha \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}),\ f \le u\} = \alpha L^+(u). \end{aligned}$$

□

Remark. L^+ is additive, i.e. $L^+(u+v) = L^+(u) + L^+(v)$, $\forall u, v \in \mathrm{Bor}^+(X, \mathbb{R})$.

Claim. Let $u, v \in \mathrm{Bor}^+(X, \mathbb{R})$. We have $L^+(u+v) \ge L^+(u) + L^+(v)$.

Proof of Claim. If $L^+(u+v) = \infty$, then the result holds. Assume that $L^+(u+v) < \infty$. This implies that $L^+(u)$ and $L^+(v)$ are also finite ($< \infty$) by Proposition 2.14 (1). Now, fix $\epsilon > 0$, it suffices to prove

$$L^+(u+v) \ge L^+(u) + L^+(v) - \epsilon.$$

By the definition of $L^+(u)$, $\exists f, g \in \mathrm{Bor}_s^+(X, \mathbb{R})$ such that $f \le u$, $g \le v$, and

$$L_s^+(f) > L^+(u) - \frac{\epsilon}{2}, \quad L_s^+(g) > L^+(v) - \frac{\epsilon}{2}.$$

Then, $f + g \in \mathrm{Bor}_s^+(X, \mathbb{R})$ is such that $f + g \le u + v$ and

$$\begin{aligned} L^+(u+v) &\ge L_s^+(f+g) = L_s^+(f) + L_s^+(g) \quad \text{by Theorem 2.10 (1)} \\ &> \left(L^+(u) - \frac{\epsilon}{2}\right) + \left(L^+(v) - \frac{\epsilon}{2}\right) = L^+(u) + L^+(v) - \epsilon. \end{aligned}$$

□

Note that we cannot do the same to prove the other direction $L^+(u+v) \le L^+(u) + L^+(v)$ by showing

$$L^+(u+v) \le L^+(u) + L^+(v) + \epsilon.$$

So we want $L^+(u) + L^+(v) \ge L^+(u+v) - \epsilon$.

Idea: pick $h \in \mathrm{Bor}_s^+(X, \mathbb{R})$ such that $h \le u+v$ and $L_s^+(h) > L^+(u+v) - \epsilon$. Now, decompose $\underline{h = f + g}$ for some $f, g \in \mathrm{Bor}_s^+(X, \mathbb{R})$ such that $f \le u$ and $g \le v$. Then,

$$L^+(u) + L^+(v) \ge L_s^+(f) + L_s^+(g) = L_s^+(h) > L^+(u+v) - \epsilon.$$

Note: We do not know how to decompose h into $f+g$ with $f \leq u$ and $g \leq v$.

Note (**Midterm 1 Info**).

- MC 4061.
- Coverage: everything before this note.
- 3 questions:
  - Q1: proof from class (on blackboard).
  - Q2 and Q3: similar to problem set problems..

2.6 Monotone Convergence Theorem (MCT)

Definition 2.7 (**Notation**).

Let u and $u_1, \dots, u_n, \dots$ be functions in $\mathrm{Bor}^+(X, \mathbb{R})$. We write $u_n \nearrow u$ if $\forall x \in X$,

$$0 \leq u_1(x) \leq u_2(x) \leq \cdots \leq u_n(x) \leq \cdots \quad \text{and} \quad \lim_{n\to\infty} u_n(x) = u(x).$$

Theorem 2.15 (**Monotone Convergence Theorem (MCT)**).

Let u and $(u_n)_{n=1}^{\infty}$ be functions in $\mathrm{Bor}^+(X, \mathbb{R})$ such that $u_n \nearrow u$. Then,

$$L^+(u) = \lim_{n\to\infty} L^+(u_n).$$

Note. MCT is one of the fundamental theorems of the Lebesgue integration theory. By MCT, we can prove the other direction of the additivity of L^+ from the previous section.

Proposition 2.16. Let $u, v \in \mathrm{Bor}^+(X, \mathbb{R})$. Then,

$$L^+(u+v) \leq L^+(u) + L^+(v).$$

Proof. Proposition 2.9 gives sequences $(f_n)_{n=1}^{\infty}$ and $(g_n)_{n=1}^{\infty}$ in $\mathrm{Bor}_s^+(X, \mathbb{R})$ s.t. $f_n \nearrow u$ and $g_n \nearrow v$.

Thus, $(f_n + g_n)_{n=1}^{\infty}$ is a sequence in $\mathrm{Bor}_s^+(X, \mathbb{R})$ such that $f_n + g_n \nearrow u + v$. Then,

$$\begin{aligned}
L^+(u+v) &= \lim_{n\to\infty} L^+(f_n + g_n) \quad \text{by MCT} \\
&= \lim_{n\to\infty} L_s^+(f_n + g_n) \quad \text{since } L^+ \text{ is an extension of } L_s^+ \\
&= \lim_{n\to\infty} (L_s^+(f_n) + L_s^+(g_n)) \quad \text{by Theorem 2.10 (1)} \\
&= \lim_{n\to\infty} (L^+(f_n) + L^+(g_n)) \quad \text{since } L^+ \text{ is an extension of } L_s^+ \\
&= \lim_{n\to\infty} L^+(f_n) + \lim_{n\to\infty} L^+(g_n).
\end{aligned}$$

By Proposition 2.14 (1), $L^+(f_n) \le L^+(u)$ and $L^+(g_n) \le L^+(v)$ for all $n \in \mathbb{N}$. Hence,

$$\lim_{n\to\infty} L^+(f_n) + \lim_{n\to\infty} L^+(g_n) \le L^+(u) + L^+(v), \quad \forall n \in \mathbb{N}.$$

Therefore, $L^+(u+v) \le L^+(u) + L^+(v)$. □

Lecture 13, 2026/06/10

Remark. To prove the MCT, we need to prove the following fact.

> *Claim.* If $u_n \nearrow u \in \mathrm{Bor}^+(X, \mathbb{R})$, then the increasing limit in $[0, \infty]$
>
> $$\Lambda := \lim_{n\to\infty} L^+(u_n) = \sup\{L^+(u_n) \mid n \in \mathbb{N}\}$$
>
> satisfies $\Lambda \ge L^+(u)$.

Why is this sufficient?

- From $u_1 \le u_2 \le \cdots \le u_n \le \cdots$, it follows from Proposition 2.14 (1) that $L^+(u_1) \le L^+(u_2) \le \cdots \le L^+(u_n) \le \cdots$, which implies the existence of Λ.
- From $u_n \nearrow u$, we also get $u_n \le u$, $\forall n \in \mathbb{N}$. Hence, $L^+(u_n) \le L^+(u)$, $\forall n \in \mathbb{N}$, which implies $L^+(u)$ is an upper bound for $\{L^+(u_n) \mid n \in \mathbb{N}\}$ and thus $L^+(u) \ge \Lambda$.
- If we can prove the claim that $\Lambda \ge L^+(u)$, then we have $\Lambda = L^+(u)$, and MCT follows.

The following two lemmas are special cases of the claim. Lemma 2.17 is the special case when $u = \chi_A$ with $A \in \mathfrak{M}$, and Lemma 2.18 is the special case when $u = f_0 \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

Lemma 2.17. Let $u_1 \le u_2 \le \cdots \le u_n \le \cdots$ be in $\mathrm{Bor}^+(X, \mathbb{R})$ and let $A \in \mathfrak{M}$, such that $u_n \nearrow \chi_A$. Consider the increasing limit

$$\Lambda := \lim_{n\to\infty} L^+(u_n) \in [0, \infty].$$

Then, $\Lambda \ge \mu(A)$ (also note that $\mu(A) = L^+(\chi_A)$).

Proof. For $A = \varnothing$, we have $\chi_A = \underline{0}$ and also $u_n = \underline{0}$ for all $n \in \mathbb{N}$. Hence, $\Lambda = 0 = \mu(A)$. Now, assume that $A \ne \varnothing$. For now, fix $\theta \in (0,1)$. For every $n \in \mathbb{N}$, let $A_n = \{x \in X \mid u_n(x) > \theta\}$. Observe that $A_n \in \mathfrak{M}$ for all $n \in \mathbb{N}$ because $A_n = A \cap u_n^{-1}((\theta, \infty))$ and $u_n^{-1}((\theta,\infty)) \in \mathfrak{M}$ by the definition of $\mathrm{Bor}(X, \mathbb{R})$. Also, observe that $A_n \subseteq A_{n+1}$ for all $n \in \mathbb{N}$. Indeed,

$$\begin{aligned}\Big[x \in A_n\Big] &\implies \Big[x \in A \text{ and } u_n(x) > \theta\Big] \\ &\implies \Big[x \in A \text{ and } u_{n+1}(x) > \theta\Big] \quad \text{since } u_n(x) \le u_{n+1}(x) \\ &\implies \Big[x \in A_{n+1}\Big].\end{aligned}$$

Moreover, $A = \bigcup_{n=1}^\infty A_n$. Indeed, $A_n \subseteq A$ for all $n \in \mathbb{N}$, hence $\bigcup_{n=1}^\infty A_n \subseteq A$. For "$\supseteq$", fix $x \in A$. Then,

$$u_n(x) \underset{n\to\infty}{\longrightarrow} \chi_A(x) = 1 > \theta.$$

Hence, $\exists n_0 \in \mathbb{N}$ such that $u_{n_0}(x) > \theta$. This implies that $x \in A_{n_0} \subseteq \bigcup_{n=1}^\infty A_n$.

Therefore, $(A_n)_{n=1}^\infty$ is an increasing chain of sets in $\mathfrak{M}$ with $\bigcup_{n=1}^\infty A_n = A$. By the continuity of μ for increasing chains,

$$\mu(A) = \lim_{n\to\infty} \mu(A_n).$$

For every $n \in \mathbb{N}$, we have $u_n \ge \theta\chi_{A_n}$. We check cases $x \in A_n$ and $x \in X \setminus A_n$ separately. If $x \in X\setminus A_n$, then $\theta\chi_{A_n}(x) = 0 \le u_n(x)$. If $x \in A_n$, then $\theta\chi_{A_n}(x) = \theta < u_n(x)$. By Proposition 2.14 (1),

$$\begin{aligned}L^+(u_n) \ge L^+(\theta\chi_{A_n}) &= \theta L^+(\chi_{A_n}) \quad \text{by Proposition 2.14 (2)} \\ &= \theta L_s^+(\chi_{A_n}) = \theta\mu(A_n).\end{aligned}$$

So, $L^+(u_n) \ge \theta\mu(A_n), \forall n \in \mathbb{N}$. Hence, $\Lambda \ge \theta\mu(A)$. Since $\theta \in (0,1)$ is arbitrary, we have $\Lambda \ge \mu(A)$. $\square$

Note. Above is a good proof for the final exam!

Lemma 2.18. Let $u_1 \le u_2 \le \cdots \le u_n \le \cdots$ be in $\mathrm{Bor}^+(X, \mathbb{R})$ and let $f_0 \in \mathrm{Bor}_s^+(X, \mathbb{R})$, such that $u_n \nearrow f_0$. Consider the increasing limit

$$\Lambda := \lim_{n\to\infty} L^+(u_n) \in [0, \infty].$$

Then, $\Lambda \ge L_s^+(f_0)$ (also note that $L_s^+(f_0) = L^+(f_0)$).

Note. This proof will not be on the final exam.

Proof. to be continued □

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Proposition 2.19. Let $u_n \nearrow u$ be in $\mathrm{Bor}^+(X, \mathbb{R})$. Then, $\Lambda = \lim_{n\to\infty} L^+(u_n) \ge L^+(u)$.

Proof. to be continued □

Theorem 2.20. The map $L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \to [0, \infty]$ satisfies and is uniquely determined by the following properties:

(1) $L^+(\chi_A) = \mu(A), \forall A \in \mathfrak{M}$.

(2) $L^+(u + v) = L^+(u) + L^+(v), \forall u, v \in \mathrm{Bor}^+(X, \mathbb{R})$.

(3) $L^+(\alpha u) = \alpha L^+(u), \forall u \in \mathrm{Bor}^+(X, \mathbb{R})$ and $\alpha \in [0, \infty)$.

(4) $\lim_{n\to\infty} L^+(u_n) = L^+(u)$ whenever u and $(u_n)_{n=1}^\infty$ are in $\mathrm{Bor}^+(X, \mathbb{R})$ such that $u_n \nearrow u$.

Proof. For the existence part, (1) follows from Proposition 2.13, (2) follows from the additivity of L^+, (3) follows from the homogeneity of L^+, and (4) is MCT.

For the uniqueness part, we consider the following lemma (which will also be useful later on). □

Lemma 2.21 (Method of the "Friendly Functions").
Let $(X, \mathfrak{M})$ be a measurable space and consider a set of functions $\mathcal{G} \subseteq \mathrm{Bor}^+(X, \mathbb{R})$ which has the following properties:

(1) $\chi_A \in \mathcal{G}, \forall A \in \mathfrak{M}$.

(2) $\Big[g_1, g_2 \in \mathcal{G}\Big] \implies \Big[g_1 + g_2 \in \mathcal{G}\Big]$.

(3) $\Big[g \in \mathcal{G},\ \alpha \in [0,\infty)\Big] \implies \Big[\alpha g \in \mathcal{G}\Big]$.

(4) $\Big[g \in \mathrm{Bor}^+(X,\mathbb{R}),\ g_1 \le g_2 \le \cdots \le g_n \le \cdots \text{ in } \mathcal{G},\text{ and } g_n \nearrow g\Big] \implies \Big[g \in \mathcal{G}\Big]$.

Under the above assumptions, $\mathcal{G} = \mathrm{Bor}^+(X,\mathbb{R})$.

Proof. For $f \in \mathrm{Bor}_s^+(X,\mathbb{R})$, we can write

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n}, \quad \text{by Lemma 2.11 (1).}$$

It follows that $f \in \mathcal{G}$ by (1), (2), and (3). Hence, $\mathrm{Bor}_s^+(X,\mathbb{R}) \subseteq \mathcal{G}$.

For every $u \in \mathrm{Bor}^+(X,\mathbb{R})$, by Proposition 2.9, $\exists (f_n)_{n=1}^\infty$ in $\mathrm{Bor}_s^+(X,\mathbb{R})$ such that $f_n \nearrow u$. Then, $u \in \mathcal{G}$ by (4) and the first part of the proof (i.e. $f_n \in \mathcal{G}, \forall n \in \mathbb{N}$). Hence, $\mathrm{Bor}^+(X,\mathbb{R}) \subseteq \mathcal{G}$.

Thus, $\mathrm{Bor}^+(X,\mathbb{R}) = \mathcal{G}$. □

Now we prove the uniqueness part of Theorem 2.20.

Proof. Suppose that there is another map $M^+ : \mathrm{Bor}^+(X,\mathbb{R}) \to [0,\infty]$ satisfying (1)-(4) in Theorem 2.20. We want to show that $M^+(u) = L^+(u)$ for all $u \in \mathrm{Bor}^+(X,\mathbb{R})$. Let

$$\mathcal{G} = \{g \in \mathrm{Bor}^+(X,\mathbb{R}) \mid L^+(g) = M^+(g)\} \subseteq \mathrm{Bor}^+(X,\mathbb{R}).$$

We check that $\mathcal{G} = \mathrm{Bor}^+(X,\mathbb{R})$ using the above lemma.

(1) $\chi_A \in \mathcal{G}$ because $L^+(\chi_A) = \mu(A) = M^+(\chi_A)$ by (1) in Theorem 2.20.

(2) If $g_1, g_2 \in \mathcal{G}$, then

$$\begin{aligned} L^+(g_1 + g_2) &= L^+(g_1) + L^+(g_2) \quad \text{by (2) in Theorem 2.20 for } L^+ \\ &= M^+(g_1) + M^+(g_2) \quad \text{since } g_1, g_2 \in \mathcal{G} \\ &= M^+(g_1 + g_2) \quad \text{by (2) in Theorem 2.20 for } M^+. \end{aligned}$$

Hence, $g_1 + g_2 \in \mathcal{G}$.

(3) Exercise.

(4) Suppose that $u \in \mathrm{Bor}^+(X, \mathbb{R})$ and $g_1 \le g_2 \le \cdots \le g_n \le \cdots$ are in $\mathcal{G}$ such that $g_n \nearrow u$. By (4) in Theorem 2.20 for L^+ and M^+,

$$\begin{aligned} L^+(u) &= \lim_{n\to\infty} L^+(g_n) \quad \text{by (4) for } L^+ \\ &= \lim_{n\to\infty} M^+(g_n) \quad \text{since } g_n \in \mathcal{G} \\ &= M^+(u) \quad \text{by (4) for } M^+. \end{aligned}$$

Hence, $u \in \mathcal{G}$.

Therefore, $\mathcal{G} = \mathrm{Bor}^+(X, \mathbb{R})$ and thus $M^+(u) = L^+(u)$ for all $u \in \mathrm{Bor}^+(X, \mathbb{R})$. □

Lecture 15, 2026/06/15

2.7 The Space of Integrable Functions, $\mathcal{L}^1(\mu)$

Remark. We want to define a map L on $\mathrm{Bor}(X, \mathbb{R})$ that also tries to become the integral with respect to μ. To do so, we will first split any $f \in \mathrm{Bor}(X, \mathbb{R})$ into its positive part f^+ and negative part f^-.

Definition 2.8 (Notation).
Define

$$f_+ := f \vee \underline{0} = \max\{f, \underline{0}\}, \quad f_- := (-f) \vee \underline{0} = \max\{-f, \underline{0}\}$$

where $\underline{0} : X \to \mathbb{R}$ is the zero function.

Note that $f_+, f_- \in \mathrm{Bor}(X, \mathbb{R})$ because $\mathrm{Bor}(X, \mathbb{R})$ is stable under $\vee$. Clearly, $f_+, f_- \ge \underline{0}$. Also, we can write explicitly:

$$f_+(x) = \begin{cases} f(x), & \text{if } f(x) \ge 0, \\ 0, & \text{if } f(x) < 0, \end{cases} \qquad f_-(x) = \begin{cases} 0, & \text{if } f(x) \ge 0, \\ -f(x), & \text{if } f(x) < 0. \end{cases}$$

We see that

$$f_+ + f_- = |f|, \quad f_+ - f_- = f.$$

Thus,

$$f_+ = \frac{f + |f|}{2}, \quad f_- = \frac{|f| - f}{2}.$$

Lemma 2.22. For $f \in \mathrm{Bor}(X, \mathbb{R})$,

$$\Big[L^+(|f|) < \infty\Big] \iff \Big[L^+(f_+) < \infty \text{ and } L^+(f_-) < \infty\Big].$$

Proof.
($\Rightarrow$): $\Big[f(t) \leq |f(t)|\Big] \implies \Big[L^+(f_+) \leq L^+(|f|) < \infty\Big]$. Similarly, $\Big[-f(t) \leq |f(t)|\Big] \implies \Big[L^+(f_-) \leq L^+(|f|) < \infty\Big]$.

($\Leftarrow$): $L^+(|f|) = L^+(f_+ + f_-) = L^+(f_+) + L^+(f_-) < \infty$. $\square$

Definition 2.9 (Integrable, $\mathcal{L}^1(\mu)$, and L).
A function $f : X \to \mathbb{R}$ is said to be **integrable** with respect to μ if $f \in \mathrm{Bor}(X, \mathbb{R})$ and $L^+(|f|) < \infty$. We denote

$$\mathcal{L}^1(\mu) := \{f : X \to \mathbb{R} \mid f \text{ is integrable with respect to } \mu\}.$$

We define a map $L : \mathcal{L}^1(\mu) \to \mathbb{R}$ by

$$L(f) := L^+(f_+) - L^+(f_-), \quad \forall f \in \mathcal{L}^1(\mu).$$

Lemma 2.23. Let $f \in \mathcal{L}^1(\mu)$. Suppose that $f = h_1 - h_2$ for some $h_1, h_2 \in \mathrm{Bor}^+(X, \mathbb{R})$ such that $L^+(h_1) < \infty$ and $L^+(h_2) < \infty$. Then, $L(f) = L^+(h_1) - L^+(h_2)$.

Proof. Since $f = f_+ - f_-$ and $f = h_1 - h_2$, we have $f_+ + h_2 = f_- + h_1$. Then,

$$L^+(f_+) + L^+(h_2) = L^+(f_+ + h_2) = L^+(f_- + h_1) = L^+(f_-) + L^+(h_1).$$

Hence, $L^+(f_+) - L^+(f_-) = L^+(h_1) - L^+(h_2)$ (numbers in $[0, \infty)$), which implies $L(f) = L^+(h_1) - L^+(h_2)$. $\square$

Theorem 2.24.
(1) $\mathcal{L}^1(\mu)$ is stable under linear combinations, and hence, is a linear subspace of $\mathrm{Bor}(X, \mathbb{R})$.
(2) The map $L : \mathcal{L}^1(\mu) \to \mathbb{R}$ is linear.

Proof.

(1) Pick some $f, g \in \mathcal{L}^1(\mu)$ and $\alpha, \beta \in \mathbb{R}$. Let $h \coloneqq \alpha f + \beta g \in \mathrm{Bor}(X, \mathbb{R})$. We want to show that $h \in \mathcal{L}^1(\mu)$, i.e. $L^+(|h|) < \infty$. Note that

$$|h| = |\alpha f + \beta g| \leq |\alpha| \cdot |f| + |\beta| \cdot |g|.$$

Then,

$$\begin{aligned} L^+(|h|) &\leq L^+(|\alpha| \cdot |f| + |\beta| \cdot |g|) \\ &\leq |\alpha| \cdot L^+(|f|) + |\beta| \cdot L^+(|g|) \quad \text{by Theorem 2.20 (2) and (3)} \\ &< \infty \quad \text{since } f, g \in \mathcal{L}^1(\mu). \end{aligned}$$

It follows that $h \in \mathcal{L}^1(\mu)$.

(2) We first verify that $L(f+g) = L(f) + L(g)$ for all $f, g \in \mathcal{L}^1(\mu)$. Write

$$\begin{aligned} f + g &= (f_+ - f_-) + (g_+ - g_-) \\ &= \underbrace{(f_+ + g_+)}_{h_1} - \underbrace{(f_- + g_-)}_{h_2}. \end{aligned}$$

Observe that $h_1, h_2 \in \mathrm{Bor}^+(X, \mathbb{R})$ with $L^+(h_1) = L^+(f_+) + L^+(g_+) < \infty$ and $L^+(h_2) < \infty$. By the previous lemma,

$$\begin{aligned} L(f+g) &= L^+(h_1) - L^+(h_2) \\ &= (L^+(f_+) + L^+(g_+)) - (L^+(f_-) + L^+(g_-)) \\ &= (L^+(f_+) - L^+(f_-)) + (L^+(g_+) - L^+(g_-)) \\ &= L(f) + L(g). \end{aligned}$$

Next, we verify that $L(\alpha f) = \alpha L(f)$ for all $f \in \mathcal{L}^1(\mu)$ and $\alpha \in \mathbb{R}$. We check cases $\alpha > 0$, $\alpha = 0$, and $\alpha < 0$ separately. If $\alpha > 0$, then

$$\begin{aligned}
L(\alpha f) &= L^+((\alpha f)_+) - L^+((\alpha f)_-) \\
&= L^+(\alpha f_+) - L^+(\alpha f_-) \quad \text{since } \alpha > 0 \\
&= \alpha L^+(f_+) - \alpha L^+(f_-) \quad \text{by Proposition 2.14 (2)} \\
&= \alpha L(f).
\end{aligned}$$

If $\alpha = 0$, then $L(\alpha f) = L(\underline{0}) = 0 = \alpha L(f)$. We assume $\alpha = -1$ for the case $\alpha < 0$. Note that $(-f)_+ = f_-$ and $(-f)_- = f_+$. Hence,

$$\begin{aligned}
L(\alpha f) = L(-f) &= L^+((-f)_+) - L^+((-f)_-) \\
&= L^+(f_-) - L^+(f_+) \\
&= -(L^+(f_+) - L^+(f_-)) = -L(f) = \alpha L(f).
\end{aligned}$$

For the general case $\alpha < 0$, write $\alpha = (-1) \cdot |\alpha|$ and use the above two cases to get $L(\alpha f) = L((-1) \cdot (|\alpha| f)) = -L(|\alpha| f) = -(|\alpha| L(f)) = \alpha L(f)$. $\square$

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Proposition 2.25.

(1) If $f \in \mathcal{L}^1(\mu)$ and also $f \in \mathrm{Bor}^+(X, \mathbb{R})$, then $L(f) = L^+(f)$.

(2) If $f, g \in \mathcal{L}^1(\mu)$ with $f \leq g$, then $L(f) \leq L(g)$.

(3) If $f \in \mathcal{L}^1(\mu)$, then $|f| \in \mathcal{L}^1(\mu)$ and $|L(f)| \leq L(|f|)$.

Proof.

(1) Note that $f_+ = f \vee \underline{0} = f$ since $f \geq \underline{0}$, and $f_- = (-f) \vee \underline{0} = \underline{0}$ since $-f \leq \underline{0}$. Hence,

$$L(f) = L^+(f_+) - L^+(f_-) = L^+(f) - L^+(\underline{0}) = L^+(f) \in [0, \infty).$$

(2) Let $f, g \in \mathcal{L}^1(\mu)$ with $f \leq g$. Then,

$$\begin{aligned} f \leq g &\implies g - f \in \mathcal{L}^1(\mu) \cap \mathrm{Bor}^+(X, \mathbb{R}) \\ &\implies L(g-f) = L^+(g-f) \in [0, \infty) \quad \text{by (1)} \\ &\implies L(g) - L(f) = L(g-f) \geq 0 \quad \text{by linearity of } L \\ &\implies L(g) \geq L(f). \end{aligned}$$

(3) Let $f \in \mathcal{L}^1(\mu)$. Note that

$$\begin{aligned} f \in \mathcal{L}^1(\mu) &\implies L^+(|f|) < \infty \\ &\implies |f| \in \mathcal{L}^1(\mu) \cap \mathrm{Bor}^+(X, \mathbb{R}) \quad \text{with } L(|f|) = L^+(|f|) < \infty \text{ by (1)} \end{aligned}$$

Observe that $-|f| \leq f \leq |f|$. By (2),

$$\begin{aligned} L(-|f|) \leq L(f) \leq L(|f|) &\implies -L(|f|) \leq L(f) \leq L(|f|) \quad \text{by linearity of } L \\ &\implies |L(f)| \leq L(|f|). \end{aligned}$$

□

Remark. For our measure space $(X, \mathfrak{M}, \mu)$, we have constructed two maps:

$$L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \to [0, \infty], \quad \text{and} \quad L : \mathcal{L}^1(\mu) \to \mathbb{R}$$

which are trying to become the integral with respect to μ.

Question: Suppose that $f \in \mathrm{Bor}(X, \mathbb{R})$ is such that both $L^+(f)$ and $L(f)$ are defined. Do we have $L(f) = L^+(f)$?

Answer: Yes! This is exactly Proposition 2.25 (1).

Now, we are ready to switch to the integral notation. If $f \in \mathrm{Bor}(X, \mathbb{R})$ and at least one of $L^+(f)$ and $L(f)$ is defined, then we denote that value by $\int_X f(x)\, d\mu(x)$, abbreviated as $\int f\, d\mu$.

Note. The integral notation can mean either $L^+(f)$ or $L(f)$ depending on the context. If $f \in \mathrm{Bor}^+(X, \mathbb{R})$, then $\int f\, d\mu = L^+(f)$. If $f \in \mathcal{L}^1(\mu)$, then $\int f\, d\mu = L(f)$. If $f \in \mathrm{Bor}(X, \mathbb{R})$ is such that both $L^+(f)$ and $L(f)$ are defined, then $\int f\, d\mu = L(f) = L^+(f)$.

There is a subtlety in the above definition of the integral. For example, Proposition 2.25 (3) becomes

$$\left|\int_X f(x)\,d\mu(x)\right| \leq \int_X |f(x)|\,d\mu(x), \quad \forall f \in \mathcal{L}^1(\mu)$$

where the RHS can be viewed as either $L^+(|f|)$ or $L(|f|)$.

2.8 Integral on A Subset

Definition 2.10 (Notation).

(1) For any $f \in \mathrm{Bor}^+(X, \mathbb{R})$ and $A \in \mathfrak{M}$, we denote

$$\int_A f\,d\mu := \int_X f \cdot \chi_A\,d\mu \in [0, \infty].$$

(2) For any $f \in \mathcal{L}^1(\mu)$ and $A \in \mathfrak{M}$, we observe that $f \cdot \chi_A \in \mathcal{L}^1(\mu)$ (see below), and we denote

$$\int_A f\,d\mu := \int_X f \cdot \chi_A\,d\mu \in \mathbb{R}.$$

Note. (1) is the integral in L^+ sense and (2) is the integral in L sense. To see that $f \cdot \chi_A \in \mathcal{L}^1(\mu)$. Since $f \in \mathcal{L}^1(\mu)$, we know $f \in \mathrm{Bor}(X, \mathbb{R})$. Also, since $A \in \mathfrak{M}$, we have $\chi_A \in \mathrm{Bor}(X, \mathbb{R})$. Hence, $f \cdot \chi_A \in \mathrm{Bor}(X, \mathbb{R})$. Moreover,

$$|f \cdot \chi_A| = |f| \cdot |\chi_A| = |f| \cdot \chi_A \leq |f|.$$

Hence,

$$L^+(|f \cdot \chi_A|) \leq L^+(|f|) < \infty \implies f \cdot \chi_A \in \mathcal{L}^1(\mu).$$

Proposition 2.26. Let $(X, \mathfrak{M}, \mu)$ be a measure space and let $h \in \mathrm{Bor}^+(X, \mathbb{R})$ Define $\nu : \mathfrak{M} \to [0, \infty]$ by

$$\nu(A) := \int_A h\,d\mu, \quad \forall A \in \mathfrak{M}.$$

Then, ν is a positive measure on $\mathfrak{M}$.

Proof. Observe that $\nu(\varnothing) = \int_X h \cdot \chi_\varnothing\,d\mu = \int_X h \cdot \underline{0}\,d\mu = 0$. Now, fix $(A_n)_{n=1}^\infty$ in $\mathfrak{M}$ with $A_i \cap A_j = \varnothing$ for all $i \neq j$. Let $U = \bigcup_{n=1}^\infty A_n$. We want to show that $\nu(U) = \sum_{n=1}^\infty \nu(A_n)$. Exercise. ☐

Proposition 2.27. Let the measure space $(X, \mathfrak{M}, \mu)$, the function $h \in \mathrm{Bor}^+(X, \mathbb{R})$, and the positive measure ν be as in Proposition 2.26. Then, for every $f \in \mathrm{Bor}^+(X, \mathbb{R})$,

$$\int_X f\, d\nu = \int_X fh\, d\mu \in [0, \infty].$$

Note. This is the equality of integrals in L^+ sense.

Proof. We use the method of the "friendly functions" with Lemma 2.21. Consider

$$\mathcal{G} := \left\{ g \in \mathrm{Bor}^+(X, \mathbb{R}) \mid \int_X g\, d\nu = \int_X gh\, d\mu \right\}.$$

Exercise: verify that $\mathcal{G}$ satisfies the four properties in Lemma 2.21.

Hence, $\mathcal{G} = \mathrm{Bor}^+(X, \mathbb{R})$ and thus $\int_X g\, d\nu = \int_X gh\, d\mu$ for all $g \in \mathrm{Bor}^+(X, \mathbb{R})$. □

Note. We may write $d\nu = h\, d\mu$. Then,

$$h = \frac{d\nu}{d\mu}$$

can be thought of as some kind of "derivative" of ν with respect to μ.

Lecture 17, 2026/06/22

2.9 Lebesgue's Dominated Convergence Theorem (LDCT)

The important LDCT addresses the issue of interchanging limit and integral.

Theorem 2.28 (**LDCT**).
Let f and $(f_n)_{n=1}^{\infty}$ be in $\mathrm{Bor}(X, \mathbb{R})$ such that $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in X$. Suppose moreover that $\exists h \in \mathrm{Bor}^+(X, \mathbb{R}) \cap \mathcal{L}^1(\mu)$ which dominates all f_n's, in the sense that $|f_n| \le h$ for all $n \in \mathbb{N}$. Then, f and $f_1, f_2, \ldots$ are all in $\mathcal{L}^1(\mu)$ and

$$\lim_{n\to\infty} \int_X f_n\, d\mu = \int_X f\, d\mu.$$

Note. The above can be written as $\lim\limits_{n\to\infty} \left(\int_X f_n\, d\mu \right) = \int_X \left(\lim\limits_{n\to\infty} f_n \right) d\mu$. Thus, it's about interchanging limit and integral. Here is a scheme of the proof:

$$\text{MCT} \overset{(1)}{\implies} \text{Reverse-MCT} \overset{(2)}{\implies} \text{LDCT}_0 \overset{(3)}{\implies} \text{LDCT}.$$

Proposition 2.29 (**Reverse-MCT**).
Let u and $(u_n)_{n=1}^{\infty}$ be in $\mathrm{Bor}^{+}(X, \mathbb{R})$ such that $u_n \searrow u$, i.e. $u_1 \geq u_2 \geq \cdots \geq u_n \geq \cdots$ and $\lim_{n\to\infty} u_n(x) = u(x)$ for all $x \in X$. Also, assume that $\int_X u_1 \, d\mu < \infty$. Then,

$$\lim_{n\to\infty} \int_X u_n \, d\mu = \int_X u \, d\mu.$$

Note. Reverse-MCT is a special case of LDCT where we can use $h = u_1$ as the dominating function.

Idea of proof. Apply MCT to $v_n \nearrow v$ where $v_n = u_1 - u_n$ for all $n \in \mathbb{N}$ and $v = u_1 - u$. Check LEARN file for details. □

Proposition 2.30 (**LDCT$_0$**).
Let $(g_n)_{n=1}^{\infty}$ be in $\mathrm{Bor}^{+}(X, \mathbb{R})$ such that $\lim_{n\to\infty} g_n(x) = 0$ for all $x \in X$. Suppose moreover that $\exists h \in \mathrm{Bor}^{+}(X, \mathbb{R}) \cap \mathcal{L}^1(\mu)$ such that $g_n \leq h$ for all $n \in \mathbb{N}$. Then,

$$\lim_{n\to\infty} \int_X g_n \, d\mu = 0.$$

Note. This is the essential part to prove LDCT.

Proof. We will use the "trick of Fatou". The idea is to create a new sequence $(u_n)_{n=1}^{\infty}$ to which we can apply the Reverse-MCT.

First, observe that for every $x \in X$, $\{g_k(x) \mid k \in \mathbb{N}\}$ is a bounded subset of $[0, \infty)$ since $g_n(x) \leq h(x)$ for all $n \in \mathbb{N}$. Hence, it makes sense to define $u_1 : X \to [0, \infty)$ by

$$u_1(x) := \sup\{g_k(x) \mid k \in \mathbb{N}\}, \quad \forall x \in X.$$

Note that $u_1 \in \mathrm{Bor}^{+}(X, \mathbb{R})$ since $\mathrm{Bor}(X, \mathbb{R})$ is stable under sup. Also, $u_1 \leq h$ by the definition of sup. This implies that

$$\int_X u_1 \, d\mu \leq \int_X h \, d\mu < \infty.$$

Next, we look at

$$\{g_k(x) \mid k \geq 2\} \subseteq \{g_k(x) \mid k \in \mathbb{N}\}$$

and define $u_2 : X \to [0, \infty)$ by

$$u_2(x) := \sup\{g_k(x) \mid k \geq 2\}, \quad \forall x \in X.$$

By the same argument, $u_2 \in \mathrm{Bor}^+(X, \mathbb{R})$ and $u_2 \leq h$. In general, for every $n \in \mathbb{N}$, we look at

$$\{g_k(x) \mid k \geq n\} \subseteq \{g_k(x) \mid k \in \mathbb{N}\}$$

which is a bounded subset of $[0, \infty)$ and define $u_n : X \to [0, \infty)$ by

$$u_n(x) := \sup\{g_k(x) \mid k \geq n\}, \quad \forall x \in X.$$

Then, $u_n \in \mathrm{Bor}^+(X, \mathbb{R})$ and $u_n \leq u_{n-1}$. Now, we have constructed the sequence $(u_n)_{n=1}^{\infty}$ to which we can apply the Reverse-MCT. Next, we check that $\lim_{n\to\infty} u_n(x) = 0$ for all $x \in X$. Fix $x \in X$. Let $\epsilon > 0$ be given. We need $n_0 \in \mathbb{N}$ be such that $u_n(x) < \epsilon$ for all $n \geq n_0$. Then,

$$\lim_{n\to\infty} g_n(x) = 0 \implies \exists n_0 \in \mathbb{N} \text{ such that } g_n(x) < \frac{\epsilon}{2} \text{ for all } n \geq n_0.$$

Hence,

$$u_{n_0}(x) = \sup\{g_k(x) \mid k \geq n_0\} \leq \frac{\epsilon}{2} < \epsilon.$$

Then, for all $n \geq n_0$,

$$u_n(x) \leq u_{n_0}(x) < \epsilon.$$

Now, observe that $g_n \leq u_n$ for all $n \in \mathbb{N}$. Indeed, we defined

$$u_n(x) = \sup\{g_n(x), g_{n+1}(x), g_{n+2}(x), \ldots\} \geq g_n(x), \quad \forall x \in X.$$

So, $0 \leq g_n \leq u_n$, which implies that

$$0 \leq \int_X g_n \, d\mu \leq \int_X u_n \, d\mu \xrightarrow[n\to\infty]{} \int_X 0 \, d\mu = 0 \quad \text{by Reverse-MCT.}$$

$\square$

Proof of Theorem 2.28. For every $n \in \mathbb{N}$, we have

$$\Big[|f_n| \le h\Big] \implies \int_X |f_n| \, d\mu \le \int_X h \, d\mu < \infty.$$

Hence, $f_n \in \mathcal{L}^1(\mu)$ for all $n \in \mathbb{N}$. Also, observe that $|f| \le h$, which holds because $|f(x)| = \lim_{n\to\infty} |f_n(x)| \le h(x)$ for all $x \in X$. Thus,

$$\int_X |f| \, d\mu \le \int_X h \, d\mu < \infty,$$

and hence $f \in \mathcal{L}^1(\mu)$. Now, it remains to show that

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Consider

$$g_n := |f_n - f| \in \mathrm{Bor}^+(X, \mathbb{R}), \quad \forall n \in \mathbb{N}.$$

Claim. The g_n's satisfy the hypotheses of LDCT_0. That is,

(1) $\lim_{n\to\infty} g_n(x) = 0$ for all $x \in X$.

(2) $\exists \tilde{h} \in \mathrm{Bor}^+(X, \mathbb{R}) \cap \mathcal{L}^1(\mu)$ such that $g_n \le \tilde{h}$ for all $n \in \mathbb{N}$.

Proof of Claim.

(1) For every $x \in X$, we have

$$g_n(x) = |f_n(x) - f(x)| \xrightarrow[n\to\infty]{} 0$$

by the hypothesis that $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in X$.

(2) For every $n \in \mathbb{N}$ and $x \in X$, we have

$$g_n(x) = |f_n(x) - f(x)| \le |f_n(x)| + |f(x)| \le h(x) + h(x) = 2h(x)$$

where

$$\int_X 2h \, d\mu = 2 \int_X h \, d\mu < \infty.$$

So, we can take $\tilde{h} = 2h \in \mathrm{Bor}^+(X, \mathbb{R}) \cap \mathcal{L}^1(\mu)$. □

By the claim, we can apply LDCT_0 to $(g_n)_{n=1}^{\infty}$ to get that

$$\lim_{n\to\infty}\int_X g_n\,d\mu = 0. \tag{*}$$

This implies the fact that

$$\lim_{n\to\infty}\int_X f_n\,d\mu = \int_X f\,d\mu$$

because we have the following inequality and then apply the squeeze theorem:

$$0 \le \left|\int_X f_n\,d\mu - \int_X f\,d\mu\right| = \left|\int_X (f_n - f)\,d\mu\right| \le \int_X |f_n - f|\,d\mu = \int_X g_n\,d\mu \xrightarrow[n\to\infty]{} 0 \quad \text{by (*)}.$$

□

2.10 Equality Almost Everywhere

Rule of Thumb: changes made on a set of measure zero do not affect integrability or the value of the integral. This motivates the following definition.

Definition 2.11 (Equality almost everywhere).
We say that two functions $f, g : X \to \mathbb{R}$ are **equal almost everywhere** with respect to μ, denoted as $f = g$ a.e.-μ, if $\exists Z \in \mathfrak{M}$ such that $\mu(Z) = 0$ and $f(x) = g(x)$ for all $x \in X \setminus Z$.

Note. f and g may not be in $\mathrm{Bor}(X, \mathbb{R})$. If $f, g \in \mathrm{Bor}(X, \mathbb{R})$, we use an equivalent description of equality almost everywhere in the following proposition.

Proposition 2.31. Let $f, g \in \mathrm{Bor}(X, \mathbb{R})$ and denote $M := \{x \in X \mid f(x) \neq g(x)\}$. Then,

(1) We have $M \in \mathfrak{M}$, and hence the quantity $\mu(M)$ is well-defined.

(2) $\Big[f = g \text{ a.e.-}\mu\Big] \iff \Big[\mu(M) = 0\Big]$.

Proof. to be filled in from LEARN file. □

Lemma 2.32. Let $f \in \mathrm{Bor}^+(X, \mathbb{R})$ such that $f = 0$ a.e.-μ. Then,

$$\int_X f\,d\mu = 0.$$

In particular, $f \in \mathcal{L}^1(\mu)$.

Proof. For every $n \in \mathbb{N}$, let $f_n = f \wedge \underline{n}$ where $\underline{n} : X \to \mathbb{R}$ is the constant function $x \mapsto n$. That is,

$$f_n(x) = \begin{cases} f(x), & \text{if } f(x) \leq n, \\ n, & \text{if } f(x) > n. \end{cases}$$

We have $f_n \in \mathrm{Bor}(X, \mathbb{R})$ beacuse $\mathrm{Bor}(X, \mathbb{R})$ is stable under $\wedge$, and hence $f_n \in \mathrm{Bor}^+(X, \mathbb{R})$. Now, we check that $f_n \nearrow f$. For every $x \in X$, we have $f_n(x) \leq f_{n+1}(x)$ for all $n \in \mathbb{N}$ and $\lim_{n\to\infty} f_n(x) = f(x)$. Hence, $f_n \nearrow f$. By MCT, we have

$$\int_X f \, d\mu = \lim_{n\to\infty} \int_X f_n \, d\mu.$$

So in order to get $\displaystyle\int_X f \, d\mu = 0$, it suffices to show that $\displaystyle\int_X f_n \, d\mu = 0$ for all $n \in \mathbb{N}$. Observe that $f_n \leq n\chi_Z$ for all $n \in \mathbb{N}$ where $Z = \{x \in X \mid f(x) \neq 0\}$. To see this, fix $x \in Z$. Then, $f_n(x) \leq n = n\chi_Z(x)$. If $x \in X \setminus Z = \{x \in X \mid f(x) = 0\}$, then $f_n(x) = 0 = n\chi_Z(x)$. Hence, $f_n \leq n\chi_Z$ for all $n \in \mathbb{N}$. Then, we have

$$0 \leq \int_X f_n \, d\mu \leq \int_X n\chi_Z \, d\mu = n \int_X \chi_Z \, d\mu = n\mu(Z) = 0.$$

Hence, $\displaystyle\int_X f_n \, d\mu = 0$ for all $n \in \mathbb{N}$. It follows that $\displaystyle\int_X f \, d\mu = 0$. In particular, $f \in \mathcal{L}^1(\mu)$. □

Lecture 19, 2026/06/26

The following proposition formalizes the rule of thumb that "changing the values of a function on a set of measure 0 does not affect integrability or the value of the integral".

Proposition 2.33. Let $f, g \in \mathrm{Bor}(X, \mathbb{R})$ such that $f = g$ a.e.-μ. Suppose that one of these functions, say f, is in $\mathcal{L}^1(\mu)$. Then, $g \in \mathcal{L}^1(\mu)$ and

$$\int_X f \, d\mu = \int_X g \, d\mu.$$

Proof. to be filled in from LEARN file. □

Remark. The converse of Lemma 2.32 is also true. That is, if $f \in \mathrm{Bor}^+(X, \mathbb{R})$ is such that $\displaystyle\int_X f \, d\mu = 0$, then $f = 0$ a.e.-μ.

Proof. Let $Z = \{x \in X \mid f(x) \neq 0\} = \{x \in X \mid f(x) > 0\}$. By Proposition 2.31, it suffices to show that $\mu(Z) = 0$. Indeed, $\forall n \in \mathbb{N}$, consider

$$Z_n = \left\{x \in X \mid f(x) \geq \frac{1}{n}\right\}.$$

Note that $Z_n \subseteq Z$ and $Z_1 \subseteq Z_2 \subseteq \cdots \subseteq Z_n \subseteq \cdots$. In particular, $Z = \bigcup_{n=1}^{\infty} Z_n$. Note that $\bigcup_{n=1}^{\infty} Z_n \subseteq Z$ is clear. Conversely, for any $x \in Z$, we have $f(x) > 0$. Hence, $\exists n \in \mathbb{N}$ such that $f(x) \geq \frac{1}{n}$, which implies that $x \in Z_n$. By the continuity of μ,

$$\mu(Z) = \lim_{n \to \infty} \mu(Z_n).$$

So, it suffices to show that $\mu(Z_n) = 0$ for all $n \in \mathbb{N}$. Fix $n \in \mathbb{N}$. Observe that $f \geq \frac{1}{n}\chi_{Z_n}$. To see this, if $x \in Z_n$, then $f(x) \geq \frac{1}{n} = \frac{1}{n}\chi_{Z_n}(x)$. If $x \in X \setminus Z_n$, then $f(x) \geq 0 = \frac{1}{n}\chi_{Z_n}(x)$. Hence,

$$0 = \int_X f \, d\mu \geq \int_X \frac{1}{n}\chi_{Z_n} \, d\mu = \frac{1}{n}\int_X \chi_{Z_n} \, d\mu = \frac{1}{n}\mu(Z_n) \geq 0.$$

Thus, $\mu(Z_n) = 0$ for all $n \in \mathbb{N}$. It follows that $\mu(Z) = 0$, and hence $f = 0$ a.e.-μ. ☐

As a consequence, if $f, g \in \mathrm{Bor}(X, \mathbb{R})$ for which $\displaystyle\int_X |f - g| \, d\mu = 0$ (integral in L^+ sense), then $f = g$ a.e.-μ.

Theorem 2.34 (a.e.-μ version of LDCT).

Let f and $(f_n)_{n=1}^{\infty}$ be in $\mathrm{Bor}(X, \mathbb{R})$ such that the following hold:

(a) $\exists Z_1 \in \mathfrak{M}$ with $\mu(Z_1) = 0$ such that $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in X \setminus Z_1$.

(b) $\exists Z_2 \in \mathfrak{M}$ with $\mu(Z_2) = 0$ and $h \in \mathrm{Bor}^+(X, \mathbb{R}) \cap \mathcal{L}^1(\mu)$ such that $|f_n(x)| \leq h(x)$ for all $n \in \mathbb{N}$ and all $x \in X \setminus Z_2$.

Then, f and $f_1, f_2, \ldots$ are all in $\mathcal{L}^1(\mu)$ and

$$\lim_{n \to \infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Proof. Let $Z = Z_1 \cup Z_2$. Then, $Z \in \mathfrak{M}$ and $\mu(Z) = 0$. Now, the hypotheses (a) and (b) still hold for Z

and $X \setminus Z = (X \setminus Z_1) \cap (X \setminus Z_2)$. Let $\tilde{f} = f \cdot \chi_{X \setminus Z}$ and for every $n \in \mathbb{N}$, let $\tilde{f}_n = f_n \cdot \chi_{X \setminus Z}$. Observe that $\tilde{f}$ and $(\tilde{f}_n)_{n=1}^{\infty}$ satisfy the hypotheses of LDCT. Hence, $\tilde{f}$ and $\tilde{f}_1, \tilde{f}_2, \ldots$ are all in $\mathcal{L}^1(\mu)$ and

$$\lim_{n \to \infty} \int_X \tilde{f}_n \, d\mu = \int_X \tilde{f} \, d\mu.$$

On the other hand, observe that $\tilde{f} = f$ a.e.-μ and $\tilde{f}_n = f_n$ a.e.-μ for all $n \in \mathbb{N}$. By Proposition 2.33, we have $f \in \mathcal{L}^1(\mu)$ and $\int_X f \, d\mu = \int_X \tilde{f} \, d\mu$. Similarly, $\forall n \in \mathbb{N}$, $f_n \in \mathcal{L}^1(\mu)$ and $\int_X f_n \, d\mu = \int_X \tilde{f}_n \, d\mu$. Thus,

$$\lim_{n \to \infty} \int_X f_n \, d\mu = \lim_{n \to \infty} \int_X \tilde{f}_n \, d\mu = \int_X \tilde{f} \, d\mu = \int_X f \, d\mu. \qquad \square$$

3 $\mathcal{L}^p$ Spaces

Lecture 20, 2026/06/29

3.1 Basics of the $\mathcal{L}^p$ Spaces

We fix a measure space $(X, \mathfrak{M}, \mu)$.

Definition 3.1 (**Notation**).

For every $f \in \mathcal{L}^1(\mu)$, we denote

$$\|f\|_1 = \int_X |f| \, d\mu \in [0, \infty).$$

Remark.

Claim. $\|\cdot\|_1$ is a **seminorm** on $\mathcal{L}^1(\mu)$. That is,

(1) Non-negative: $\|f\|_1 \geq 0, \forall f \in \mathcal{L}^1(\mu)$.

(2) Sub-Additive: $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1, \forall f, g \in \mathcal{L}^1(\mu)$.

(3) Homogeneous: $\|\alpha f\|_1 = |\alpha| \|f\|_1, \forall f \in \mathcal{L}^1(\mu), \forall \alpha \in \mathbb{R}$.

Proof.

(1) Since $\|f\|_1 \in [0, \infty)$ by definition, we have $\|f\|_1 \geq 0$.

(2) By the triangle inequality, we have $|f + g| \leq |f| + |g|$ for all $f, g \in \mathcal{L}^1(\mu)$. Integrating both sides, we have

$$\begin{aligned} \|f + g\|_1 = \int_X |f + g| \, d\mu &\leq \int_X (|f| + |g|) \, d\mu \\ &= \int_X |f| \, d\mu + \int_X |g| \, d\mu = \|f\|_1 + \|g\|_1. \end{aligned}$$

(3) This follows immediately from the definition of $\|\cdot\|_1$ and the fact that $|\alpha f| = |\alpha||f|$ for all $f \in \mathcal{L}^1(\mu)$ and $\alpha \in \mathbb{R}$. $\square$

Remark. For $\|\cdot\|_1$ to be a norm, we would need to have

$$\left[\|f\|_1 = 0\right] \implies \left[f = \underline{0}\right]$$

By Lemma 2.32 and the remark after Proposition 2.33, we have

$$\left[\|f\|_1 = 0\right] \iff \left[f = \underline{0} \text{ a.e.-}\mu\right] \quad \text{for all } f \in \mathcal{L}^1(\mu).$$

But in general, this does not imply that $f = \underline{0}$ (i.e. $f(x) = 0$ for all $x \in X$).

Example. Take $f = \chi_A$ where $A \in \mathfrak{M}$. Then,

$$\begin{aligned} f = \underline{0} \text{ a.e.-}\mu &\iff \mu\left(\{x \in X \mid f(x) \neq 0\}\right) = 0 \\ &\iff \mu(A) = 0. \end{aligned}$$

On the other hand, $f = \underline{0} \iff A = \varnothing$. So, if there exists $\varnothing \neq A \in \mathfrak{M}$ with $\mu(A) = 0$, then $\|f\|_1$ is not a norm.

Definition 3.2 ($\mathcal{L}^p$ **Space**).
Let $p \in [1, \infty)$ be a number. The $\mathcal{L}^p$ **space** is denoted by

$$\mathcal{L}^p(\mu) = \left\{f \in \operatorname{Bor}(X, \mathbb{R}) \mid \int_X |f|^p \, d\mu < \infty\right\}$$

where the integral is considered in the L^+ sense. For $f \in \mathcal{L}^p(\mu)$, we define

$$\|f\|_p = \left(\int_X |f|^p \, d\mu\right)^{1/p} \in [0, \infty).$$

Note. In the special case $p = 1$, above definition gives $\mathcal{L}^1(\mu)$ and $\|\cdot\|_1$.

Proposition 3.1. Let $p \in [1, \infty)$. $\mathcal{L}^p(\mu)$ is a linear subspace of $\operatorname{Bor}(X, \mathbb{R})$.

Proof. check LEARN file for proof $\square$

Remark.

(1) Pick $p \in (1, \infty)$ and consider $\|\cdot\|_p$ on the vector space $\mathcal{L}^p(\mu)$. We see that Non-negativity is immediate, and Homogeneity was verified in the proof of the above proposition. We are left with Sub-Additivity:

$$\|f+g\|_p \leq \|f\|_p + \|g\|_p, \quad \forall f, g \in \mathcal{L}^p(\mu).$$

It turns out that this does hold! It is called **Minkowski's Inequality**.

(2) For $f \in \mathcal{L}^p(\mu)$, we have

$$\Big[\|f\|_p = 0\Big] \iff \Big[f = \underline{0} \text{ a.e.-}\mu\Big].$$

This holds because

$$\Big[\|f\|_p = 0\Big] \iff \left[\int_X |f|^p \, d\mu = 0\right] \overset{(*)}{\iff} \Big[|f|^p = 0 \text{ a.e.-}\mu\Big] \iff \Big[f = \underline{0} \text{ a.e.-}\mu\Big]$$

where $(*)$ follows from Lemma 2.32 and the remark after Proposition 2.33. Similar to the $\mathcal{L}^1(\mu)$ case, this does not imply that $f(x) = 0$ for all $x \in X$, i.e. $\|\cdot\|_p$ is not a norm in general.

3.2 Conjugate Exponents

In order to prove Minkowski's Inequality for an exponent $p \in (1, \infty)$, we need to follow the idea of playing the spaces $\mathcal{L}^p(\mu)$ against a space $\mathcal{L}^q(\mu)$ where $q \in (1, \infty)$ is the **conjugate exponent** of p.

Definition 3.3 (Conjugate Exponents).

Two numbers $p, q \in (1, \infty)$ are called **conjugate exponents** if

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Note. Given $p \in (1, \infty)$, the conjugate exponent $q := \frac{p}{p-1}$ is unique.

Proposition 3.2 (Hölder's Inequality).

Let $p, q \in (1, \infty)$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. If $f \in \mathcal{L}^p(\mu)$ and $g \in \mathcal{L}^q(\mu)$, then $f \cdot g \in \mathcal{L}^1(\mu)$ and

$$\|f \cdot g\|_1 \leq \|f\|_p \cdot \|g\|_q.$$

As a consequence, we also have

$$\left|\int_X f \cdot g \, d\mu\right| \leq \|f\|_p \cdot \|g\|_q.$$

Remark. This is like a generalization of the Cauchy-Schwarz inequality and introduces a "**duality**" between $\mathcal{L}^p(\mu)$ and $\mathcal{L}^q(\mu)$. That is, $\forall f \in \mathcal{L}^p(\mu)$, we associate a linear function $\varphi_f : \mathcal{L}^q(\mu) \to \mathbb{R}$ defined by

$$\varphi_f(g) := \int_X f \cdot g \, d\mu, \quad \forall g \in \mathcal{L}^q(\mu).$$

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Remark. In the special case $p = q = 2$, Hölder's Inequality claims that

$$\left[f, g \in \mathcal{L}^2(\mu)\right] \implies \left[f \cdot g \in \mathcal{L}^1(\mu)\right].$$

We introduce the inner product notation

$$\langle f, g \rangle := \int_X f \cdot g \, d\mu, \quad \forall f, g \in \mathcal{L}^2(\mu).$$

Thus, Hölder's Inequality gives us the **Cauchy-Schwarz** inequality in $\mathcal{L}^2(\mu)$:

$$|\langle f, g \rangle| \leq \|f\|_2 \cdot \|g\|_2, \quad \forall f, g \in \mathcal{L}^2(\mu).$$

Lemma 3.3. Let $p, q \in (1, \infty)$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. For every $a, b \in [0, \infty)$, we have

$$a \cdot b \leq \frac{1}{p} a^p + \frac{1}{q} b^q.$$

Proof. check LEARN file for proof □

Lemma 3.4. Let $p, q \in (1, \infty)$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. Consider two functions $f \in \mathcal{L}^p(\mu)$

and $g \in \mathcal{L}^q(\mu)$. Suppose that $\|f\|_p = \|g\|_q = 1$. Then, it follows that $f \cdot g \in \mathcal{L}^1(\mu)$ and

$$\left|\int_X f \cdot g \, d\mu\right| \le 1.$$

Proof. Note that

$$\|f\|_p = 1 \implies \left(\int_X |f|^p \, d\mu\right)^{1/p} = 1 \implies \int_X |f|^p \, d\mu = 1.$$

Similarly, $\int_X |g|^q \, d\mu = 1$. By the previous lemma with $a = |f(x)|$ and $b = |g(x)|$, we have

$$|f(x)| \cdot |g(x)| \le \frac{1}{p}|f(x)|^p + \frac{1}{q}|g(x)|^q, \quad \forall x \in X.$$

Then,

$$\begin{aligned} |f \cdot g| &\le \frac{1}{p}|f|^p + \frac{1}{q}|g|^q \\ \implies \int_X |f \cdot g| \, d\mu &\le \int_X \left(\frac{1}{p}|f|^p + \frac{1}{q}|g|^q\right) d\mu \\ &= \frac{1}{p}\int_X |f|^p \, d\mu + \frac{1}{q}\int_X |g|^q \, d\mu = \frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = 1. \end{aligned}$$

□

Proof of Hölder's Inequality.

Denote $\|f\|_p = \alpha$ and $\|g\|_q = \beta$ for $\alpha, \beta \in [0, \infty)$.

Case 1: $\alpha = 0$ or $\beta = 0$. Say $\alpha = 0$ WLOG. Then, $\|f\|_p = 0 \implies f = \underline{0}$ a.e.-μ. Then,

$$\mu\left(\{x \in X \mid f(x) \neq 0\}\right) = 0.$$

So,

$$\begin{aligned} \{x \in X \mid (f \cdot g)(x) \neq 0\} \subseteq \{x \in X \mid f(x) \neq 0\} &\implies \mu\left(\{x \in X \mid (f \cdot g)(x) \neq 0\}\right) = 0 \\ &\implies f \cdot g = \underline{0} \text{ a.e.-}\mu \\ &\implies \|f \cdot g\|_1 = 0 = \alpha \cdot \beta = \|f\|_p \cdot \|g\|_q. \end{aligned}$$

Case 2: Both α and β are non-zero. Then, let $\tilde{f} = \frac{f}{\alpha}$ and $\tilde{g} = \frac{g}{\beta}$. Then, $\tilde{f} \in \mathcal{L}^p(\mu)$ with

$$\left\|\tilde{f}\right\|_p = \frac{1}{\alpha}\|f\|_p = 1.$$

Similarly, $\tilde{g} \in \mathcal{L}^q(\mu)$ with $\|\tilde{g}\|_q = 1$. By Lemma 3.4, we have $\tilde{f} \cdot \tilde{g} \in \mathcal{L}^1(\mu)$ and

$$\left\|\tilde{f} \cdot \tilde{g}\right\|_1 = \int_X |\tilde{f} \cdot \tilde{g}| \, d\mu \leq 1.$$

But since $f = \alpha\tilde{f}$ and $g = \beta\tilde{g}$, we have $f \cdot g = \alpha\beta(\tilde{f} \cdot \tilde{g})$, which implies that $f \cdot g \in \mathcal{L}^1(\mu)$ with

$$\begin{aligned}\|f \cdot g\|_1 = \left\|\alpha\beta(\tilde{f} \cdot \tilde{g})\right\|_1 &= |\alpha\beta|\left\|\tilde{f} \cdot \tilde{g}\right\|_1 \\ &\leq \alpha\beta \cdot 1 = \|f\|_p \cdot \|g\|_q.\end{aligned}$$

□

3.3 Minkowski's Inequality

Definition 3.4 (Notation).

Let f be a function in $\mathcal{L}^p(\mu)$ for some $p \in (1, \infty)$. Define a function $\varphi_f : \mathcal{L}^q(\mu) \to \mathbb{R}$ by

$$\varphi_f(g) := \int_X f \cdot g \, d\mu, \quad \forall g \in \mathcal{L}^q(\mu).$$

Remark. Let f be a function in $\mathcal{L}^p(\mu)$ for some $p \in (1, \infty)$.

(1) φ_f is well-defined by Hölder's Inequality, which assures that $f \cdot g \in \mathcal{L}^1(\mu)$.

(2) φ_f is linear and has the property that

$$|\varphi_f(g)| \leq \|f\|_p \cdot \|g\|_q, \quad \forall g \in \mathcal{L}^q(\mu).$$

(3) The property in (2) implies that the set $\left\{|\varphi_f(g)| \mid g \in \mathcal{L}^q(\mu), \|g\|_q \leq 1\right\} \subseteq [0, \infty)$ is bounded from above, with $\|f\|_p$ being an upper bound. Then, we can consider the number

$$\lambda(f) := \sup\left\{|\varphi_f(g)| \mid g \in \mathcal{L}^q(\mu), \|g\|_q \leq 1\right\} \in [0, \infty)$$

and we know that $\lambda(f) \leq \|f\|_p$.

Lemma 3.5. Let $f \in \mathcal{L}^p(\mu)$ and consider the number $\lambda(f) \in [0, \infty)$ defined above. Then, $\lambda(f) = \|f\|_p$.

Proposition 3.6 (**Minkowski's Inequality**).
Let $f_1, f_2 \in \mathcal{L}^p(\mu)$ for some $p \in (1, \infty)$. Then,

$$\|f_1 + f_2\|_p \leq \|f_1\|_p + \|f_2\|_p.$$

Proof. By Lemma 3.5, it is equivalent to prove that $\lambda(f_1 + f_2) \leq \lambda(f_1) + \lambda(f_2)$. Note that

$$\begin{aligned}\lambda(f_1 + f_2) &\coloneqq \sup\left\{|\varphi_{f_1+f_2}(g)| \mid g \in \mathcal{L}^q(\mu), \|g\|_q \leq 1\right\} \\ &= \sup\left\{\left|\int_X (f_1 + f_2) \cdot g \, d\mu\right| \mid g \in \mathcal{L}^q(\mu), \|g\|_q \leq 1\right\}.\end{aligned}$$

Then, it suffices to check that $\left|\int_X (f_1 + f_2) \cdot g \, d\mu\right| \leq \lambda(f_1) + \lambda(f_2)$ $\forall g \in \mathcal{L}^q(\mu)$ with $\|g\|_q \leq 1$. And indeed, for $g \in \mathcal{L}^q(\mu)$ with $\|g\|_q \leq 1$, we have

$$\begin{aligned}\left|\int_X (f_1 + f_2) \cdot g \, d\mu\right| &= \left|\int_X f_1 \cdot g \, d\mu + \int_X f_2 \cdot g \, d\mu\right| \\ &\leq \left|\int_X f_1 \cdot g \, d\mu\right| + \left|\int_X f_2 \cdot g \, d\mu\right| \\ &\leq \lambda(f_1) + \lambda(f_2).\end{aligned}$$

□

Now, we prove Lemma 3.5.

Proof of Lemma 3.5. We saw in the previous remark (3) that $\lambda(f) \leq \|f\|_p$. For $\lambda(f) \geq \|f\|_p$, we consider a function $g_0 \in \mathcal{L}^q(\mu)$ such that $\|g_0\|_q \leq 1$ and $\varphi_f(g_0) = \|f\|_p$. Using that g_0, we will have that $\lambda(f) = \sup\left\{|\varphi_f(g)| \mid g \in \mathcal{L}^q(\mu), \|g\|_q \leq 1\right\} \geq |\varphi_f(g_0)| = \|f\|_p$.

To construct such g_0, we consider the following motivation. Think of the case $p = q = 2$. Then, we have $\varphi_f(g) = \int_X f \cdot g \, d\mu = \langle f, g \rangle$ for $f, g \in \mathcal{L}^2(\mu)$. Given $f \in \mathcal{L}^2(\mu)$, we want to find $g_0 \in \mathcal{L}^2(\mu)$ such that $\|g_0\|_2 = 1$ and $\langle f, g_0 \rangle = \|f\|_2 \cdot \|g_0\|_2$. This is the equality case of the Cauchy-Schwarz Inequality, which occurs when $g_0 = \alpha f$ for some $\alpha \in \mathbb{R}$. For general $p \in (1, \infty)$ with $p \neq q$, we cannot do this directly. But observe that $h = |f|^{\frac{p}{q}} \in \mathcal{L}^q(\mu)$ and hence $h^q = |f|^p$. So,

$$\int_X h^q \, d\mu = \int_X |f|^p \, d\mu < \infty \implies h \in \mathcal{L}^q(\mu).$$

Claim (1)**.** $\exists u \in \mathrm{Bor}(X, \mathbb{R})$ such that $|u(x)| = 1$ for all $x \in X$ and $f = u \cdot |f|$.

Proof of Claim (1). Let $u : X \to \mathbb{R}$ be defined by

$$u(x) = \begin{cases} 1, & \text{if } f(x) \geq 0, \\ -1, & \text{if } f(x) < 0. \end{cases}$$

Exercise: verify u. □

Claim (2)**.** Let $h = u \cdot |f|^{\frac{p}{q}}$ with u as in Claim (1). Then, $h \in \mathcal{L}^q(\mu)$ and $\|h\|_q = \|f\|_p^{p-1}$ and

$$\int_X f \cdot h \, d\mu = \|f\|_p^p.$$

Proof. **Exercise**: verify Claim (2).

Idea for $\int_X f \cdot h \, d\mu = \|f\|_p^p$: we have

$$f \cdot h = (u \cdot |f|) \cdot \left(u \cdot |f|^{\frac{p}{q}}\right) = u^2 \cdot |f|^{1+\frac{p}{q}} = |f|^{p\left(\frac{1}{p}+\frac{1}{q}\right)} = |f|^p.$$

Then, $\int_X f \cdot h \, d\mu = \int_X |f|^p \, d\mu = \|f\|_p^p$. □

If $\|f\|_p = 0$, then use $g_0 = \underline{0}$ and we are done. So, suppose that $\|f\|_p \neq 0$. Let h be as in Claim (2) and define $g_0 = \frac{1}{\|f\|_p^{p-1}} h \in \mathcal{L}^q(\mu)$. Then,

$$\|g_0\|_q = \frac{1}{\|f\|_p^{p-1}} \|h\|_q = \frac{1}{\|f\|_p^{p-1}} \cdot \|f\|_p^{p-1} = 1.$$

And,

$$\int_X f \cdot g_0 \, d\mu = \frac{1}{\|f\|_p^{p-1}} \int_X f \cdot h \, d\mu = \frac{1}{\|f\|_p^{p-1}} \cdot \|f\|_p^p = \|f\|_p \implies \varphi_f(g_0) = \|f\|_p.$$ □

Corollary 3.7. The map $\|\cdot\|_p$ is a seminorm on the vector space $\mathcal{L}^p(\mu)$.

Proof. Exercise. $\square$

3.4 Null-Space of a Norm and $L^p(\mu)$ vs. $\mathcal{L}^p(\mu)$ (Appendix)

Definition 3.5 (**Null-Space**).

Let $\mathcal{V}$ be a vector space over $\mathbb{R}$ and $\mathcal{V} \ni v \mapsto \|v\| \in [0, \infty)$ be a seminorm. The set

$$\mathcal{N} := \{v \in \mathcal{V} \mid \|v\| = 0\} \subseteq \mathcal{V}$$

is called the **null-space** of the seminorm $\|\cdot\|$.

Remark. $\mathcal{N}$ is a linear subspace of $\mathcal{V}$. That is,

(1) The zero vector $0_{\mathcal{V}}$ of $\mathcal{V}$ is in $\mathcal{N}$.

(2) $\left[v_1, v_2 \in \mathcal{N}\right] \implies \left[v_1 + v_2 \in \mathcal{N}\right]$.

(3) $\left[v \in \mathcal{N}, \alpha \in \mathbb{R}\right] \implies \left[\alpha v \in \mathcal{N}\right]$.

Proof. Exercise. $\square$

Note. Observe that $\|\cdot\|$ is a norm on $\mathcal{V} \iff$ its null-space $\mathcal{N} = \{0_{\mathcal{V}}\}$.

Definition 3.6 (**Quotient by a Linear Space**).

Let $\mathcal{V}$ be a vector space over $\mathbb{R}$ and let $\mathcal{W} \subseteq \mathcal{V}$ be a linear subspace. Consider the following equivalence relation $\sim$ on $\mathcal{V}$: for $v_1, v_2 \in \mathcal{V}$,

$$v_1 \sim v_2 \iff v_1 - v_2 \in \mathcal{W}.$$

Let $\mathcal{Q}$ be the collection of all equivalence classes with respect to $\sim$. Then, we define the **quotient space** to be $\mathcal{Q} = \mathcal{V}/\mathcal{W}$.

Exercise: Check that $\sim$ is an equivalence relation.

Remark. There is a surjective linear map from $\mathcal{V}$ to $\mathcal{Q}$ defined by

$$\mathcal{V} \ni v \mapsto \hat{v} \in \mathcal{Q}.$$

It is convenient if we don't have to constantly remind ourselves how the vectors in $\mathcal{Q}$ are defined. So, we will use the notation $\hat{v}$ to denote the equivalence class of $v \in \mathcal{V}$. Thus, $\mathcal{Q} = \{\hat{v} \mid v \in \mathcal{V}\}$, where for $v_1, v_2 \in \mathcal{V}$, we have

$$\hat{v}_1 = \hat{v}_2 \iff v_1 - v_2 \in \mathcal{W}$$

and where "hats respect linear combinations":

$$(\widehat{\alpha_1 v_1 + \alpha_2 v_2}) = \alpha_1 \hat{v}_1 + \alpha_2 \hat{v}_2, \quad \forall v_1, v_2 \in \mathcal{V}, \forall \alpha_1, \alpha_2 \in \mathbb{R}.$$

Remark (**How to get a norm from a seminorm?**).
Let $\mathcal{V}$ be a vector space over $\mathbb{R}$ and let $\mathcal{N}$ be the null-space of a seminorm $\|\cdot\|$ on $\mathcal{V}$. Consider the quotient space $\mathcal{Q} = \mathcal{V}/\mathcal{N}$. Then, we can define a norm for elements in $\mathcal{Q}$ as follows:

- Take an element $\xi \in \mathcal{Q}$.
- Write it in the form $\xi = \hat{v}$ for some $v \in \mathcal{V}$.
- Define the norm of ξ as $\|\xi\|_{\mathcal{Q}} := \|v\| \in [0, \infty)$.

There are three things we need to check:

(1) If $\xi = \hat{v}_1 = \hat{v}_2$ for some $v_1, v_2 \in \mathcal{V}$, then $\|v_1\| = \|v_2\|$.

Proof. If $\xi = \hat{v}_1 = \hat{v}_2$, then $v_1 - v_2 \in \mathcal{N}$, which means $\|v_1 - v_2\| = 0$. By the triangle inequality,

$$\|v_1\| = \|v_1 - v_2 + v_2\| \leq \|v_1 - v_2\| + \|v_2\| = 0 + \|v_2\| = \|v_2\|.$$

Similarly, $\|v_2\| \leq \|v_1\|$, which implies that $\|v_1\| = \|v_2\|$. □

(2) The map $\mathcal{Q} \ni \xi \mapsto \|\xi\|_{\mathcal{Q}} \in [0, \infty)$ inherits the seminorm properties of $\|\cdot\|$ on $\mathcal{V}$.

Proof.

- Non-negativity: $\|\xi\|_{\mathcal{Q}} = \|v\| \geq 0$ for any $\xi = \hat{v} \in \mathcal{Q}$.

- Homogeneity: take $\alpha \in \mathbb{R}$ and $\xi = \hat{v} \in \mathcal{Q}$. Then,

$$\|\alpha\xi\|_{\mathcal{Q}} = \|\alpha\hat{v}\|_{\mathcal{Q}} = \|\widehat{\alpha v}\|_{\mathcal{Q}} = \|\alpha v\| = |\alpha|\|v\| = |\alpha|\|\hat{v}\|_{\mathcal{Q}} = |\alpha|\|\xi\|_{\mathcal{Q}}.$$

- Sub-additivity: take $\xi_1 = \hat{v}_1, \xi_2 = \hat{v}_2 \in \mathcal{Q}$. Then,

$$\|\xi_1 + \xi_2\|_{\mathcal{Q}} = \|\hat{v}_1 + \hat{v}_2\|_{\mathcal{Q}} = \left\|\widehat{v_1 + v_2}\right\|_{\mathcal{Q}} = \|v_1 + v_2\| \leq \|v_1\| + \|v_2\| = \|\xi_1\|_{\mathcal{Q}} + \|\xi_2\|_{\mathcal{Q}}.$$

$\square$

(3) The map $\mathcal{Q} \ni \xi \mapsto \|\xi\|_{\mathcal{Q}} \in [0, \infty)$ is a norm on $\mathcal{Q}$.

Proof. We checked the three properties of a seminorm in (2). It remains to check that $\|\xi\|_{\mathcal{Q}} = 0 \implies \xi = 0_{\mathcal{Q}}$. Take $\xi = \hat{v} \in \mathcal{Q}$ such that $\|\xi\|_{\mathcal{Q}} = 0$. Then, $\|v\| = 0$, which means $v \in \mathcal{N}$. Note that $v - 0_{\mathcal{V}} = v \in \mathcal{N}$, which implies that $\hat{v} = \hat{0}_{\mathcal{V}} = 0_{\mathcal{Q}}$. Hence, $\xi = 0_{\mathcal{Q}}$. $\square$

Now, we return to our usual setting, where we fix a measure space $(X, \mathfrak{M}, \mu)$. Define

$$\mathcal{N} := \{f \in \mathrm{Bor}(X, \mathbb{R}) \mid f = \underline{0} \text{ a.e.-}\mu\} \subseteq \mathrm{Bor}(X, \mathbb{R}).$$

Proposition 3.8. Let $p \in [1, \infty)$. Consider the space of functions $\mathcal{L}^p(\mu)$ with the seminorm $\|\cdot\|_p$. Then, $\mathcal{N} \subseteq \mathcal{L}^p(\mu)$ and $\mathcal{N}$ is the null-space of the seminorm $\|\cdot\|_p$ on $\mathcal{L}^p(\mu)$.

Proof. The proof is covered in the discussion in Remark 3.1 (2). $\square$

Definition 3.7 (The Space $L^p(\mu)$).
For every $p \in [1, \infty)$, we denote the quotient space of $\mathcal{L}^p(\mu)$ as

$$L^p(\mu) := \mathcal{L}^p(\mu)/\mathcal{N}.$$

Note. Following the procedure in the previous remark, the seminorm $\|\cdot\|_p$ on $\mathcal{L}^p(\mu)$ gives a norm $\|\cdot\|_p$ on $L^p(\mu)$. Thus, $(L^p(\mu), \|\cdot\|_p)$ is a normed vector space.

Remark. Let $p \in [1, \infty)$. From previous discussion in this subsection, we can say that $L^p(\mu)$ is "a space of objects with hats":

$$L^p(\mu) = \{\hat{f} \mid f \in \mathcal{L}^p(\mu)\},$$

where for $f, g \in \mathcal{L}^p(\mu)$, we have

$$\hat{f} = \hat{g} \iff f - g \in \mathcal{N} \iff f - g = \underline{0} \text{ a.e.-}\mu \iff f = g \text{ a.e.-}\mu.$$

Also,

- Hats respect linear combinations:

$$(\widehat{\alpha_1 f_1 + \alpha_2 f_2}) = \alpha_1 \hat{f}_1 + \alpha_2 \hat{f}_2, \quad \forall f_1, f_2 \in \mathcal{L}^p(\mu), \forall \alpha_1, \alpha_2 \in \mathbb{R}.$$

- The seminorm $\|\cdot\|_p$ on $\mathcal{L}^p(\mu)$ is transferred to the quotient space $L^p(\mu)$ by letting

$$\left\|\hat{f}\right\|_p := \|f\|_p, \quad \forall f \in \mathcal{L}^p(\mu).$$

  Then, $\|\cdot\|_p$ is a norm on $L^p(\mu)$.

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3.5 $\mathcal{L}^p(\mu)$ - Completeness

We will prove that the normed vector space $(L^p(\mu), \|\cdot\|_p)$ is complete, i.e. it is a Banach space.

We need to revisit the notion of completeness in the framework of a seminorm which may not be a norm. Fix $(\mathcal{V}, \|\cdot\|)$ where $\mathcal{V}$ is a vector space over $\mathbb{R}$ and the map $\|\cdot\| : \mathcal{V} \to [0, \infty)$ is a seminorm.

Definition 3.8 (Convergence).
Let $(v_n)_{n=1}^\infty$ be a sequence of vectors in $\mathcal{V}$ and let $v \in \mathcal{V}$. We say that $(v_n)_{n=1}^\infty$ **converges** to v if $\lim_{n\to\infty} \|v_n - v\| = 0$.

Remark. In the framework of a seminorm, the limit v is not unique. If $\lim_{n\to\infty} \|v_n - v\| = 0 = \lim_{n\to\infty} \|v_n - v'\|$, then

$$\|v - v'\| \leq \|v - v_n\| + \|v_n - v'\| \ \ \forall n \in \mathbb{N} \overset{n\to\infty}{\Longrightarrow} \|v - v'\| = 0 \not\Longrightarrow v = v'.$$

Definition 3.9 (**Cauchy Sequence, Geometric Sequence**).

Let $(v_n)_{n=1}^{\infty}$ be a sequence of vectors in $\mathcal{V}$.

(1) We say that $(v_n)_{n=1}^{\infty}$ is a **Cauchy sequence** if for every $\epsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that whenever $m, n \geq n_0$, we have $\|v_n - v_m\| < \epsilon$.

(2) We say that $(v_n)_{n=1}^{\infty}$ is a $\frac{1}{2}$-**geometric sequence** if $\|v_n - v_{n+1}\| < \frac{1}{2^n}$ for all $n \in \mathbb{N}$.

Remark. One can check the following:

- If $(v_n)_{n=1}^{\infty}$ is convergent to some $v \in \mathcal{V}$, then it is Cauchy.
- If $(v_n)_{n=1}^{\infty}$ is $\frac{1}{2}$-geometric, then it is Cauchy.
- If $(v_n)_{n=1}^{\infty}$ is Cauchy, then it is possible to select indices $1 \leq n(1) < n(2) < \ldots$ such that the subsequence $(v_{n(k)})_{k=1}^{\infty}$ is $\frac{1}{2}$-geometric.

Definition 3.10 (**Completeness**).

A semi-normed space $(\mathcal{V}, \|\cdot\|)$ is said to be **complete** if every Cauchy sequence in $\mathcal{V}$ converges to some $v \in \mathcal{V}$.

Remark (**Completeness Criterion**).

Let $(\mathcal{V}, \|\cdot\|)$ be a semi-normed space such that every $\frac{1}{2}$-geometric sequence in $\mathcal{V}$ converges to some $v \in \mathcal{V}$. Then, $(\mathcal{V}, \|\cdot\|)$ is complete.

Also, we have the following fact: Let $(\mathcal{V}, \|\cdot\|)$ be a semi-normed space so that whenever $(w_n)_{n=1}^{\infty}$ is a $\frac{1}{2}$-geometric sequence in $\mathcal{V}$ with $w_1 = 0_{\mathcal{V}}$, then $\exists w \in \mathcal{V}$ such that $(w_n)_{n=1}^{\infty}$ converges to w. Then, $(\mathcal{V}, \|\cdot\|)$ is complete.

This fact implies the completeness critierion above.

Now, we fix a measure space $(X, \mathfrak{M}, \mu)$ and an exponent $p \in [1, \infty)$. We will prove that the semi-normed space $(\mathcal{L}^p(\mu), \|\cdot\|_p)$ is complete.

Lemma 3.9 (**Convergence Mechanism**).

Let $f_1, f_2, \ldots, f_n, \ldots$ be a sequence of functions from X to $\mathbb{R}$. Consider the functions $h_n : X \to$

$[0,\infty)$ defined by

$$h_1 = |f_1|, \quad h_2 = |f_1| + |f_2 - f_1|, \quad h_3 = |f_1| + |f_2 - f_1| + |f_3 - f_2|,$$
$$\dots, h_n = |f_1| + \sum_{m=2}^{n} |f_m - f_{m-1}|, \quad \dots$$

Observe that $0 \le h_1 \le h_2 \le \cdots \le h_n \le \dots$ and let

$$Z := \left\{ x \in X \mid \lim_{n\to\infty} h_n(x) = \infty \right\}.$$

Then, for every $x \in X \setminus Z$, the sequence $(f_n(x))_{n=1}^{\infty}$ is convergent in $\mathbb{R}$.

Note. This lemma tells us how to find a candidate for limit. It makes sense to consider a function $f : X \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0, & \text{if } x \in Z, \\ \lim_{n\to\infty} f_n(x), & \text{if } x \in X \setminus Z. \end{cases}$$

Proof. Pick $x \in X \setminus Z$ and verify that $(f_n(x))_{n=1}^{\infty}$ is convergent in $\mathbb{R}$. Since $x \notin Z$, we have $\lim_{n\to\infty} h_n(x) = \ell \in [0,\infty)$ (MCT). Note that for $m < n$ in $\mathbb{N}$, we have

$$|f_m(x) - f_n(x)| \le |f_m(x) - f_{m+1}(x)| + |f_{m+1}(x) - f_{m+2}(x)| + \cdots + |f_{n-1}(x) - f_n(x)| = h_n(x) - h_m(x).$$

□

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Proposition 3.10. Let $(f_n)_{n=1}^{\infty}$ be a sequence of functions in $\mathcal{L}^p(\mu)$ such that $f_1 = \underline{0}$ and $\|f_n - f_{n+1}\|_p < \frac{1}{2^n}$ for all $n \in \mathbb{N}$. Applying the convergence mechanism to $(f_n)_{n=1}^{\infty}$, we obtain a function $f : X \to \mathbb{R}$. Then, $f \in \mathcal{L}^p(\mu)$ and $\lim_{n\to\infty} \|f_n - f\|_p = 0$.

Proof. We use the help functions $(h_n)_{n=1}^{\infty} : X \to [0,\infty)$ defined in the convergence mechanism.

Claim (1). For every $n \in \mathbb{N}$, we have $h_n \in \mathcal{L}^p(\mu)$ and $\|h_n\|_p \le 1$.

Proof. check LEARN file for proof $\square$

Claim (2). Let $Z = \{x \in X \mid \lim_{n\to\infty} h_n(x) = \infty\}$. Then, $Z \in \mathfrak{M}$ and $\mu(Z) = 0$. Also, let $h : X \to [0, \infty)$ be defined by

$$h(x) = \begin{cases} 0, & \text{if } x \in Z, \\ \lim_{n\to\infty} h_n(x), & \text{if } x \in X \setminus Z. \end{cases}$$

Then, $h \in \mathcal{L}^p(\mu)$ and $\|h\|_p \leq 1$.

Proof. We have $0 = h_1^p \leq h_2^p \leq \cdots \leq h_n^p \leq \dots$ with $\displaystyle\int_X h_n^p \, d\mu = \|h_n\|_p^p \leq 1$ for all $n \in \mathbb{N}$. Applying the "addendum to MCT", we have

$$\left\{x \in X \mid \lim_{n\to\infty} h_n^p(x) = \infty\right\} \in \mathfrak{M} \text{ and } \mu\left(\left\{x \in X \mid \lim_{n\to\infty} h_n^p(x) = \infty\right\}\right) = 0.$$

But $\left\{x \in X \mid \lim\limits_{n\to\infty} h_n^p(x) = \infty\right\} = Z$ as $\lim\limits_{n\to\infty} h_n^p(x) = \infty \iff \lim\limits_{n\to\infty} h_n(x) = \infty$. So, $Z \in \mathfrak{M}$ and $\mu(Z) = 0$. Moreover,

$$\lim_{n\to\infty} \int_X h_n^p \, d\mu = \int_X h^p \, d\mu$$

as $\lim\limits_{n\to\infty} h_n^p(x) = h^p(x)$ a.e.-μ. Since $\lim\limits_{n\to\infty} \displaystyle\int_X h_n^p \, d\mu \leq 1 < \infty$, we have

$$\int_X h^p \, d\mu \leq 1 < \infty \implies h \in \mathcal{L}^p(\mu) \text{ and } \|h\|_p \leq 1.$$

$\square$

Claim (3). $f \in \mathrm{Bor}(X, \mathbb{R})$.

Proof. Note that $f(x) = \lim\limits_{n\to\infty} f_n(x) \cdot \chi_{X\setminus Z}(x)$ for all $x \in X$. Since $f_n \in \mathrm{Bor}(X, \mathbb{R})$ and $\chi_{X\setminus Z} \in \mathrm{Bor}(X, \mathbb{R})$, we have $f_n \cdot \chi_{X\setminus Z} \in \mathrm{Bor}(X, \mathbb{R})$ for all $n \in \mathbb{N}$. Thus, $f \in \mathrm{Bor}(X, \mathbb{R})$ as $\mathrm{Bor}(X, \mathbb{R})$ is closed under pointwise limits. $\square$

Claim (4). Fix $n_0 \in \mathbb{N}$ aand consider $g := f_{n_0} \cdot \chi_{X\setminus Z} - f \in \mathrm{Bor}(X, \mathbb{R})$. Then, $g \in \mathcal{L}^p(\mu)$ and $\|g\|_p \leq \frac{1}{2^{n_0-1}}$.

Proof. For every $n \in \mathbb{N}$, let $g_n = f_{n_0} \cdot \chi_{X \setminus Z} - f_n \cdot \chi_{X \setminus Z}$. Then, $g(x) = \lim_{n\to\infty} g_n(x)$ for all $x \in X$. Observe that for $n > n_0$, we have

$$\begin{aligned}\|g_n\|_p &= \left\| f_{n_0} \cdot \chi_{X\setminus Z} - f_n \cdot \chi_{X\setminus Z} \right\|_p = \left\| (f_{n_0} - f_n) \cdot \chi_{X\setminus Z} \right\|_p \\ &= \left\| f_{n_0} - f_n \right\|_p \\ &\le \left\| f_{n_0} - f_{n_0+1} \right\|_p + \left\| f_{n_0+1} - f_{n_0+2} \right\|_p + \cdots + \|f_{n-1} - f_n\|_p \\ &< \frac{1}{2^{n_0}} + \frac{1}{2^{n_0+1}} + \cdots + \frac{1}{2^{n-1}} < \frac{1}{2^{n_0-1}}.\end{aligned}$$

Now, let's apply LDCT for the convergence $\lim_{n\to\infty} |g_n(x)|^p = |g(x)|^p$ for all $x \in X$. Using the dominating function h^p with h from Claim (2), we have $|g_n|^p \le h^p$ (exercise) and $\int_X h^p \, d\mu \le 1$. Then, by LDCT, we have

$$\int_X |g|^p \, d\mu = \lim_{n\to\infty} \int_X |g_n|^p \, d\mu \implies \|g\|_p = \lim_{n\to\infty} \|g_n\|_p \le \frac{1}{2^{n_0-1}}.$$

□

Claim (5). $f \in \mathcal{L}^p(\mu)$.

Proof. check LEARN file for proof □

Claim (6). $\lim_{n\to\infty} \|f_n - f\|_p = 0$.

Proof. check LEARN file for proof □

□

Corollary 3.11. The semi-normed vector space $(\mathcal{L}^p(\mu), \|\cdot\|_p)$ is complete.

Proof. This follows from Proposition 3.10 and the completeness criterion in a previous remark. □

Corollary 3.12. The normed vector space $(L^p(\mu), \|\cdot\|_p)$ is a Banach space.

Proof. check LEARN file for proof ☐

Definition 3.11 (Almost Everywhere Convergence).
Let $(X, \mathfrak{M}, \mu)$ be a measure space, and let f and $(f_n)_{n=1}^{\infty}$ be functions in $\mathrm{Bor}(X, \mathbb{R})$. We say that $(f_n)_{n=1}^{\infty}$ converges to f **almost everywhere with respect to** μ (a.e.-μ) if $\exists Z \in \mathfrak{M}$ with $\mu(Z) = 0$ such that $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in X \setminus Z$.

Note. Upon examining the proof of Proposition 3.10, we see that the function f considered there also has the property that $(f_n)_{n=1}^{\infty}$ converges to f a.e.-μ. Indeed, the convergence $f_n(x) \to f(x)$ may only fail for points $x \in Z$ with $\mu(Z) = 0$.

Corollary 3.13. Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $p \in [1, \infty)$, and let $(f_n)_{n=1}^{\infty}$ be a sequence of functions in $\mathcal{L}^p(\mu)$ such that $\|f_n - f_{n+1}\|_p < \frac{1}{2^n}$ for all $n \in \mathbb{N}$. Then, $\exists f \in \mathcal{L}^p(\mu)$ such that $\lim_{n\to\infty} f_n = f$ a.e.-μ.

Proof. This follows from the proof of Proposition 3.10 and the note above. ☐

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3.6 Bounded Linear Functions on $\mathcal{L}^2(\mu)$

Fix a measure space $(X, \mathfrak{M}, \mu)$ and look at the semi-normed vector space $(\mathcal{L}^2(\mu), \|\cdot\|_2)$.

Note. We will refer a linear function $\varphi : \mathcal{L}^2(\mu) \to \mathbb{R}$ as a **linear functional**.

Definition 3.12 (Bounded Linear Functional).
Let $\varphi : \mathcal{L}^2(\mu) \to \mathbb{R}$ be a linear functional. We say that φ is **bounded** if $\exists c \geq 0$ such that

$$|\varphi(f)| \leq c \cdot \|f\|_2, \quad \forall f \in \mathcal{L}^2(\mu).$$

Remark. If $f, g \in \mathcal{L}^2(\mu)$, then $f \cdot g \in \mathcal{L}^1(\mu)$ and satisfies the inequality

$$\left| \int_X f \cdot g \, d\mu \right| \leq \|f\|_2 \cdot \|g\|_2.$$

Previously, we also used the inner product notation $\langle f,g\rangle := \int_X f\cdot g\,d\mu \in \mathbb{R}$ for all $f,g \in \mathcal{L}^2(\mu)$. Then, we can write the inequality above as

$$|\langle f,g\rangle| \le \|f\|_2 \cdot \|g\|_2.$$

This is called the **Cauchy-Schwarz Inequality** for $\mathcal{L}^2(\mu)$-functions. Also,

$$\langle f,f\rangle = \int_X f^2\,d\mu = \|f\|_2^2, \quad \forall f \in \mathcal{L}^2(\mu).$$

Moreover, we have a recipe for constructing bounded linear functionals on $\mathcal{L}^2(\mu)$. For every $f \in \mathcal{L}^2(\mu)$, we define a linear functional $\varphi_f : \mathcal{L}^2(\mu) \to \mathbb{R}$ by

$$\varphi_f(g) := \langle f,g\rangle = \int_X f\cdot g\,d\mu, \quad \forall g \in \mathcal{L}^2(\mu).$$

Note that φ_f is bounded because

$$|\varphi_f(g)| \le c\cdot \|g\|_2, \quad \forall g \in \mathcal{L}^2(\mu), \text{ where } c = \|f\|_2.$$

Theorem 3.14 (**Riesz**).
Let $\varphi : \mathcal{L}^2(\mu) \to \mathbb{R}$ be a bounded linear functional. Then, $\exists f_0 \in \mathcal{L}^2(\mu)$ such that $\varphi = \varphi_{f_0}$, i.e.

$$\varphi(g) = \langle f_0, g\rangle, \quad \forall g \in \mathcal{L}^2(\mu).$$

Lemma 3.15 (**The Parallelogram Law**).
For every $f,g \in \mathcal{L}^2(\mu)$, we have

$$\|f+g\|_2^2 + \|f-g\|_2^2 = 2\left(\|f\|_2^2 + \|g\|_2^2\right).$$

Proof. **Exercise**. □

Lemma 3.16. Let $\varphi : \mathcal{L}^2(\mu) \to \mathbb{R}$ be a bounded linear functional not identically 0. Consider the set $\mathcal{U} := \{u \in \mathcal{L}^2(\mu) \mid \varphi(u) = 1\}$. Then,

(1) $\mathcal{U} \neq \varnothing$.

(2) Let $\alpha_0 := \inf\{\|u\|_2 \mid u \in \mathcal{U}\}$. Then, $\alpha_0 > 0$.

(3) $\exists u_0 \in \mathcal{U}$ such that $\|u_0\|_2 = \alpha_0$.

Proof.

(1) Since φ is not identically equal to 0, we can pick $f \in \mathcal{L}^2(\mu)$ such that $\varphi(f) \neq 0$. Let $u = \lambda f$ where $\lambda = \frac{1}{\varphi(f)}$. Then, $\varphi(u) = \varphi(\lambda f) = \lambda \cdot \varphi(f) = 1$, so $u \in \mathcal{U}$ and hence $\mathcal{U} \neq \varnothing$.

(2) Let $c \in [0, \infty)$ be such that $|\varphi(f)| \leq c \cdot \|f\|_2$ for all $f \in \mathcal{L}^2(\mu)$. Note that $c \neq 0$, as $c = 0$ would force φ to be identically 0. We will prove that $\alpha_0 \geq \frac{1}{c} > 0$. Pick $u \in \mathcal{U}$. Then, $1 = |\varphi(u)| \leq c \cdot \|u\|_2$ and hence $\|u\|_2 \geq \frac{1}{c} > 0$.

(3) Since $\alpha_0 = \inf\{\|u\|_2 \mid u \in \mathcal{U}\}$, $\forall n \in \mathbb{N}$, $\exists u_n \in \mathcal{U}$ such that

$$\alpha_0 \leq \|u_n\|_2 < \left(1 + \frac{1}{n}\right)\alpha_0.$$

Exercise: By using the parallelogram law, prove that

$$\|u_m - u_n\|_2^2 \leq 6\alpha_0^2\left(\frac{1}{m} + \frac{1}{n}\right), \quad \forall m, n \in \mathbb{N}.$$

Hint: Observe that $\frac{u_m + u_n}{2} \in \mathcal{U}$.

For now, assume the above exercise holds.

Claim (1). $(u_n)_{n=1}^{\infty}$ is a Cauchy sequence in $(\mathcal{L}^2(\mu), \|\cdot\|_2)$.

Proof. Let $\epsilon > 0$ be given. Pick $n_0 \in \mathbb{N}$ such that $\sqrt{\frac{12\alpha_0^2}{n_0}} < \epsilon$. Then, for $m, n \geq n_0$, we have

$$\begin{aligned}\|u_m - u_n\|_2^2 \leq 6\alpha_0^2\left(\frac{1}{m} + \frac{1}{n}\right) &\leq 6\alpha_0^2\left(\frac{1}{n_0} + \frac{1}{n_0}\right)\\ &= \frac{12\alpha_0^2}{n_0} < \epsilon^2 \implies \|u_m - u_n\|_2 < \epsilon.\end{aligned}$$

□

By the completeness of $(\mathcal{L}^2(\mu), \|\cdot\|_2)$, $\exists u_0 \in \mathcal{L}^2(\mu)$ such that $\lim_{n\to\infty} \|u_n - u_0\|_2 = 0$.

Claim (2). $u_0 \in \mathcal{U}$ and $\|u_0\|_2 = \alpha_0$.

Proof. There are two things to verify:

(i) $\varphi(u_0) = 1$.

(ii) $\|u_0\|_2 = \alpha_0$.

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$\square$

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Lemma 3.17. Let $\varphi : \mathcal{L}^2(\mu) \to \mathbb{R}$ be a bounded linear functional not identically 0. Consider $\mathcal{U} := \{u \in \mathcal{L}^2(\mu) \mid \varphi(u) = 1\}$ and let $u_0 \in \mathcal{U}$ be such that $\|u_0\|_2 = \alpha_0$ (as in Lemma 3.16). Let $\mathcal{V} := \{v \in \mathcal{L}^2(\mu) \mid \varphi(v) = 0\} = \mathrm{Ker}(\varphi)$. Then, $\langle u_0, v\rangle = 0$ for all $v \in \mathcal{V}$.

Proof. check LEARN file for proof $\square$

Now, we are ready to prove Theorem 3.14.

Proof of Theorem 3.14. The case when $\varphi(g) = 0$ for all $g \in \mathcal{L}^2(\mu)$ is clear, as we can simply take $f_0 = \underline{0} \in \mathcal{L}^2(\mu)$. Now, assume that φ is not identically 0. By Lemma 3.16, we can pick $u_0 \in \mathcal{L}^2(\mu)$ such that $\varphi(u_0) = 1$ and $\|u_0\|_2 = \alpha_0 := \inf\{\|u\|_2 \mid u \in \mathcal{U}\}$.

Claim. If $f_0 = \frac{1}{\alpha_0^2} u_0$, then $\varphi(g) = \langle f_0, g\rangle$ for all $g \in \mathcal{L}^2(\mu)$.

Proof. For $g \in \mathcal{L}^2(\mu)$, let $\varphi(g) = \lambda \in \mathbb{R}$ and look at $v = g - \lambda u_0$. Then,

$$\varphi(v) = \varphi(g - \lambda u_0) = \varphi(g) - \lambda \cdot \varphi(u_0) = \lambda - \lambda \cdot 1 = 0 \implies v \in \mathrm{Ker}(\varphi).$$

By Lemma 3.17, we have $\langle u_0, v\rangle = 0$. Therefore,

$$\begin{aligned} 0 = \langle u_0, v\rangle &= \langle u_0, g - \lambda u_0\rangle \\ &= \langle u_0, g\rangle - \lambda \langle u_0, u_0\rangle \\ &= \langle u_0, g\rangle - \lambda \alpha_0^2. \end{aligned}$$

Thus, $\langle u_0, g\rangle = \lambda\alpha_0^2 \implies \left\langle \frac{u_0}{\alpha_0^2}, g\right\rangle = \lambda = \varphi(g)$. Therefore, $\varphi(g) = \langle f_0, g\rangle$ for all $g \in \mathcal{L}^2(\mu)$. □

Done. □

4 Radon-Nikodym Theorem

4.1 Radon-Nikodym Theorem in the Special Case $\nu \le \mu$

Fix a measurable space $(X, \mathfrak{M})$.

Definition 4.1 (Notation).

Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be positive measures. We write $\nu \le \mu$ if $\nu(A) \le \mu(A)$ for all $A \in \mathfrak{M}$.

Proposition 4.1. Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be positive measures such that $\nu \le \mu$. Then,

$$\int_X f\, d\nu \le \int_X f\, d\mu, \quad \forall f \in \mathrm{Bor}^+(X, \mathbb{R}).$$

Proof. Use the method of the friendly functions. That is, we let

$$\mathcal{G} = \left\{ g \in \mathrm{Bor}^+(X, \mathbb{R}) \mid \int_X g\, d\nu \le \int_X g\, d\mu \right\}.$$

We need to prove that $\mathcal{G} = \mathrm{Bor}^+(X, \mathbb{R})$.

Exercise: check the four properties in Lemma 2.21. $\square$

Remark. Suppose we are given a positive measure $\mu : \mathfrak{M} \to [0, \infty]$ and a function $h \in \mathrm{Bor}^+(X, \mathbb{R})$. By Proposition 2.26, we can define a positive measure $\nu : \mathfrak{M} \to [0, \infty]$ by

$$\nu(A) := \int_A h\, d\mu = \int_X h\chi_A\, d\mu, \quad \forall A \in \mathfrak{M}.$$

We used the notation $d\nu = h\, d\mu$. Now, let's assume that h is such that $h(x) \le 1$ for all $x \in X$. Then, $\forall A \in \mathfrak{M}$, we have $h\chi_A \le \chi_A$ and hence

$$\nu(A) = \int_X h\chi_A\, d\mu \le \int_X \chi_A\, d\mu = \mu(A) \implies \nu \le \mu.$$

Theorem 4.2 (**A Special Case of the Radon-Nikodym Theorem**).
Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be finite positive measures such that $\nu \leq \mu$. Then, $\exists h \in \mathrm{Bor}^+(X, \mathbb{R})$ such that $h(x) \leq 1$ for all $x \in X$ and such that $d\nu = h\, d\mu$, i.e.

$$\nu(A) = \int_A h\, d\mu, \quad \forall A \in \mathfrak{M}.$$

Moreover, h is uniquely determined up to a.e.-μ equality. That is, if $\tilde{h} \in \mathrm{Bor}^+(X, \mathbb{R})$ is such that $d\nu = \tilde{h}\, d\mu$, then $h = \tilde{h}$ a.e.-μ.

Remark.

(1) Easy to see that the converse holds (by the previous remark).

(2) The function h obtained by the theorem is called the **R-N derivative** of υ with respect to μ and is denoted by $h = \frac{d\nu}{d\mu}$.

(3) The R-N derivative is uniquely determined up to μ-a.e. equality. That is, if $h, \tilde{h} \in \mathrm{Bor}^+(X, \mathbb{R})$ are such that $d\nu = h\, d\mu = \tilde{h}\, d\mu$, i.e. $\int_A h\, d\mu = \nu(A) = \int_A \tilde{h}\, d\mu$ for all $A \in \mathfrak{M}$, then $h = \tilde{h}$ a.e.-μ (see A3 Q2).

(4) The difficult part of this theorem is the existence of h. We will prove it by using an $\mathcal{L}^2$-space.

Lemma 4.3. Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be finite positive measures such that $\nu \leq \mu$. Then, $\mathcal{L}^2(\mu) \subseteq \mathcal{L}^2(\nu)$ and for $f \in \mathcal{L}^2(\mu) \subseteq \mathcal{L}^2(\nu)$, we have $\|f\|_{2,\nu} \leq \|f\|_{2,\mu}$.

Proof. Let $f \in \mathcal{L}^2(\mu)$. Then, $\int_X f^2\, d\mu < \infty$. By Proposition 4.1,

$$\int_X f^2\, d\nu \leq \int_X f^2\, d\mu$$

so we have $\int_X f^2\, d\nu < \infty$ and hence $f \in \mathcal{L}^2(\nu)$. Also, this is the same as $\|f\|_{2,\nu}^2 \leq \|f\|_{2,\mu}^2$ and hence $\|f\|_{2,\nu} \leq \|f\|_{2,\mu}$. ☐

Lemma 4.4. Let $\nu : \mathfrak{M} \to [0,\infty]$ be finite positive measures. Then, $\mathcal{L}^2(\nu) \subseteq \mathcal{L}^1(\nu)$ (similar to A4 Q1). Moreover, for $f \in \mathcal{L}^2(\nu) \subseteq \mathcal{L}^1(\nu)$, we have

$$\left|\int_X f \, d\nu\right| \leq \sqrt{\nu(X)} \cdot \|f\|_{2,\nu}.$$

Proof. The proof of $\mathcal{L}^2(\nu) \subseteq \mathcal{L}^1(\nu)$ is similar to A4 Q1.

For the inequality, note that $\underline{1} \in \mathcal{L}^2(\nu)$ and $\left\|\underline{1}\right\|_{2,\nu} = \sqrt{\int_X \underline{1}^2 \, d\nu} = \sqrt{\nu(X)}$. Then, for every $f \in \mathcal{L}^2(\nu)$, we have $f \cdot \underline{1} \in \mathcal{L}^1(\nu)$ with $\left|\int_X f \cdot \underline{1} \, d\nu\right| \leq \|f\|_{2,\nu} \cdot \left\|\underline{1}\right\|_{2,\nu} = \sqrt{\nu(X)} \cdot \|f\|_{2,\nu}$ by Cauchy-Schwarz inequality. $\square$

Proof of Theorem 4.2. Scratch work: the defined h will have $\int_X \chi_A \, d\nu = \int_X h\chi_A \, d\mu$ for all $A \in \mathfrak{M}$. By Proposition 2.27 and A3 Q1, we have

$$\int_X g \, d\nu = \int_X gh \, d\mu, \quad \forall g \in \mathrm{Bor}^+(X, \mathbb{R}).$$

Claim (1). Define $\varphi : \mathcal{L}^2(\mu) \to \mathbb{R}$ by $\varphi(g) = \int_X g \, d\nu$ for all $g \in \mathcal{L}^2(\mu)$. Then, φ is a bounded linear functional on $\mathcal{L}^2(\mu)$.

Proof. Note that $\mathcal{L}^2(\mu) \overset{\text{Lemma 4.3}}{\subseteq} \mathcal{L}^2(\nu) \overset{\text{Lemma 4.4}}{\subseteq} \mathcal{L}^1(\nu)$. So for $g \in \mathcal{L}^2(\mu)$, it makes sense to define $\varphi(g) = \int_X g \, d\nu$ and $\int_X g \, d\nu < \infty$, for all $g \in \mathcal{L}^2(\mu) \subseteq \mathcal{L}^1(\nu)$. Also, φ is linear because of the linearity of the integral with respect to ν. Moreover for $g \in \mathcal{L}^2(\mu)$, we have

$$|\varphi(g)| = \left|\int_X g \, d\nu\right| \overset{\text{Lemma 4.4}}{\leq} \sqrt{\nu(X)} \cdot \|g\|_{2,\nu} \overset{\text{Lemma 4.3}}{\leq} \sqrt{\nu(X)} \cdot \|g\|_{2,\mu}.$$

$\square$

Claim (2). There exists $f_0 \in \mathcal{L}^2(\mu)$ such that $\int_X g \, d\nu = \int_X f_0 g \, d\mu$ for all $g \in \mathcal{L}^2(\mu)$.

Proof. Applying the Riesz Theorem to the linear functional φ from Claim (1). $\square$

Claim (3). Let f_0 be as in Claim (2). Let $Z_- = \{x \in X \mid f_0(x) < 0\}$ and $Z_+ = \{x \in X \mid f_0(x) > 1\}$. Then, $Z_-, Z_+ \in \mathfrak{M}$ and $\mu(Z_-) = \mu(Z_+) = 0$.

Proof. check LEARN □

Conclusion: Let $Z = Z_- \cup Z_+$ and define $h = f_0(1 - \chi_Z)$. That is,

$$h(x) = \begin{cases} f_0(x), & \text{if } 0 \leq f_0(x) \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then, $h \in \mathrm{Bor}^+(X, \mathbb{R})$ with $h(x) \leq 1$ for all $x \in X$. Moreover, note that $h = f_0$ a.e.-μ because $\mu(Z_- \cup Z_+) = 0$. Let $g = \chi_A \in \mathcal{L}^2(\mu)$ for $A \in \mathfrak{M}$. Then, we have

$$\int_X \chi_A \, d\nu = \int_X h\chi_A \, d\mu \quad \text{as desired.}$$

□

Lecture 29, 2026/07/22

4.2 Connecting Equation, and Radon-Nikodym Theorem for $\nu \ll \mu$

Goal: Replace the condition $\nu \leq \mu$ with more general condition $\nu \ll \mu$ (i.e. ν is absolutely continuous with respect to μ).

Definition 4.2 (**Notation**).
Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be positive measures. We can define $\mu + \nu : \mathfrak{M} \to [0, \infty]$ by

$$(\mu + \nu)(A) := \mu(A) + \nu(A), \quad \forall A \in \mathfrak{M}.$$

Remark. It is easy to see that $\mu + \nu$ is a positive measure. Moreover,

$$\int_X f \, d(\mu + \nu) = \int_X f \, d\mu + \int_X f \, d\nu, \quad \forall f \in \mathrm{Bor}^+(X, \mathbb{R}).$$

Proposition 4.5 (**Trick of the Connecting Equation**).
Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be finite positive measures. Then, $\exists h_1, h_2 \in \mathrm{Bor}^+(X, \mathbb{R})$ such that

$$h_1(x) + h_2(x) = 1, \quad \forall x \in X$$

and such that

$$\int_X gh_1\, d\mu = \int_X gh_2\, d\nu, \quad \forall g \in \mathrm{Bor}^+(X, \mathbb{R}).$$

Proof. Observe that $\nu \le \mu + \nu$. By RNT special case, we can find $h \in \mathrm{Bor}^+(X, \mathbb{R})$ such that $h(x) \le 1$ for all $x \in X$ and $d\nu = h\, d(\mu + \nu)$, i.e. $\int_X g\, d\nu = \int_X gh\, d(\mu + \nu)$ for all $g \in \mathrm{Bor}^+(X, \mathbb{R})$. Choose $h_1 = h$, and $h_2 = 1 - h = 1 - h_1 \in \mathrm{Bor}^+(X, \mathbb{R})$. Clearly, $h_1 + h_2 = 1$. We can rewrite the equation:

$$\int_X g\, d\nu = \int_X gh_1\, d(\mu + \nu) = \int_X gh_1\, d\mu + \int_X gh_1\, d\nu.$$

Also since $1 = h_1 + h_2$, we have

$$\begin{aligned} \int_X g\, d\nu &= \int_X g(h_1 + h_2)\, d\nu = \int_X gh_1\, d\nu + \int_X gh_2\, d\nu \\ \implies \int_X gh_1\, d\nu + \int_X gh_2\, d\nu &= \int_X gh_1\, d\mu + \int_X gh_1\, d\nu. \end{aligned} \tag{*}$$

Use the method of the friendly function. Let

$$\mathcal{G} = \left\{ g \in \mathrm{Bor}^+(X, \mathbb{R}) : \int_X gh_1\, d\mu = \int_X gh_2\, d\nu \right\}.$$

We check the four properties in Lemma 2.21.

(1) Let $A \in \mathfrak{M}$ and set $g = \chi_A$. Observe that since ν is finite and h_1 is bounded, $\int_X \chi_A h_1\, d\nu < \infty$. Then, we can cancel $\int_X \chi_A h_1\, d\nu$ from both sides of $(*)$ to get

$$\int_X \chi_A h_1\, d\mu = \int_X \chi_A h_2\, d\nu.$$

(2), (3), (4) are exercises.

Then, $\mathcal{G} = \mathrm{Bor}^+(X, \mathbb{R})$ and hence $\int_X gh_1\, d\mu = \int_X gh_2\, d\nu$ for all $g \in \mathrm{Bor}^+(X, \mathbb{R})$. ☐

Lemma 4.6. Let μ, ν, h_1, h_2 be as in the previous proposition. Consider

$$S = \{x \in X \mid h_1(x) = 1\} = \{x \in X \mid h_2(x) = 0\}.$$

Then, $\mu(S) = 0$.

Proof. Set $g = \chi_S$ in the connecting equation $\int_X gh_1 \, d\mu = \int_X gh_2 \, d\nu$. Then, we have

$$\begin{aligned}
\int_X \chi_S h_1 \, d\mu &= \int_X \chi_S h_2 \, d\nu \\
\implies \int_S h_1 \, d\mu &= \int_S h_2 \, d\nu \\
\implies \int_S 1 \, d\mu &= \int_S 0 \, d\nu \\
\implies \mu(S) &= 0.
\end{aligned}$$

□

Definition 4.3 (**Absolute Continuity**).

Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be positive measures. We say that ν is **absolutely continuous** with respect to μ, denoted by $\nu \ll \mu$, if

$$\Big[A \in \mathfrak{M} \text{ and } \mu(A) = 0\Big] \implies \Big[\nu(A) = 0\Big].$$

Remark. It is clear that

$$\big[\nu \leq \mu\big] \implies \big[\nu \ll \mu\big].$$

Remark. We saw that if $\mu : \mathfrak{M} \to [0, \infty]$ is a positive measure, then we can find ν satisfying $\nu \leq \mu$ by choosing $h \in \mathrm{Bor}^+(X, \mathbb{R})$ with $h(x) \leq 1$ for all $x \in X$ and setting

$$\nu(A) = \int_A h \, d\mu = \int_X h\chi_A \, d\mu, \quad \forall A \in \mathfrak{M}.$$

If h is not bounded, we don't have $\nu \leq \mu$ (exercise).

But we do have $\nu \ll \mu$.

Proof. Let $\nu(A) = \int_A h\, d\mu$ for some $h \in \mathrm{Bor}^+(X, \mathbb{R})$. If $A \in \mathfrak{M}$ is such that $\mu(A) = 0$, then

$$\nu(A) = \int_A h\, d\mu = \int_X h\chi_A\, d\mu = 0.$$

$\square$

Theorem 4.7 (**Radon-Nikodym Theorem**).

Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be finite positive measures such that $\nu \ll \mu$. Then, $\exists h \in \mathrm{Bor}^+(X, \mathbb{R})$ such that $d\nu = h\, d\mu$, i.e. $\nu(A) = \int_A h\, d\mu$ for all $A \in \mathfrak{M}$.

Moreover, the function h is uniquely determined up to μ-a.e. equality. That is, if $\tilde{h} \in \mathrm{Bor}^+(X, \mathbb{R})$ is such that $d\nu = \tilde{h}\, d\mu$, then $h = \tilde{h}$ a.e.-μ.

Proof. We will prove that $\exists h \in \mathrm{Bor}^+(X, \mathbb{R})$ such that

$$\int_X f\, d\nu = \int_X fh\, d\mu, \quad \forall f \in \mathrm{Bor}^+(X, \mathbb{R}).$$

Then, for every $A \in \mathfrak{M}$, take $f = \chi_A$ to get

$$\nu(A) = \int_X \chi_A\, d\nu = \int_X \chi_A h\, d\mu = \int_A h\, d\mu.$$

Let h_1, h_2 be the functions satisfying the connecting equation. Let $A = \{x \in X : h_1(x) < 1\}$ and $S = \{x \in X : h_1(x) = 1\}$. Also, $X = A \sqcup S$. Note that $\forall x \in A$, we have $h_2(x) > 0$. Given $f \in \mathrm{Bor}^+(X, \mathbb{R})$, set

$$g(x) = \begin{cases} \frac{f(x)}{h_2(x)}, & \text{if } x \in A, \\ 0, & \text{if } x \in S. \end{cases}$$

Note that $g \in \mathrm{Bor}^+(X, \mathbb{R})$. Now, applying the connecting equation to g gives

$$\int_X gh_1\, d\mu = \int_X gh_2\, d\nu \implies \int_A f\, d\nu = \int_A f\frac{h_1}{h_2}\, d\mu.$$

Let

$$h(x) = \begin{cases} \frac{h_1(x)}{h_2(x)}, & \text{if } x \in A, \\ 0, & \text{if } x \in S. \end{cases}$$

Then, $\displaystyle\int_A f\,d\nu = \int_A fh\,d\mu$. Since $h = 0$ on S, so $\displaystyle\int_S fh\,d\mu = 0$. Also since $\mu(S) = 0$ and $\nu \ll \mu$, we have $\nu(S) = 0$ and hence $\displaystyle\int_S f\,d\nu = 0$. Therefore, we have

$$\int_X f\,d\nu = \int_X fh\,d\mu, \quad \forall f \in \mathrm{Bor}^+(X, \mathbb{R}).$$

For uniqueness, suppose $\tilde{h} \in \mathrm{Bor}^+(X, \mathbb{R})$ is such that $d\nu = \tilde{h}\,d\mu$. Since ν is finite, we have $\displaystyle\int_X \tilde{h}\,d\mu < \infty$. Let $P = \{x \in X : h(x) - \tilde{h}(x) \geq 0\}$ and $N = \{x \in X : h(x) - \tilde{h}(x) < 0\}$. So we can write

$$\int_X fh\,d\mu = \int_X f\tilde{h}\,d\mu \implies \int_X f(h - \tilde{h})\,d\mu = 0, \quad \text{for } f = \chi_P.$$

Then, letting $f = \chi_P$ gives

$$\int_X \chi_P(h - \tilde{h})\,d\mu = \int_P (h - \tilde{h})\,d\mu = \int_X (h - \tilde{h})^+\,d\mu = 0.$$

Similarly, letting $f = \chi_N$ gives

$$\int_X (h - \tilde{h})^-\,d\mu = 0.$$

Both $(h - \tilde{h})^+$ and $(h - \tilde{h})^-$ are zero μ-a.e. and hence $h = \tilde{h}$ a.e.-μ. $\square$

Lecture 30, 2026/07/24

4.3 Lebesgue Decomposition Theorem

Definition 4.4 (**Concentrated and Mutually Singular Measures**).

Let $(X, \mathfrak{M})$ be a measureable space.

(1) Let $\mu : \mathfrak{M} \to [0, \infty]$ be a positive measure and let $P \in \mathfrak{M}$. We say that μ is **concentrated on** P if $\mu(X \setminus P) = 0$.

(2) Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be positive measures. We say that μ and ν are **mutually singular**, denoted by $\mu \perp \nu$, if $\exists P, Q \in \mathfrak{M}$ with $P \cap Q = \varnothing$ such that μ is concentrated on P and ν is concentrated on Q.

Theorem 4.8 (**Lebesgue Decomposition Theorem**).

Let $(X, \mathfrak{M})$ be a measurable space. Let $\mu, \nu : \mathfrak{M} \to [0, \infty]$ be finite positive measures. Then, $\exists$ finite positive measures $\nu_1, \nu_2 : \mathfrak{M} \to [0, \infty]$, uniquely determined, such that

$$\nu = \nu_1 + \nu_2, \quad \text{where } \nu_1 \ll \mu \text{ and } \nu_2 \perp \mu.$$

Note. ν_1 is called the **absolutely continuous part** of ν with respect to μ, and ν_2 is called the **singular part** of ν with respect to μ. The relation $\nu = \nu_1 + \nu_2$ is called the **Lebesgue decomposition** of ν with respect to μ.

Proof. For uniqueness, see Problem P14.1. For existence, we use the connecting equation:

$$\int_X gh_1 \, d\mu = \int_X gh_2 \, d\nu, \quad \forall g \in \mathrm{Bor}^+(X, \mathbb{R})$$

with $h_1 + h_2 = \underline{1}$. Recall that $S = \{x \in X : h_1(x) = 1\} = \{x \in X : h_2(x) = 0\}$ has $\mu(S) = 0$.

Idea: for every $A \in \mathfrak{M}$, we let

$$\nu_1(A) = \nu(A \cap (X \setminus S)), \quad \nu_2(A) = \nu(A \cap S).$$

We can check that ν_1, ν_2 are finite positive measures. Moreover, $\nu = \nu_1 + \nu_2$. Indeed, for every $A \in \mathfrak{M}$,

$$\begin{aligned} \nu_1(A) + \nu_2(A) &= \nu(A \cap (X \setminus S)) + \nu(A \cap S) \\ &= \nu\Big((A \cap (X \setminus S)) \cup (A \cap S)\Big) = \nu(A). \end{aligned}$$

Claim (1). $\nu_2 \perp \mu$.

Proof. From the definition of ν_2, we have ν_2 is concentrated on $P = S$. Indeed,

$$\nu_2(X \setminus S) = \nu((X \setminus S) \cap S) = \nu(\varnothing) = 0.$$

But μ is concentrated on $Q = X \setminus S$. Indeed,

$$\mu(X \setminus Q) = \mu(S) = 0.$$

Since $P \cap Q = S \cap (X \setminus S) = \emptyset$, we have $\nu_2 \perp \mu$. $\square$

Claim (2). $\nu_1 \ll \mu$.

Proof. Let $A \in \mathfrak{M}$ with $\mu(A) = 0$. We want to show that $\nu_1(A) = 0$. That is, $\nu(A \cap (X \setminus S)) = 0$. Let $g = \chi_A$ in the connecting equation. Then, we have

$$\int_X \chi_A h_1 \, d\mu = \int_X \chi_A h_2 \, d\nu.$$

The LHS is 0 because $\chi_A h_1 = \underline{0}$ a.e.-μ (since $\mu(A) = 0$). Hence, we have $\chi_A h_2 \in \mathrm{Bor}^+(X, \mathbb{R})$ with

$$\int_X \chi_A h_2 \, d\nu = 0.$$

It follows that $\chi_A h_2 = \underline{0}$ a.e.-ν since $f = \underline{0}$ a.e.-$\nu \iff \displaystyle\int_X f \, d\nu = 0$. Let

$$Z = \{x \in X \mid \chi_A(x) h_2(x) \neq 0\}.$$

Then, $\nu(Z) = 0$. But,

$$\begin{aligned} Z &= \{x \in X \mid \chi_A(x) \neq 0\} \cap \{x \in X \mid h_2(x) \neq 0\} \\ &= A \cap (X \setminus S). \end{aligned}$$

Hence, $\nu(A \cap (X \setminus S)) = \nu(Z) = 0$ and hence $\nu_1(A) = 0$. Therefore, $\nu_1 \ll \mu$. $\square$

We are done. $\square$

Lecture 31, 2026/07/27

4.4 Conditional Expectations

Let $(X, \mathfrak{M}, \mu)$ be a probability space. An important application of the Radon-Nikodym theorem is that it allows us to define conditional expectation of a function $f \in \mathcal{L}^1(\mu)$ onto a sub-σ-algebra $\mathfrak{N} \subseteq \mathfrak{M}$.

Definition 4.5 (Notation).
Let $(X, \mathfrak{M})$ be a measurable space and let $\mathfrak{N}$ be a σ-algebra of subsets of X such that $\mathfrak{N} \subseteq \mathfrak{M}$. We define

$$\mathrm{Bor}\left((X, \mathfrak{M}), \mathbb{R}\right) = \{f : X \to \mathbb{R} \mid f^{-1}(B) \in \mathfrak{M}, \forall B \in \mathcal{B}_{\mathbb{R}}\},$$
$$\mathrm{Bor}\left((X, \mathfrak{N}), \mathbb{R}\right) = \{f : X \to \mathbb{R} \mid f^{-1}(B) \in \mathfrak{N}, \forall B \in \mathcal{B}_{\mathbb{R}}\}.$$

Remark. Clearly, $\mathrm{Bor}\left((X, \mathfrak{N}), \mathbb{R}\right) \subseteq \mathrm{Bor}\left((X, \mathfrak{M}), \mathbb{R}\right)$.

Example. Let $X = [0,1]$ and $\mathfrak{M} = \mathcal{B}_{[0,1]}$. We partition $[0,1]$ by $[0,1] = J_1 \cup J_2 \cup J_3$ where

$$J_1 = \left[0, \frac{1}{3}\right], \quad J_2 = \left(\frac{1}{3}, \frac{1}{2}\right], \quad J_3 = \left(\frac{1}{2}, 1\right].$$

Let

$$\mathfrak{N} = \{\varnothing, J_1, J_2, J_3, J_1 \cup J_2, J_1 \cup J_3, J_2 \cup J_3, X\}.$$

Then, $\mathfrak{N} \subseteq \mathfrak{M}$ is a sub-σ-algebra.
Let us compare the spaces of functions $\mathrm{Bor}\left((X, \mathfrak{M}), \mathbb{R}\right)$ and $\mathrm{Bor}\left((X, \mathfrak{N}), \mathbb{R}\right)$. Observe that $\mathrm{Bor}\left((X, \mathfrak{M}), \mathbb{R}\right)$ is a large infinite-dimensional vector space (for instance, it contains all continuous functions $f : [0,1] \to \mathbb{R}$). On the other hand,

$$\mathrm{Bor}\left((X, \mathfrak{N}), \mathbb{R}\right) = \mathrm{Span}\left\{\chi_{J_1}, \chi_{J_2}, \chi_{J_3}\right\}$$

is just a 3-dimensional space.

Proof. **Exercise**: see Problem P14.2. "$\supseteq$" is immediate. For "$\subseteq$", prove (e.g. by contradiction) that $f \in \mathrm{Bor}\left((X, \mathfrak{N}), \mathbb{R}\right) \implies f$ is constant on each of J_1, J_2, J_3. $\square$

Lemma 4.9. Let $(X, \mathfrak{M}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $\mathfrak{N} \subseteq \mathfrak{M}$ be a sub-σ-algebra. Let $\mu_0 : \mathfrak{N} \to [0, \infty]$ be the set-function obtained by restricting μ to $\mathfrak{N}$ (i.e. $\mu_0(N) = \mu(N)$ for all $N \in \mathfrak{N}$). Then, μ_0 is a finite positive measure on $\mathfrak{N}$. Also,

$$\int_X g \, d\mu_0 = \int_X g \, d\mu, \quad \forall g \in \mathrm{Bor}^+\left((X, \mathfrak{N}), \mathbb{R}\right).$$

Proof. Use the method of the friendly function. Let

$$\mathcal{G} = \left\{ g \in \mathrm{Bor}^+\left((X, \mathfrak{N}), \mathbb{R}\right) \mid \int_X g \, d\mu_0 = \int_X g \, d\mu \right\}.$$

Exercise: check the four properties in Lemma 2.21. Then, $\mathcal{G} = \mathrm{Bor}^+\left((X, \mathfrak{N}), \mathbb{R}\right)$ and hence $\int_X g \, d\mu_0 = \int_X g \, d\mu$ for all $g \in \mathrm{Bor}^+\left((X, \mathfrak{N}), \mathbb{R}\right)$. □

Definition 4.6 (Conditional Expectation).
Let $(X, \mathfrak{M}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $\mathfrak{N} \subseteq \mathfrak{M}$ be a sub-σ-algebra. Let $\mu_0 : \mathfrak{N} \to [0, \infty]$ be the restriction of μ to $\mathfrak{N}$. Consider the new measure space $(X, \mathfrak{N}, \mu_0)$.
Let $f \in \mathcal{L}^1(X, \mathfrak{M}, \mu)$. We say that $g : X \to \mathbb{R}$ is a **conditional expectation** of f onto $\mathfrak{N}$ if $g \in \mathcal{L}^1(X, \mathfrak{N}, \mu_0)$ and

$$\int_N g \, d\mu_0 = \int_N f \, d\mu, \quad \forall N \in \mathfrak{N}.$$

Note. g in the definition is only determined up to μ_0-a.e. equality. That is, if g_1, g_2 are both conditional expectations of f onto $\mathfrak{N}$, then $g_1 = g_2$ a.e.-μ_0.

Proposition 4.10. Let $(X, \mathfrak{M}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $\mathfrak{N} \subseteq \mathfrak{M}$ be a sub-σ-algebra. Let $f \in \mathcal{L}^1(X, \mathfrak{M}, \mu)$ be such that $f(x) \geq 0$ for all $x \in X$. Then, f has a conditional expectation g onto $\mathfrak{N}$ such that $g(x) \geq 0$ for all $x \in X$.

Proof. We look at $\mu : \mathfrak{M} \to [0, \infty]$ and $\nu : \mathfrak{M} \to [0, \infty]$ defined by $d\nu = f \, d\mu$. That is,

$$\nu(A) = \int_A f \, d\mu, \quad \forall A \in \mathfrak{M}.$$

Then, ν is a finite positive measure on $\mathfrak{M}$ by Proposition 2.26. Also, $\nu \ll \mu$ because if $A \in \mathfrak{M}$ is such that $\mu(A) = 0$, then

$$f \chi_A = \underline{0} \text{ a.e.-}\mu \implies \int_X f \chi_A \, d\mu = 0 \implies \nu(A) = 0.$$

Now, let us restrict to $\mathfrak{N}$. Let $\mu_0, \nu_0 : \mathfrak{N} \to [0, \infty]$ be the restrictions of μ, ν to $\mathfrak{N}$ respectively. That is, $\mu_0(N) = \mu(N)$ and $\nu_0(N) = \nu(N)$ for all $N \in \mathfrak{N}$. Then, μ_0, ν_0 are finite positive measures on $\mathfrak{N}$ and

$\nu_0 \ll \mu_0$. Indeed, if $\mu_0(N) = 0$ for some $N \in \mathfrak{N}$, then

$$\mu(N) = 0 \implies \nu(N) = 0 \implies \nu_0(N) = 0.$$

Applying the Radon-Nikodym theorem to get a R-N derivative $g \in \mathrm{Bor}^+(X, \mathbb{R})$ s.t. $d\nu_0 = g\, d\mu_0$. That is,

$$\nu_0(N) = \int_N g\, d\mu_0, \quad \forall N \in \mathfrak{N}.$$

Then, for every $N \in \mathfrak{N}$, we have

$$\int_N g\, d\mu_0 = \nu_0(N) = \nu(N) = \int_N f\, d\mu.$$

Note that $g \in \mathcal{L}^1(X, \mathfrak{N}, \mu_0)$ because for $N = X$,

$$\int_X |g|\, d\mu_0 = \int_X g\, d\mu_0 = \nu_0(X) = \nu(X) = \int_X f\, d\mu < \infty$$

since $f \in \mathcal{L}^1(X, \mathfrak{M}, \mu)$. Therefore, g is a conditional expectation of f onto $\mathfrak{N}$. □

Corollary 4.11. Let $(X, \mathfrak{M}, \mu)$ be a measure space with $\mu(X) < \infty$, let $\mathfrak{N} \subseteq \mathfrak{M}$ be a sub-σ-algebra, and let $f \in \mathcal{L}^1(X, \mathfrak{M}, \mu)$. Then,

(1) f has a conditional expectation onto $\mathfrak{N}$.

(2) If g and g' are both conditional expectations of f onto $\mathfrak{N}$, then $\exists Z \in \mathfrak{N}$ with $\mu(Z) = 0$ such that $g(x) = g'(x)$ for all $x \in X \setminus Z$.

Proof. Exercise. □

Definition 4.7 (Notation).

The conditional expectation g of f onto $\mathfrak{N}$ is often denoted by

$$g = \mathbb{E}[f \mid \mathfrak{N}] \in \mathcal{L}^1(X, \mathfrak{N}, \mu_0).$$

Lecture 32, 2026/07/29

Idea: the conditional expectation g "mimics" f on integrals over $N \in \mathfrak{N}$.

Example. Recall the previous example with $X = [0,1]$, $\mathfrak{M} = \mathcal{B}_{[0,1]}$, and

$$\mathfrak{N} = \{\varnothing, J_1, J_2, J_3, J_1 \cup J_2, J_1 \cup J_3, J_2 \cup J_3, X\} = \sigma\text{-Alg}\{J_1, J_2, J_3\}$$

where $J_1 = \left[0, \frac{1}{3}\right]$, $J_2 = \left(\frac{1}{3}, \frac{1}{2}\right]$, and $J_3 = \left(\frac{1}{2}, 1\right]$. Then, $\mathfrak{N}$ is a finite collection of sets, i.e. $|\mathfrak{N}| = 2^3 = 8$. Let $f \in \mathrm{Bor}^+\left((X, \mathfrak{M}), \mathbb{R}\right)$ be $f(x) = x^2$, $\forall x \in [0,1]$. We want to compute the conditional expectation $g = \mathbb{E}\left[f \mid \mathfrak{N}\right]$. However, we are missing the measure μ.to compute the conditional expectation $g = \mathbb{E}\left[f \mid \mathfrak{N}\right]$.
Let us accept that there exists a probability measure $\mu : \mathcal{B}_{[0,1]} \to [0,1]$, uniquely determined, such that

$$\mu\big([a,b]\big) = b - a, \quad \forall 0 \leq a \leq b \leq 1.$$

This is called the **Lebesgue measure**. In particular, we have

$$\mu(\{a\}) = 0, \quad \forall a \in [0,1].$$

This implies that $\mu(J) = \max(J) - \min(J)$ for any interval $J \subseteq [0,1]$. For our case, we have

$$\mu(J_1) = \frac{1}{3}, \quad \mu(J_2) = \frac{1}{6}, \quad \mu(J_3) = \frac{1}{2}.$$

Important property of μ for a continuous function$f : [0,1] \to \mathbb{R}$:

$$\int_X f \, d\mu = \int_0^1 f(x)\, dx \quad \text{(Riemann integral).}$$

Moreover,

$$\int_J f \, d\mu = \int_{\min(J)}^{\max(J)} f(x)\, dx \quad \forall J \subseteq [0,1].$$

Now, we can compute $g = \mathbb{E}\left[f \mid \mathfrak{N}\right]$. Recall that $g \in \mathrm{Bor}^+\left((X, \mathfrak{N}), \mathbb{R}\right)$ is such that

$$\int_N g \, d\mu = \int_N x^2 \, d\mu(x), \quad \forall N \in \mathfrak{N}.$$

Also, $\mathrm{Bor}^+\left((X,\mathfrak{N}),\mathbb{R}\right) = \mathrm{Span}\left\{\chi_{J_1},\ \chi_{J_2},\ \chi_{J_3}\right\}$. Hence, we can write

$$g = \alpha_1\chi_{J_1} + \alpha_2\chi_{J_2} + \alpha_3\chi_{J_3}, \quad \text{for some } \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}.$$

For example, to find α_1, we take $N = J_1$ to get

$$\int_{J_1} g\,d\mu = \int_{J_1} x^2\,d\mu(x) = \int_0^{1/3} x^2\,dx = \frac{x^3}{3}\Big|_0^{1/3} = \frac{1}{81}.$$

On the other hand,

$$\begin{aligned}\int_{J_1} g\,d\mu = \int_X g\chi_{J_1}\,d\mu &= \int_X (\alpha_1\chi_{J_1} + \alpha_2\chi_{J_2} + \alpha_3\chi_{J_3})\chi_{J_1}\,d\mu \\ &= \int_X \alpha_1\chi_{J_1}^2\,d\mu = \alpha_1\int_X \chi_{J_1}\,d\mu = \alpha_1\mu(J_1) = \frac{\alpha_1}{3}.\end{aligned}$$

Then, $\frac{\alpha_1}{3} = \frac{1}{81} \implies \alpha_1 = \frac{1}{27}$. Similarly, we can compute α_2 and α_3 to get

$$\alpha_2 = \frac{19}{108}, \quad \alpha_3 = \frac{7}{24}.$$

Therefore,

$$g = \mathbb{E}\left[f \mid \mathfrak{N}\right] = \frac{1}{27}\chi_{J_1} + \frac{19}{108}\chi_{J_2} + \frac{7}{24}\chi_{J_3}.$$

5 Construction of Measures

Example. Let $X = \mathbb{R}$ with its usual metric. Proceed like in A1 (and P1.3, P1.6),

$$\mathcal{J} = \{\varnothing\} \cup \{(a,b] \mid a < b, a, b \in \mathbb{R}\} \cup \{(-\infty, b] \mid b \in \mathbb{R}\} \cup \{(a, \infty) \mid a \in \mathbb{R}\} \cup \{\mathbb{R}\}.$$

Consider the algebra of subsets of $\mathbb{R}$:

$$\mathfrak{A} = \left\{ A \subseteq \mathbb{R} \mid \exists k \in \mathbb{N} \text{ and } J_1, \dots, J_k \in \mathcal{J} \text{ with } J_i \cap J_j = \varnothing \text{ for } i \neq j \text{ and } E = \bigcup_{i=1}^{k} J_i \right\}.$$

It is easy to see that $\sigma\text{-Alg}(\mathfrak{A}) = \mathcal{B}_{\mathbb{R}}$. Suppose we have $G : \mathbb{R} \to \mathbb{R}$ an increasing map (i.e. $a \leq b \implies G(a) \leq G(b)$). Then, assign $\mu_0\big((a,b]\big) = G(b) - G(a)$. Also,

$$\mu_0\big((-\infty, b]\big) = G(b) - \lim_{s \to -\infty} G(s) \in [0, \infty], \quad \mu_0\big((a, \infty)\big) = \lim_{t \to \infty} G(t) - G(a) \in [0, \infty],$$
$$\mu_0(\mathbb{R}) = \lim_{t \to \infty} G(t) - \lim_{s \to -\infty} G(s) \in [0, \infty].$$

Then, μ_0 extends by plain additivity to become $\mu_0 : \mathcal{E} \to [0, \infty]$ an additive set-function.
Question: Is μ_0 a pre-measure?
If yes, then Caratheodory extends μ_0 to a measure $\mu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ (Lesbegue-Stieltjes measure).

Lecture 33, 2026/07/31

5.1 Pre-Measure and Caratheodory's Extension Theorem

Definition 5.1 (Pre-Measure).
Let X be a non-empty set, let $\mathfrak{A}$ be an algebra of subsets of X, and let $\mu_0 : \mathfrak{A} \to [0, \infty]$ be an additive set-function. We say that μ_0 is a **pre-measure** if it satisfies "pre-sigma-additivity":

$$\text{If } (A_n)_{n=1}^{\infty} \text{ are from } \mathfrak{A} \text{ with } A_i \cap A_j = \varnothing \text{ for } i \neq j \text{ and } \bigcup_{n=1}^{\infty} A_n \in \mathfrak{A},$$
$$\text{then } \mu_0\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

Note. Pre-sigma-additivity is, clearly, necessary if we want to extend μ_0 to a positive measure $\mu : \mathfrak{M} \to [0, \infty]$ where $\mathfrak{M} = \sigma\text{-Alg}(\mathfrak{A})$.

Theorem 5.1 (**Caratheodory's Extension Theorem**).
Let X be a non-emptyset, let $\mathfrak{A}$ be an algebra of subsets of X, and let $\mathfrak{M} = \sigma\text{-Alg}(\mathfrak{A})$. Suppose that $\mu_0 : \mathfrak{A} \to [0, \infty]$ is a pre-measure. Then, $\exists$ a positive measure $\mu : \mathfrak{M} \to [0, \infty]$ which extends μ_0, i.e. $\mu(A) = \mu_0(A)$ for all $A \in \mathfrak{A}$.

We will only discuss the blueprint of the proof.

Remark (**What was Caratheodory's idea?**).
From $\mathfrak{A}$ you overshoot all the way up to 2^X, and then trim down back to $\mathfrak{M} = \sigma\text{-Alg}(\mathfrak{A})$.

Overshoot to 2^X: extend μ_0 to a set function $\mu^* : 2^X \to [0, \infty]$ called an **outer measure**:

Trim-down: Find a σ-algebra $\mathcal{G} \subseteq 2^X$ such that $\mathcal{G} \supseteq \mathfrak{A}$ and such that $\tilde{\mu} = \mu^*|_{\mathcal{G}}$ is a positive measure on $\mathcal{G}$.

Why does the overshoot-and-trim-down idea work?

Fact 1: $\left[\mathcal{G} \text{ is a } \sigma\text{-algebra and } \mathcal{G} \supseteq \mathfrak{A}\right] \implies \left[\mathcal{G} \supseteq \mathfrak{M}\right]$.

Fact 2: Let $\mu : \mathfrak{M} \to [0, \infty]$ be the restriction of $\tilde{\mu}$ from $\mathcal{G}$ to $\mathfrak{M}$. Then, μ is a positive measure on $\mathfrak{M}$.

Fact 3: The measure $\mu : \mathfrak{M} \to [0, \infty]$ found in Fact 2 extends the pre-measure $\mu_0 : \mathfrak{A} \to [0, \infty]$.

Proof. We have $\mu(A) = \tilde{\mu}(A) = \mu^*(A) = \mu_0(A)$ for all $A \in \mathfrak{A}$. □

Definition 5.2 (**Outer Measure**).
Let X be a non-empty set. A function $\mu^* : 2^X \to [0, \infty]$ is called an **outer measure** if it satisfies the following three conditions:

OM1 $\mu^*(\varnothing) = 0$.

OM2 μ^* is an increasing set-function, i.e. $E \subseteq F \implies \mu^*(E) \leq \mu^*(F)$ for all $E, F \subseteq X$.

OM3 μ^* is countably subadditive, i.e. $\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu^*(E_n)$ for every family $(E_n)_{n=1}^{\infty}$ of subsets of X.

Proposition 5.2. Let X be a non-empty set, let $\mathfrak{A}$ be an algebra of subsets of X, and let $\mu_0 : \mathfrak{A} \to [0, \infty]$ be a pre-measure. Define a function $\mu^* : 2^X \to [0, \infty]$ by

$$\mu^*(E) = \inf\left\{\sum_{n=1}^{\infty} \mu_0(A_n) \mid (A_n)_{n=1}^{\infty} \text{ in } \mathfrak{A} \text{ with } \bigcup_{n=1}^{\infty} A_n \supseteq E\right\}, \quad \forall E \subseteq X.$$

Then,

(1) μ^* is an outer measure.

(2) μ^* extends μ_0, i.e. $\mu^*(A) = \mu_0(A)$ for all $A \in \mathfrak{A}$.

Note. This is the overshoot part.

Proposition 5.3 (**The Trim-Down Step of Caratheodory, Part 1**).
Let X be a non-empty set, and let $\mu^* : 2^X \to [0, \infty]$ be an outer measure. We say that a subset $G \subseteq X$ is "good for μ" if it satisfies:

$$\text{whenever } E \subseteq X \text{ and } F \subseteq X \setminus G, \text{ we have } \mu^*(E \cup F) = \mu^*(E) + \mu^*(F).$$

Then,

(1) The collection of sets $\mathcal{G} := \{G \subseteq X \mid G \text{ is good for } \mu^*\} \subseteq 2^X$ is a σ-algebra.

(2) The restriction $\tilde{\mu} = \mu^*|_{\mathcal{G}}$ is a positive measure on $\mathcal{G}$.

Proposition 5.4 (**The Trim-Down Step of Caratheodory, Part 2**).
Consider the framework and notation from Proposition 5.2. Let μ^* be the outer measure of that proposition. Then, every set $A \in \mathfrak{A}$ is good for μ^*.

Lecture 34, 2026/08/04

5.2 Lebesgue-Stieltjes Measure on $\mathbb{R}$

Example. Recall the previous example: let $X = \mathbb{R}$ with its usual metric. Proceed like in problems of Lecture 1 on learn.

$$\mathcal{J} = \{\emptyset\} \cup \{(a, b] \mid a < b, a, b \in \mathbb{R}\} \cup \{(-\infty, b] \mid b \in \mathbb{R}\} \cup \{(a, \infty) \mid a \in \mathbb{R}\} \cup \{\mathbb{R}\}.$$

Recall that $\mathcal{J}$ is a semi-algebra of subsets of $\mathbb{R}$. Consider the algebra of subsets of $\mathbb{R}$:

$$\mathfrak{A} = \left\{ A \subseteq \mathbb{R} \mid \exists k \in \mathbb{N} \text{ and } J_1, \dots, J_k \in \mathcal{J} \text{ with } J_i \cap J_j = \varnothing \text{ for } i \neq j \text{ and } E = \bigcup_{i=1}^{k} J_i \right\}$$

with σ-Alg$(\mathfrak{A}) = \mathcal{B}_{\mathbb{R}}$.

Syppose we are given $G : \mathbb{R} \to \mathbb{R}$ an increasing map (i.e. $a \leq b \implies G(a) \leq G(b)$). Denote $\ell_+ := \lim_{t\to\infty} G(t) \in \mathbb{R} \cup \{\infty\}$ and $\ell_- := \lim_{s\to-\infty} G(s) \in \mathbb{R} \cup \{-\infty\}$.

Define $\mu_0 : \mathfrak{A} \to [0, \infty]$ by

$$\begin{aligned}
\mu_0\big((a, b]\big) &= G(b) - G(a) \in [0, \infty], \quad \forall a < b \text{ in } \mathbb{R}, \\
\mu_0\big((-\infty, b]\big) &= G(b) - \ell_- \in [0, \infty], \quad \forall b \in \mathbb{R}, \\
\mu_0\big((a, \infty)\big) &= \ell_+ - G(a) \in [0, \infty], \quad \forall a \in \mathbb{R}, \\
\mu_0(\mathbb{R}) &= \ell_+ - \ell_- \in [0, \infty].
\end{aligned}$$

Note that for $(a, c] = (a, b] \cup (b, c]$ in $\mathcal{J}$, $\mu_0\big((a, c]\big) = \mu_0\big((a, b]\big) + \mu_0\big((b, c]\big)$.

From P1.5 and P1.6, we can extend μ_0 to a finitely additive set-function on $\mathfrak{A}$, which we still denote by μ_0. So, we have $\mu_0 : \mathfrak{A} \to [0, \infty]$.

Definition 5.3 (**Cadlag**).

We say that the map $G : \mathbb{R} \to \mathbb{R}$ from the previous example is **cadlag** if it is continuous from the right at every $b \in \mathbb{R}$, i.e.

$$\lim_{s\to b^+} G(s) = G(b), \quad \forall b \in \mathbb{R}.$$

More precisely, for every $a \in \mathbb{R}$,

$$\lim_{t\to a^+} G(t) \text{ exists and is } \leq G(a).$$

Proposition 5.5. Suppose the increasing function $G : \mathbb{R} \to \mathbb{R}$ from the previous example is cadlag. Then, the additive set-function $\mu_0 : \mathfrak{A} \to [0, \infty]$ constructed there is a pre-measure, and hence extends to a positive measure $\mu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$.

Definition 5.4 (**Lebesgue-Stieltjes Measure**).

The resulting measure $\mu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ from the previous proposition is called the **Lebesgue-Stieltjes measure** on $\mathcal{B}_{\mathbb{R}}$ defined by G.

Note. μ is uniquely determined by the requirement that $\mu\big((a,b]\big) = G(b) - G(a)$ for all $a < b$ in $\mathbb{R}$.

Proof. Let $\mu, \tilde{\mu} : \mathcal{B}_{\mathbb{R}} \to [0,\infty]$ be two positive measures such that $\mu\big((a,b]\big) = G(b) - G(a) = \tilde{\mu}\big((a,b]\big)$ for all $a < b$ in $\mathbb{R}$. Then, for $a < b$ in $\mathbb{R}$, we have

$$\mu\big((a,b)\big) = \lim_{n\to\infty} \mu\left(\left(a, b - \frac{1}{n}\right]\right) = \lim_{n\to\infty} G\left(b - \frac{1}{n}\right) - G(a) = G(b) - G(a) = \tilde{\mu}\big((a,b)\big).$$

From $\mu\big((a,b)\big) = \tilde{\mu}\big((a,b)\big)$ $\forall a < b$ in $\mathbb{R}$, we conclude further that $\mu(U) = \tilde{\mu}(U)$ $\forall U \subseteq \mathbb{R}$ open (write U as a contable disjoint union of open intervals). FInally, we have $\mu(B) = \tilde{\mu}(B)$ $\forall B \in \mathcal{B}_{\mathbb{R}}$ by using regularity considerations from P2.5. $\square$

Remark. How do we see that relevance of the cadlag property? Think in reverse, from μ towards G.

Given $s_n \searrow b$ in $\mathbb{R}$ (i.e. $s_n \geq s_{n+1}$ and $\lim_{n\to\infty} s_n = b$), pick $a < b$ and write $(a,b] = \bigcap_{n=1}^{\infty} (a, s_n]$. The continuity of μ along decreasing chain gives

$$\mu\big((a,b]\big) = \lim_{n\to\infty} \mu\left((a, s_n]\right) = \lim_{n\to\infty} G(s_n) - G(a) \implies G(b) = \lim_{n\to\infty} G(s_n).$$

How about a sequence $t_n \nearrow a$ in $\mathbb{R}$?

Pick $b > a$ and write $\bigcap_{n=1}^{\infty} (t_n, b] = [a,b]$. Then,

$$G(b) - G(t_n) = \lim_{n\to\infty} \mu\left((t_n, b]\right) = \mu([a,b]) = \mu(\{a\}) + \mu((a,b]) = G(b) - G(a) + \mu(\{a\}).$$

This gives

$$\lim_{n\to\infty} G(t_n) = G(a) - \mu(\{a\}) \leq G(a).$$

Note: we write "$<$" precisely when a is an atom of μ (i.e. $\mu(\{a\}) > 0$).

Example. Pick $a < b$ in $\mathbb{R}$. Let

$$G(t) = \begin{cases} 0, & t \leq a, \\ t - a, & a \leq t \leq b, \\ b - a, & t \geq b. \end{cases}$$

The resulting μ is the Lebesgue measure on $[a, b]$.

Note. For the final exam, know the idea of Caratheodory's extension theorem, i.e. the overshoot and trim-down idea and be able to explain each. However, there won't be detailed proofs or definitions in this part.

END OF PMATH 450!